

SOLUTIONS MANUAL FOR

Understanding Advanced
Statistical Methods

—by—

Peter Westfall and Kevin S.S. Henning



CRC Press
Taylor & Francis Group

SOLUTIONS MANUAL FOR

Understanding Advanced Statistical Methods

_____ by _____

Peter Westfall and Kevin S.S. Henning



CRC Press

Taylor & Francis Group

Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an informa business

CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

© 2014 by Taylor & Francis Group, LLC
CRC Press is an imprint of Taylor & Francis Group, an Informa business

No claim to original U.S. Government works

Printed on acid-free paper
Version Date: 20130712

International Standard Book Number-13: 978-1-4822-2600-3 (Ancillary)

This book contains information obtained from authentic and highly regarded sources. Reasonable efforts have been made to publish reliable data and information, but the author and publisher cannot assume responsibility for the validity of all materials or the consequences of their use. The authors and publishers have attempted to trace the copyright holders of all material reproduced in this publication and apologize to copyright holders if permission to publish in this form has not been obtained. If any copyright material has not been acknowledged please write and let us know so we may rectify in any future reprint.

Except as permitted under U.S. Copyright Law, no part of this book may be reprinted, reproduced, transmitted, or utilized in any form by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying, microfilming, and recording, or in any information storage or retrieval system, without written permission from the publishers.

For permission to photocopy or use material electronically from this work, please access www.copyright.com (<http://www.copyright.com/>) or contact the Copyright Clearance Center, Inc. (CCC), 222 Rosewood Drive, Danvers, MA 01923, 978-750-8400. CCC is a not-for-profit organization that provides licenses and registration for a variety of users. For organizations that have been granted a photocopy license by the CCC, a separate system of payment has been arranged.

Trademark Notice: Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation without intent to infringe.

Visit the Taylor & Francis Web site at
<http://www.taylorandfrancis.com>

and the CRC Press Web site at
<http://www.crcpress.com>

Contents

1. Introduction: Probability, Statistics, and Science	1
2. Random Variables and Their Probability Distributions	5
3. Probability Calculation and Simulation	13
4. Identifying Distributions	19
5. Conditional Distributions and Independence	26
6. Marginal Distributions, Joint Distributions, Independence, and Bayes' Theorem	30
7. Sampling from Populations and Processes	36
8. Expected Value and the Law of Large Numbers	41
9. Functions of Random Variables: Their Distributions and Expected Values	48
10. Distributions of Totals	52
11. Estimation: Unbiasedness, Consistency, and Efficiency	60
12. Likelihood Function and Maximum Likelihood Estimates	64
13. Bayesian Statistics	73
14. Frequentist Statistical Methods	79
15. Are Your Results Explainable by Chance Alone?	82
16. Chi-Squared, Student's t, and F-Distributions, with Applications	89
17. Likelihood Ratio Tests	94
18. Sample Size and Power	99
19. Robustness and Nonparametric Methods	105
Appendix. SAS Code for Selected Problems	107

1. Introduction: Probability, Statistics, and Science

Solutions to Exercises

1. $Y_1 = Y_0(1 + R_1)$ implies that $Y_1 = Y_0 + Y_0R_1$, which in turn implies $Y_1 - Y_0 = Y_0R_1$, and finally that $R_1 = (Y_1 - Y_0)/Y_0$.
2. To show: $Y_{10} = Y_9(1 + R_{10})$. But $R_{10} = (Y_{10} - Y_9)/Y_9$, so $1 + R_{10} = Y_{10}/Y_9$, implying the desired result.
3. I tried $\lambda = 0.01, 0.5, 1.0, 2.0, 4.0$. here are the first few values generated:

$\lambda = 0.01$	$\lambda = 0.5$	$\lambda = 1.0$	$\lambda = 2.0$	$\lambda = 4.0$
0	0	1	2	5
0	0	0	0	5
0	0	0	2	3
0	1	1	1	4
0	1	3	2	6
0	0	1	1	5
0	1	0	1	1
0	0	0	2	6
0	0	1	1	5
0	2	1	6	7
0	0	1	2	5
0	0	3	2	1
0	0	0	2	4
0	1	0	5	4

According to the problem description and the data generated, $\lambda = 0.5$ seem reasonable. Larger λ predict too many sales, and $\lambda = 0.01$ predicts too few sales.

4.
 - A. (i) Luxury car sales processes (buyer behavior, advertising, dealership reputation); (ii) design: go to the dealership and ask for their records, measurement: the count of sales of luxury cars in a given day; (iii) the numeric values of sales that you will collect, (iv) the Poisson probability distribution, (v) the numbers produced by the Poisson distribution for different λ .
 - B. If it produces sales data that look like sales data that you would actually see, for some values of the parameters. By this criterion, the model with $\lambda = 0.5$ was good, or at least better than the other choices.
 - C. (i), (ii) and (iii) are reality; (iv) and (v) are the model.
5. On average, is more than one car sold per day?
6. Here are some data* produced by a normal distribution with mean = 0 and standard deviation = 1. 0.295754035, -0.422794528, -0.19995241, -0.538682343, -0.820380137, 0.303595016. These numbers don't look anything like car sales. For one thing, they aren't integers. For another, some are negative.

7.

A. The first few:

Bernoulli Outcome	Stock Movement
1	Up
0	Down
0	Down
1	Up
1	Up
0	Down
1	Up
1	Up
1	Up
1	Up
0	Down
0	Down
0	Down
0	Down
1	Up
1	Up

B. (i) financial markets; (ii) design: go to the financial pages on the internet, measurement: either “up” or “down”; (iii) the list of “up” or “down” determinations that you get after collecting the data; (iv) the recoded Bernoulli distribution, (v) the values produced by the computer using recoded Bernoulli outcomes. (i) – (iii) are real, (iv), (v) are model.

C. With a p of 0.57, I would expect 57%, or 570.

D. Observed 567, not 570. Since this is random, you expect potentially different values every time, much like a flip of 11 coins.

8. 73 out of 1000 for an estimate of 0.073.

9.

18. Here is the pdf:

y	$p(y)$
0	π_0
1	π_1
2	π_2
3	π_3
...	...
Total:	1.00

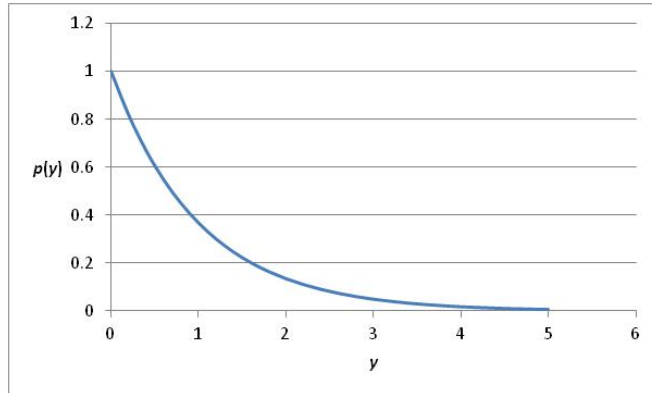
It is better to use π 's because the actual probabilities are unknown (*model has unknown parameters*).

19. The set of outcomes of data produced by the deterministic model is just a set comprised of a single number, $\{3.10\}$, and this does not match reality where the set of driving times has many possible (infinitely many) values.

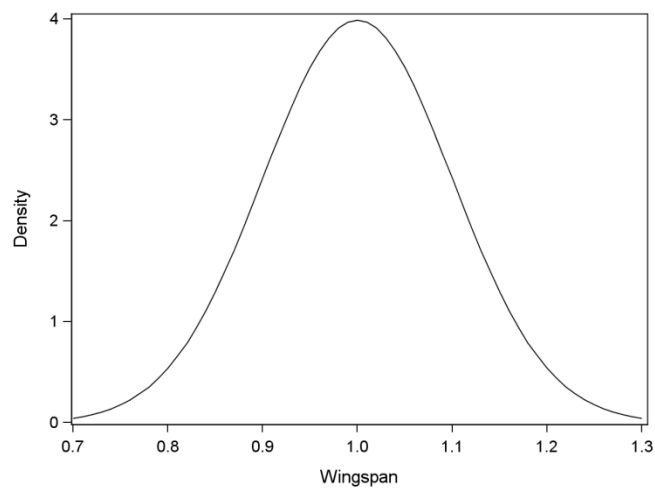
2. Random Variables and Their Probability Distributions

Solutions to Exercises

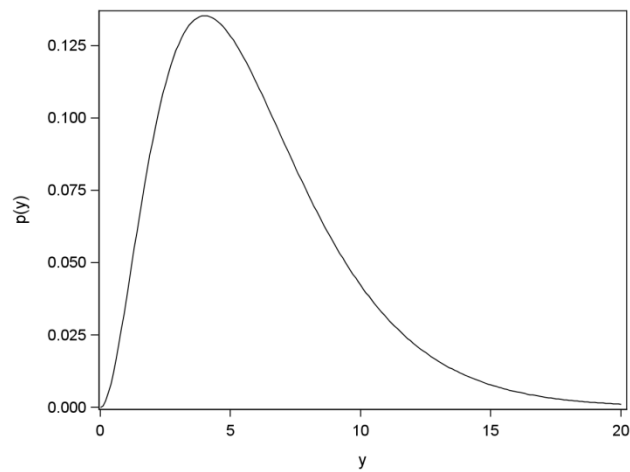
1.



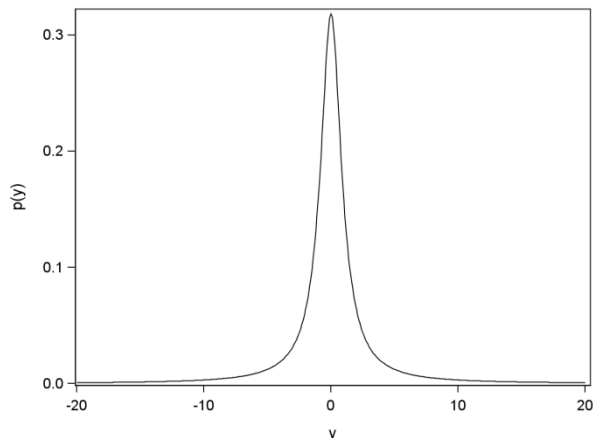
2.



3.



4.



5. (i) Y = questionnaire response on a 1 – 5 scale where people are ambivalent. (ii) Y = questionnaire response on a 1 – 5 scale where people select the low numbers (1, 2, 3) more often. (iii) Y = questionnaire response on a 1 – 5 scale where people select the high numbers (3, 4, 5) more often. (iv) Y = Height of an adult male. (v) Y = Time you have to wait in line. (vi) Y = grade point average of a college student.

6. Function: $p(y) = 0.3^y \times 0.7^{1-y}$, for $y = 0, 1$. List:

y	$p(y)$
0	0.7
1	0.3
Total	1.0

7. At the time of writing:

Wikipedia	This book
$f(k, p)$	$p(y \pi)$
K	y
p	π

8. $p(y | \theta) = \pi_1^{I(y=\text{red})} \pi_2^{I(y=\text{gray})} \pi_3^{I(y=\text{green})}$, for $y \in \{\text{red, gray, green}\}$. When $y = \text{red}$ you get $p(y | \theta) = \pi_1^1 \pi_2^0 \pi_3^0 = \pi_1$. When $y = \text{gray}$ you get $p(y | \theta) = \pi_1^0 \pi_2^1 \pi_3^0 = \pi_2$. When $y = \text{green}$ you get $p(y | \theta) = \pi_1^0 \pi_2^0 \pi_3^1 = \pi_3$. These are the probabilities shown in Table 2.3.

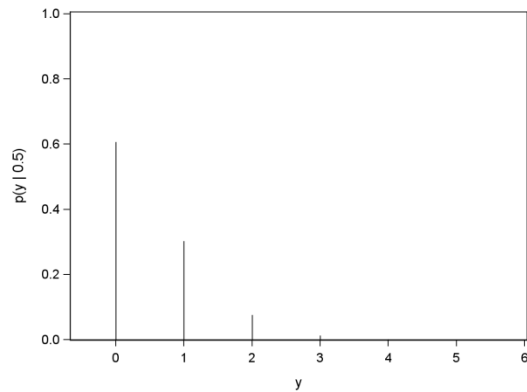
9. From Table 2.4:

y	$p(y \lambda=0.5)$
0	$e^{-0.5} = 0.6065$
1	$0.5e^{-0.5} = 0.3033$
2	$0.5^2 e^{-0.5} / 2 = 0.0758$
3	$0.5^3 e^{-0.5} / 6 = 0.0126$
4	$0.5^4 e^{-0.5} / 24 = 0.0016$

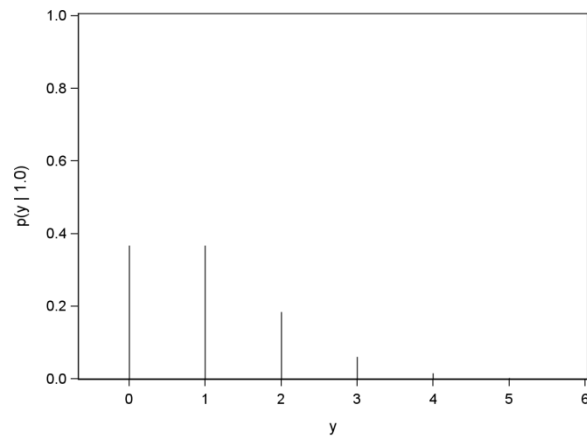
...

Total: 1.00

Graph for $\lambda = 0.5$:



Graph for $\lambda = 1.0$:



Similar: They both produce values that are 0, 1, 2, 3, or Different: The $\lambda = 1.0$ pdf produces fewer zeros and more 2's, 3's, ...

10. Similar: Used for count data or 0, 1, 2, 3, ... data. Different: NB more flexible than Poisson.

11. A. 4, 5, 3, 7, 9, 8, 7, 7, 7, 6 B. 2.72, 2.72, 5.44, 2.72, 1.36, 2.72, 1.36, 2.72, 2.72, 1.36
 C. 0.543, 0.476, 0.774, 0.339, 0.507, 0.511, 0.483, 0.219, 0.455, 0.390.
 D. 120.23, 129.21, 140.56, 147.22, 129.01, 133.32, 135.22, 136.43, 134.00, 139.61

12.

A. Simulated data setsⁱ:

	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10
Fish 0	29	33	40	38	37	35	34	47	40	28

1	34	36	26	37	40	36	40	26	31	42
2	24	23	21	21	16	20	18	21	21	21
3	8	8	11	4	5	8	8	4	6	7
4	5	0	1	0	1	1	0	1	1	2
5	0	0	1	0	0	0	0	1	1	0
6	0	0	0	0	1	0	0	0	0	0

The original data set looks like it could have been produced this way, so the model is good.

B. 4/10 or 40%. In 100 scoops, it is likely to see more than four pupfish, even though this was not seen in the original data.

13.

A. Based on the 10 random samples, the actual data look qualitatively similar to data produced by the truncated normal distribution.

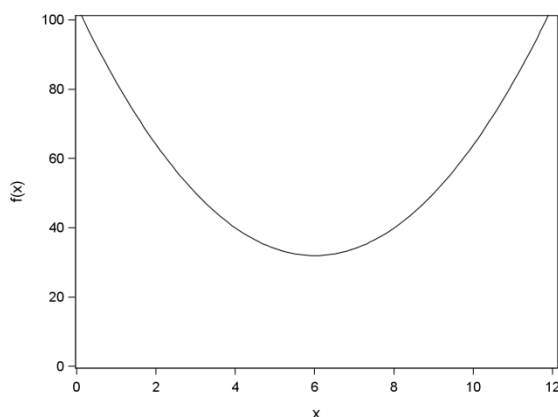
	Simulated Sample									
	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>	<u>10</u>
315	314	309	307	320	309	308	308	301	312	
306	307	306	309	309	308	312	307	304	316	
304	312	306	318	315	310	309	306	316	315	
314	308	319	306	306	310	313	314	315	310	
307	307	305	305	308	310	311	314	313	311	
300	309	305	319	306	315	307	313	315	311	
308	311	306	306	308	319	316	307	315	304	
306	311	317	303	312	307	314	315	308	312	
307	311	306	317	320	313	310	315	310	309	
312	311	306	306	315	303	311	311	312	305	
305	308	307	313	312	310	317	308	303	314	
308	305	299	319	306	304	314	310	312	307	
306	310	315	306	314	312	311	311	317	315	
308	314	307	312	310	309	311	308	318	311	
309	309	311	309	310	311	307	307	316	312	
313	310	303	310	308	319	312	307	311	312	
318	316	313	309	304	314	309	303	310	312	
310	317	307	311	308	320	311	316	310	309	
311	312	315	306	312	307	314	307	304	313	
306	308	318	308	307	312	309	311	314	309	
310	303	305	303	304	308	305	309	314	312	
313	305	306	312	311	302	323	302	311	317	
312	303	305	312	312	305	305	305	302	315	
316	306	310	311	305	306	309	308	316	309	
307	302	313	314	302	317	311	321	311	310	

303 311 313 317 309 302 308 313 310 316
 316 302 305 310 311 310 312 306 312 309
 312 318 308 312 314 308 311 309 304 305
 309 314 304 310 300 303 307 317 314 316
 304 313 316 313 305 309 314 312 311 312

B. Among the 300 simulations, the minimum was 299 and the maximum was 323. So differences from target as large as 13 are possible even when the process is in control.

14.

A. The graphⁱⁱ:



B.

$$f'(x) = \frac{\partial \{(2-x)^2 + (10-x)^2\}}{\partial x} \quad (\text{by definition})$$

$$= \frac{\partial (2-x)^2}{\partial x} + \frac{\partial (10-x)^2}{\partial x} \quad (\text{by D3})$$

$$= 2(x-2) + 2(x-10) \quad (\text{by D8})$$

$$= 4x - 24 \quad (\text{by algebra and arithmetic}).$$

Then:

$$f'(x_{\min}) = 0 \quad (\text{the slope is zero at the minimum})$$

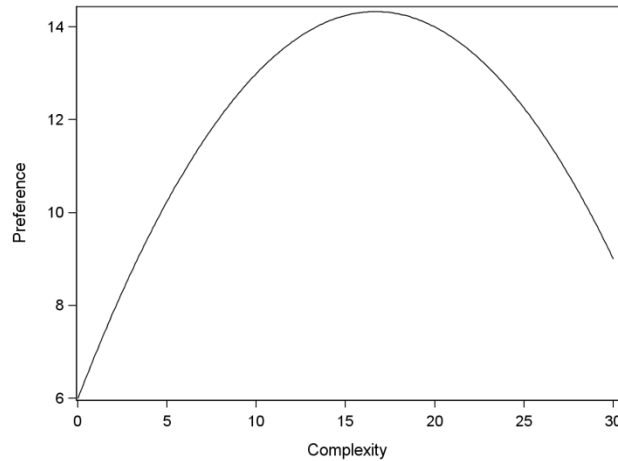
$$\Leftrightarrow 4x_{\min} - 24 = 0 \quad (\text{by substitution, using the calculated derivative from above})$$

$$\Leftrightarrow x_{\min} = 24/4 = 6.0 \quad (\text{by algebra}).$$

- C. The sum of squares function given in the chapter is minimized by the sample average. In the case the sample average is $(2 + 10)/2 = 6.0$, as shown by the calculus solution.

15.

- A. The deterministic model predicts that all 10 people have preference $6 + 15 - 0.03(15)^2 = 14.25$. This is silly. People are different. The model is bad because it produces data with no variability, that do not look like the variable data that are actually observed.
- B. The graphⁱⁱⁱ:



C.

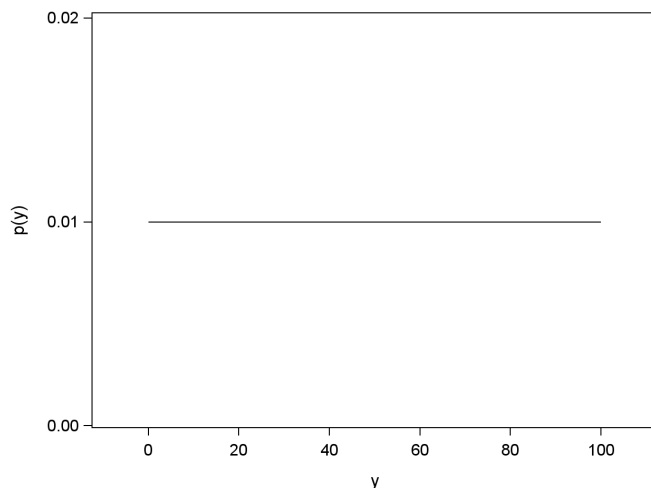
$$\begin{aligned}
 f'(x) &= \frac{\partial \{6 + x - 0.03x^2\}}{\partial x} && \text{(by definition)} \\
 &= \frac{\partial 6}{\partial x} + \frac{\partial x}{\partial x} - 0.03 \frac{\partial x^2}{\partial x} && \text{(by D2, and D3)} \\
 &= 0 + 1 - 0.03(2x) && \text{(by D1 and D4)} \\
 &= 1 - 0.06x && \text{(by algebra and arithmetic).}
 \end{aligned}$$

Then:

$$\begin{aligned}
 f'(x_{\min}) &= 0 && \text{(the slope is zero at the minimum)} \\
 \Leftrightarrow 1 - 0.06x_{\min} &= 0 && \text{(by substitution, using the calculated derivative from above)} \\
 \Leftrightarrow x_{\min} = 1/0.06 = 16.67 &&& \text{(by algebra).}
 \end{aligned}$$

16.

- A. Continuous, since the sample space $S = \{y : 0 < y < 100\}$ is a continuum.
- B. Here is the graph^{iv}:



- C. Numbers between 0 and 100 with many decimals (theoretically infinitely many decimals). The frequency of numbers near 0 will be the same as the frequency of numbers near 50 or any other value between 0 and 100.

17. Stretching the range to 30.00 to 140.00 and using an increment 0.02 yields a sum of 0.99049, as shown here in the first few rows of the spreadsheet corresponding to Figure 2.12.

30.00	0.021333812	0.062777027	0.001339273	2.67855E-05	0.990492776
30.02	0.021333812	0.062935169	0.001342647	2.68529E-05	
30.04	0.021333812	0.063093637	0.001346028	2.69206E-05	
30.06	0.021333812	0.063252433	0.001349415	2.69883E-05	

18. A. Discrete; probabilities sum to 1.0; it's a pdf. B. Discrete; probabilities sum to more than 1.0; it's not a pdf. C. Discrete; probabilities sum to 1.0; it's a pdf. D. Discrete; some probabilities are negative; it's not a pdf. E. Discrete; probabilities sum to 1.0; it's a pdf. F. Discrete; probabilities sum to 1.0; it's a pdf. G. Continuous; area under curve is not 1.0; it's not a pdf. H. Continuous; area under curve is 1.0; it's a pdf. I. Continuous; area under curve is 1.0; it's a pdf. J. Continuous; area under curve is not 1.0; it's not a pdf. K. Continuous; area under curve is not 1.0; it's not a pdf. L. Continuous; area under curve is 1.0; it's a pdf.

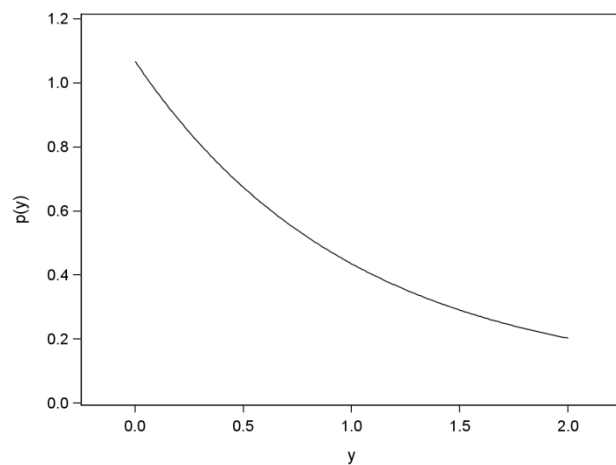
19.

- A. The area under the curve is:

$$\int_0^2 (a + e^{-y}) dy = \int_0^2 a dy + \int_0^2 e^{-y} dy = ay \Big|_0^2 + (-e^{-y}) \Big|_0^2 = (2a - 0) + \{-e^{-2} - (-e^{-0})\} = 2a + 1 - e^{-2}.$$

Since the area has to be 1.0, $2a + 1 - e^{-2} = 1$, implying $a = e^{-2}/2$.

- B. The graph^v:

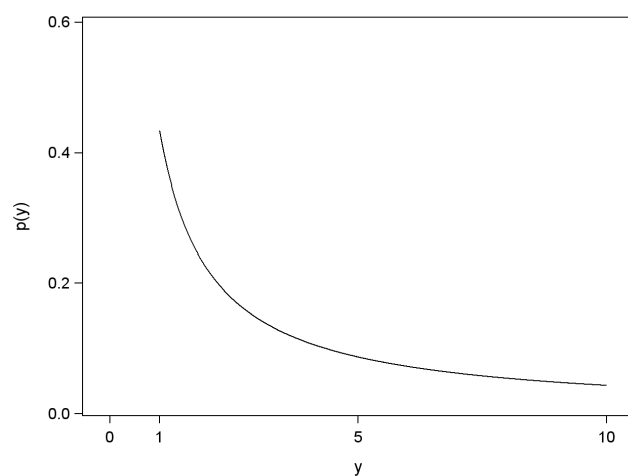


20.

A. The area under the curve is: $\int_1^{10} (a/y) dy = a \int_1^{10} (1/y) dy = a(\ln(y)) \Big|_1^{10} = a\{\ln(10) - \ln(1)\} = a \ln(10)$.

Since the area has to be 1.0, $a \ln(10) = 1$, implying $a = 1/\ln(10)$.

B. Here is the graph:



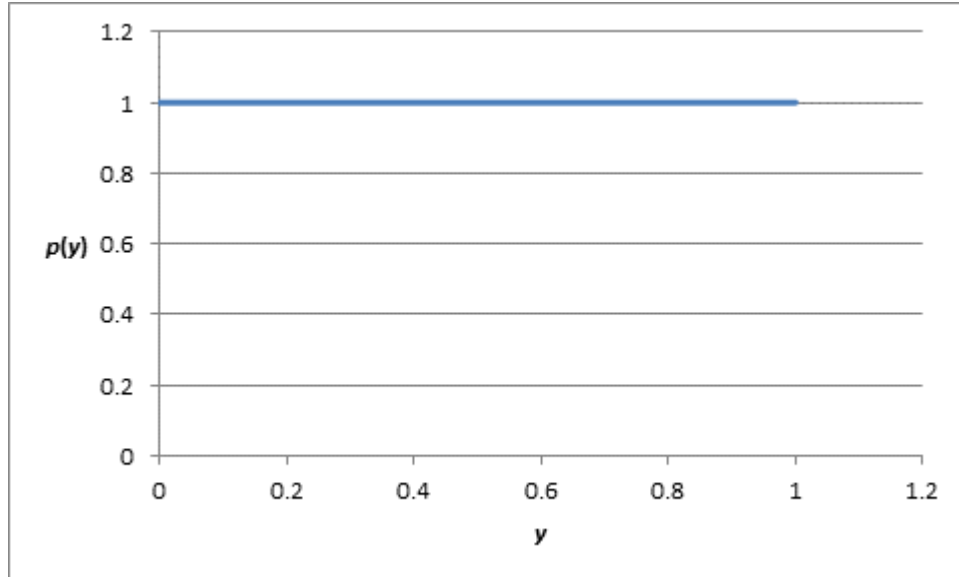
3. Probability Calculation and Simulation

Solutions to Exercises

1. The data values are between 0.00 and 1.00 with many decimals (theoretically infinitely many decimals). The frequency of numbers near 0.00 is nearly the same as the frequency of numbers near 0.50 or any other value between 0.00 and 1.00.

2.

A.



B. The graph is flat, showing that the true frequency of numbers near 0.00 will be exactly the same as the frequency of numbers near 0.50 or any other value between 0.00 and 1.00.

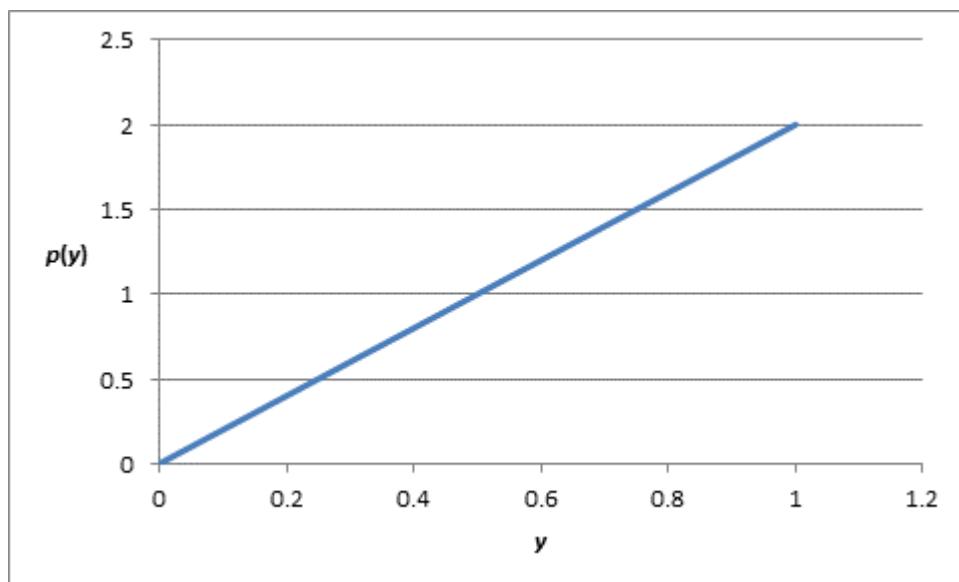
C. $\int_0^{0.5} 1 dy = y \Big|_0^{0.5} = 0.5 - 0 = 0.5$

D. $\int_{0.2}^1 1 dy = y \Big|_{0.2}^1 = 1 - 0.2 = 0.8$

E. $\int_{0.2}^{0.5} 1 dy = y \Big|_{0.2}^{0.5} = 0.5 - 0.2 = 0.3$

3.

A.



B. (i) $2y \geq 0$ for $0 < y < 1$; (ii) $\int_0^1 2y dy = y^2 \Big|_0^1 = 1^2 - 0^2 = 1.0$

C. The data values are between 0.00 and 1.00 with many decimals (theoretically infinitely many decimals). Unlike the uniform distribution, where all values between 0 and 1 are equally frequent, the frequency of triangular numbers near 0.00 is much less than the frequency of numbers near 0.50, which is in turn much less than the frequency of numbers near 1.0.

D. Well, it doesn't mean that there is a 150% probability that Y will be equal to 0.75. Instead, it tells you that the probability of observing a Y within ± 0.05 of 0.75 is approximately $0.10 \times 1.50 = 0.15$.

E. $\int_0^{0.5} 2y dy = y^2 \Big|_0^{0.5} = 0.5^2 - 0^2 = 0.25$

F. $\int_{0.2}^1 2y dy = y^2 \Big|_{0.2}^1 = 1^2 - 0.2^2 = 0.96$

G. $\int_{0.2}^{0.5} 2y dy = y^2 \Big|_{0.2}^{0.5} = 0.5^2 - 0.2^2 = 0.21$

4.

A. 5005 out of 10000 simulated $U(0, 1)$ values are < 0.5 , estimated probability is 0.5005. This is close to the true value 0.5 found in Exercise 2C.

B. 7979 out of 10000 simulated $U(0, 1)$ values are > 0.2 , estimated probability is 0.7979. This is close to the true value 0.8 found in Exercise 2D.

C. 2984 out of 10000 simulated $U(0, 1)$ values are between 0.2 and 0.5, estimated probability is 0.2984. This is close to the true value 0.3 found in Exercise 2E.

5. First off, the cdf of the pdf is $P(y) = y^2$, for $0 < y < 1$. So the inverse cdf is $y = p^{1/2}$.

A. 2499 out of 10000 simulated triangular values are < 0.5 , estimated probability is 0.2499. This is close to the true value 0.25 found in Exercise 3E.

B. 9633 out of 10000 simulated triangular values are > 0.2 , estimated probability is 0.9633. This is close to the true value 0.96 found in Exercise 3F.

C. 2132 out of 10000 simulated triangular values are between 0.2 and 0.5, estimated probability is 0.2132. This is close to the true value 0.21 found in Exercise 3G.

6.

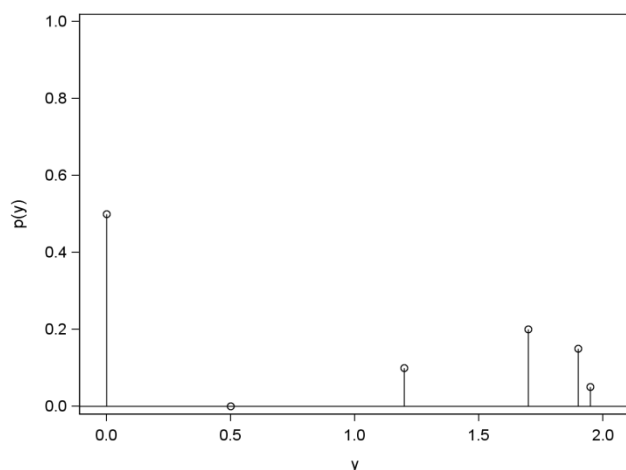
A. $A = \{y : y > 3\} = \{4, 5\}$; $\Pr(Y \in A) = \sum_{y \in A} p(y) = \sum_{y \in \{4, 5\}} p(y) = p(4) + p(5) = 0.10 + 0.30 = 0.40$.

B. $A = \{y : y \geq 3\} = \{3, 4, 5\}$; $\Pr(Y \in A) = \sum_{y \in A} p(y) = \sum_{y \in \{3, 4, 5\}} p(y) = p(3) + p(4) + p(5) = 0.20 + 0.10 + 0.30 = 0.60$.

C. Use $U \sim U(0, 1)$; if $U < 0.25$ then $Y = 1$; if $0.25 \leq U < 0.40$ then $Y = 2$; if $0.40 \leq U < 0.60$ then $Y = 3$; if $0.60 \leq U < 0.70$ then $Y = 4$; if $0.70 \leq U < 1.00$ then $Y = 5$. 5990 of 10000 simulated Y are ≥ 3 ; estimated probability is 0.5990. This is close to the true value 0.60 found in Exercise 6B.

7.

A. Here is the graph^{vi}.



B. All $p(y)$ are ≥ 0 and they all add to 1.0.

C. 0.60

D. 0.05

E. 0.30

F. 0.45

G. Same as $\Pr(0.70 < Y < 1.70)$, 0.10.

H. Same as $\Pr(Y < 0.7) + \Pr(Y > 1.7)$, 0.70.

8.

A. $\int_{50}^{100} 0.0002y dy = 0.0001y^2 \Big|_{50}^{100} = 0.0001(100^2) - 0.0001(50^2) = 0.75$

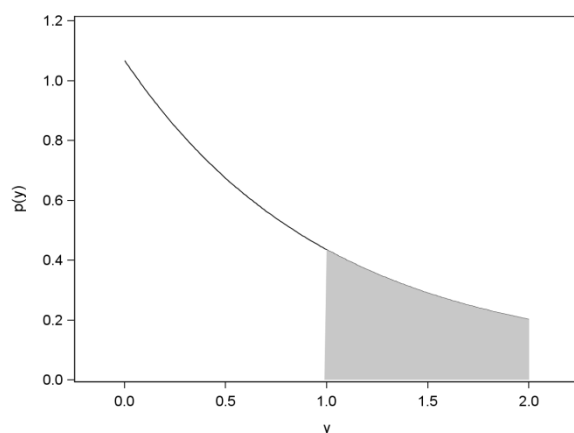
B. Larger triangle: Base = 100, Height = $0.0002(100) = 0.02$, Area = $(1/2)(100)(0.02) = 1.00$.
 Smaller triangle: Base = 50, Height = $0.0002(50) = 0.01$, Area = $(1/2)(50)(0.01) = 0.25$.
 Difference = 0.75.

9.

- A. $\Pr(Y > 90) = \int_{90}^{100} 0.01 dy = 0.01y \Big|_{90}^{100} = 0.01(100) - 0.01(90) = 0.10 - 0.09 = 0.01$. Thus you expect 10% of the 1000 values, or 100, to be > 90 .
- B. Base = 10, Height = 0.01, Area = $10 \times 0.01 = 0.10$; still expect 100 out of 1000.

10. As shown in Exercise 19 of Chapter 2, $a = e^{-2}/2$.

A. Here is the graph:



The probability is given by

$$\begin{aligned} \int_1^2 (e^{-2}/2 + e^{-y}) dy &= \int_1^2 (e^{-2}/2) dy + \int_1^2 e^{-y} dy = (e^{-2}/2)y \Big|_1^2 + (-e^{-y}) \Big|_1^2 \\ &= (e^{-2} - e^{-2}/2) + \{-e^{-2} - (-e^{-1})\} = e^{-2}/2 + e^{-1} - e^{-2} = e^{-1} - e^{-2}/2 = 0.3002. \end{aligned}$$

B. 0.3002.

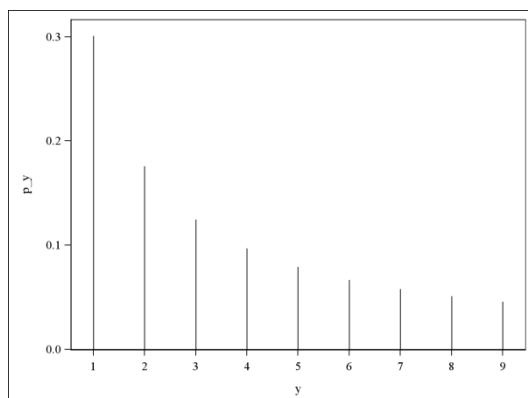
C.

$$\begin{aligned} \int_{0.5}^{1.5} (e^{-2}/2 + e^{-y}) dy &= \int_{0.5}^{1.5} (e^{-2}/2) dy + \int_{0.5}^{1.5} e^{-y} dy = (e^{-2}/2)y \Big|_{0.5}^{1.5} + (-e^{-y}) \Big|_{0.5}^{1.5} \\ &= e^{-2}/2 + e^{-0.5} - e^{-1.5} = 0.4511. \end{aligned}$$

11. Use $a = 1/\ln(10)$ as shown in Exercise 20 for Chapter 2 and follow the methods shown in the solution to Exercise 10.

12.

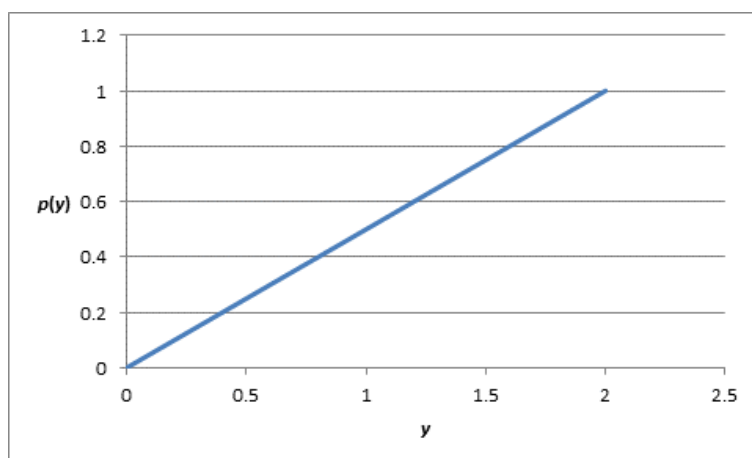
A.



- B. $\log_{10}(1 + 1/1) + \log_{10}(1 + 1/2) + \dots + \log_{10}(1 + 1/9) = \log_{10}((1 + 1)/1) + \log_{10}((2 + 1)/2) + \dots + \log_{10}((9 + 1)/9) = \log_{10}(2/1) + \log_{10}(3/2) + \dots + \log_{10}(10/9) = \log_{10}((2/1) \times (3/2) \times \dots \times (10/9)) = \log_{10}(10) = 1.0$. Or just compute the numbers and add them up.
- C. 0.301 D. 0.0 E. 0.301 F. 0.301 G. 0.301 H. 0.699

13.

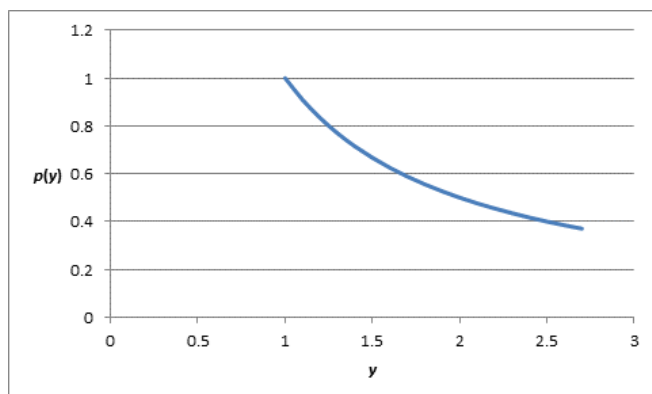
A.



- B. Area = 1.0 and the function is non-negative; it's a pdf. C. Area = $(1/2)(1.70)(1.70/2) = 0.7225$ D. 0 E. $(1/2)(1.90)(1.90/2) - (1/2)(0.50)(0.50/2) = 0.84$ F. 0.84 G. 0.60 H. 0.40

14. Note the cdf is $\int_1^y (1/t)dt = \ln(t)|_1^y = \ln(y) - \ln(1) = \ln(y)$ for $1 < y < e$.

A.



B. Area = 1.0 and the function is non-negative; it's a pdf C. $\ln(1.7)$ D. 0 E. $\ln(1.90)$
 F. $\ln(1.90)$ G. $\ln(1.70)$ H. $1 - \ln(1.70)$.

15.

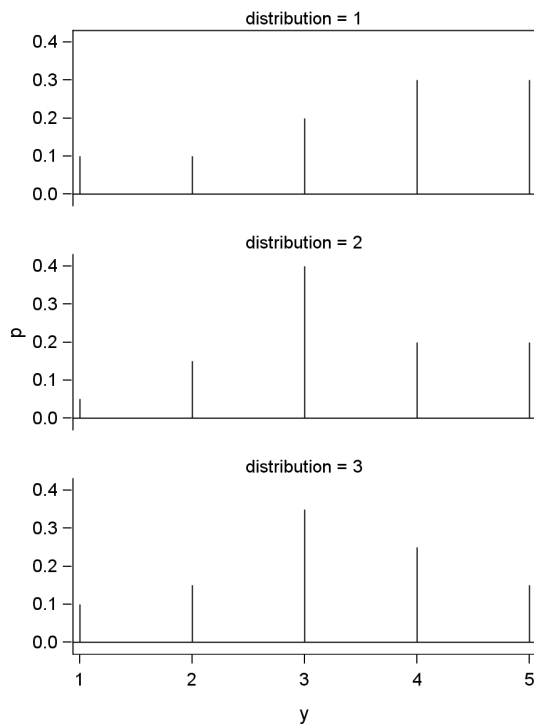
- A. $\int_{10}^{\infty} \frac{1}{\sqrt{2\pi}5} \exp\left\{-\frac{(y-20)^2}{2(5^2)}\right\} dy, =1-\text{NORM.DIST}(10,20,5,\text{TRUE}), 0.97725$
- B. $\int_{-\infty}^{10} \frac{1}{\sqrt{2\pi}5} \exp\left\{-\frac{(y-20)^2}{2(5^2)}\right\} dy, =\text{NORM.DIST}(10,20,5,\text{TRUE}), 0.02275$
- C. $\int_{20}^{\infty} \frac{1}{\sqrt{2\pi}(25)} \exp\left\{-\frac{(y-20)^2}{2(25)}\right\} dy, =1-\text{NORM.DIST}(20,20,\text{SQRT}(25),\text{TRUE}), 0.5000$
- D. $\int_{-\infty}^{10} \frac{1}{\sqrt{2\pi}(5)} \exp\left\{-\frac{(y-10)^2}{2(5)}\right\} dy, =\text{NORM.DIST}(10,10,\text{SQRT}(5),\text{TRUE}), 0.5000$
- E. $\int_{10}^{\infty} \frac{1}{\sqrt{2\pi}(100)} \exp\left\{-\frac{(y-20)^2}{2(100)}\right\} dy, =1-\text{NORM.DIST}(10,20,\text{SQRT}(100),\text{TRUE}), 0.8413$

4. Identifying Distributions

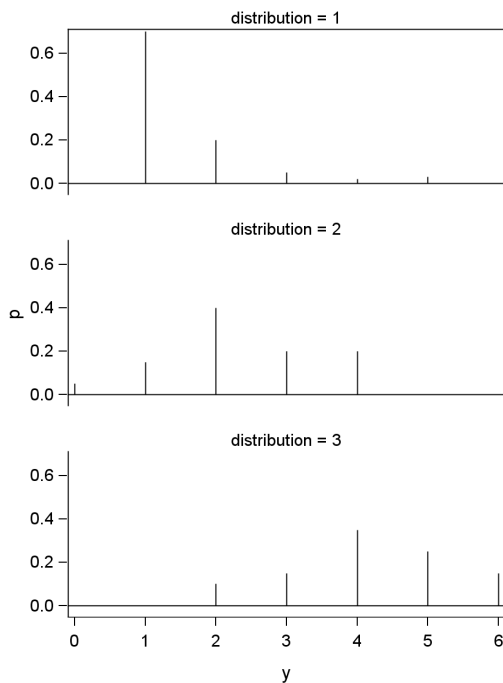
Solutions to Exercises

1.

A. Here are three possibilities^{vii}:



B. Here are some:

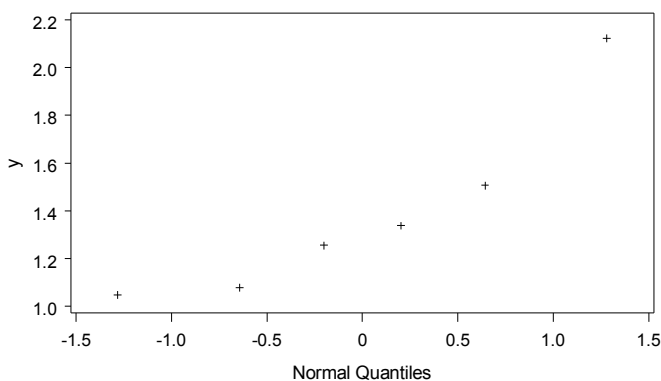


C.

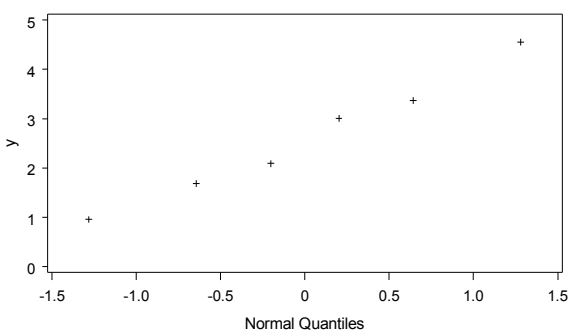
y	$p(y)$
1	π_1
2	π_2
3	π_3
4	π_4
5	π_5
Total:	1.00

Who knows what the parameters are? In this specification they can be anything, hence the model is more believable than any of the models above where the parameters are assumed equal to the given values.

- Letting $Y = \text{return}$, a more realistic model is $Y \sim p(y)$, where $p(y)$ can be any distribution. It's more believable because we aren't assuming that $p(y)$ is a normal distribution, with all its symmetries, or any other constrained distribution type.
- Letting $Y = \text{wait time}$, a more realistic model is $Y \sim p(y)$, where $p(y)$ can be any distribution. It's more believable because we aren't assuming that $p(y)$ is a normal distribution, with all its symmetries, or any other constrained distribution type.
- Adding areas of rectangles, $(.5)(.667) + (.5)(.333) + (.5)(.333) + (.5)(.333) + (.5)(.333) = 1.0$. Without dividing by 0.5, the area would be 0.5.
- Here is one normal q-q plot of data produced by a normal distribution^{viii}:

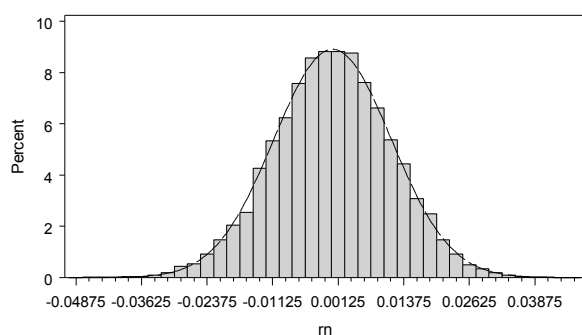


Here is another:



The deviation of the original data from the line is easily explainable by chance alone.

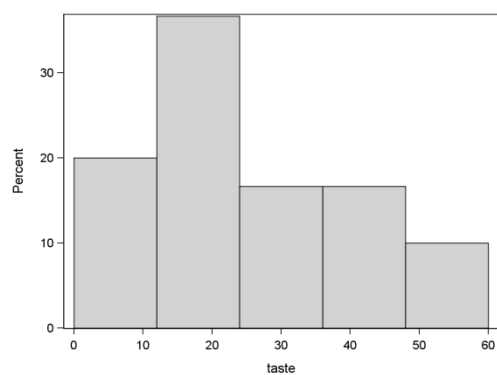
6. Here is the graph^{ix}:



The normal curve looks like a slightly better fit than Figure 4.8, but it's hard to say for sure. The q-q plot is better.

7.

A.



Looks like a right skewed distribution.

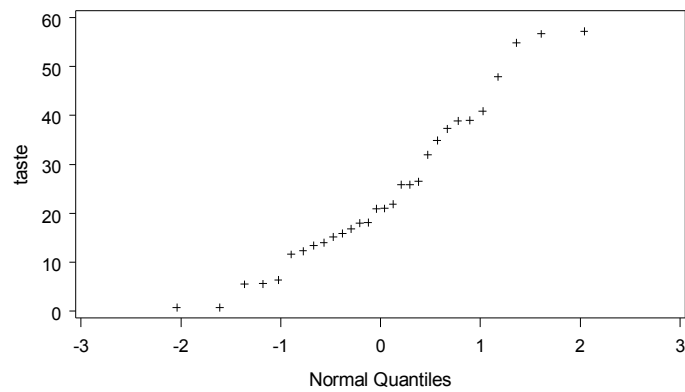
B.

Ordered

Value, i	$y_{(i)}$	$p = (i - 0.5)/n$	$\text{NORM.INV}(p, 24.53333, 16.25538)$
1	0.7	0.017	-10.059

2	0.7	0.050	-2.204
3	5.5	0.083	2.052
4	5.6	0.117	5.160
5	6.4	0.150	7.686
6	11.6	0.183	9.859
7	12.3	0.217	11.797
8	13.4	0.250	13.569
9	14.0	0.283	15.220
10	15.2	0.317	16.779
11	15.9	0.350	18.270
12	16.8	0.383	19.710
13	18.0	0.417	21.113
14	18.1	0.450	22.491
15	20.9	0.483	23.854
16	21.0	0.517	25.213
17	21.9	0.550	26.576
18	25.9	0.583	27.954
19	25.9	0.617	29.357
20	26.5	0.650	30.797
21	32.0	0.683	32.288
22	34.9	0.717	33.847
23	37.3	0.750	35.497
24	38.9	0.783	37.269
25	39.0	0.817	39.208
26	40.9	0.850	41.381
27	47.9	0.883	43.907
28	54.9	0.917	47.014
29	56.7	0.950	51.271
30	57.2	0.983	59.126

C.

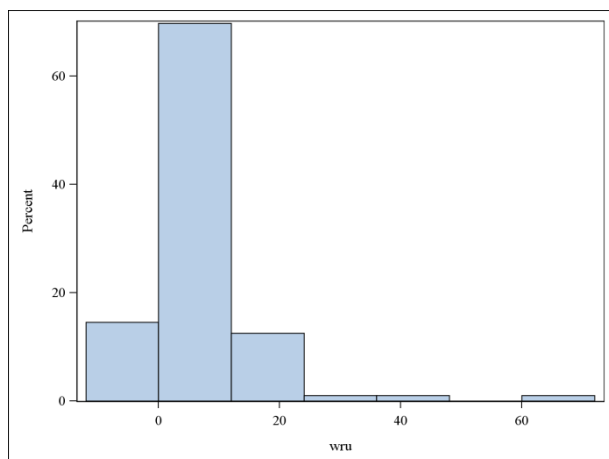


The upward bend corroborates the right skew appearance of the histogram, although there do not appear to be any notable outliers.

- D. While there is skewness in the original data, it is explainable by chance alone where data are simulated from a normal distribution: Similar bends and skewness appearances sometimes appear, even when data are simulated from a normal distribution.

8.

A.

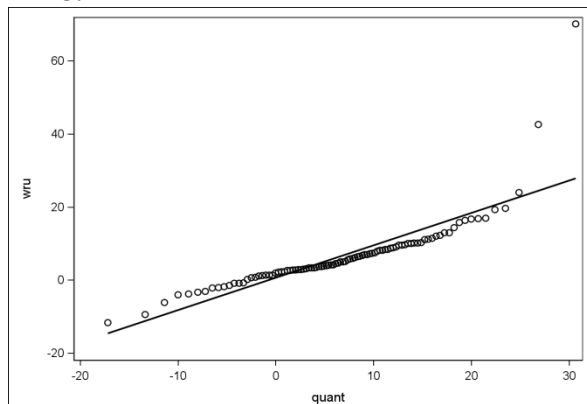


The histogram shows extreme positive skewness. Outliers are apparent on the high side.

B.

<i>Obs</i>	<i>i</i>	<i>p</i>	<i>wru</i>	<i>quant</i>
1	1	0.00521	-11.6	-17.2061
2	2	0.01563	-9.3	-13.3972
3	3	0.02604	-6.0	-11.4224
4	4	0.03646	-3.9	-10.0299
...	...	...	...	...
96	96	0.99479	70.2	30.6461

C.

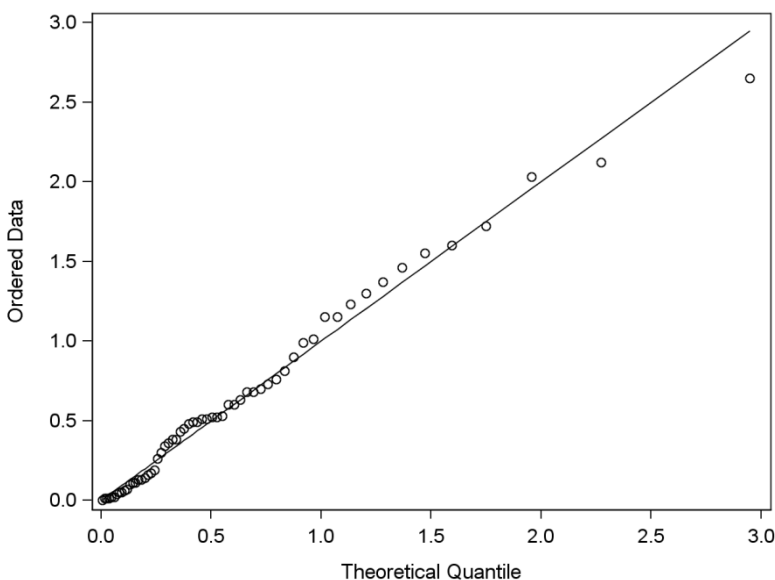


The q-q plot deviates markedly from the line, suggesting significant skewness.

D. Unlike Exercise 8, the pattern in this q-q plot does not appear explainable by chance alone.

9.

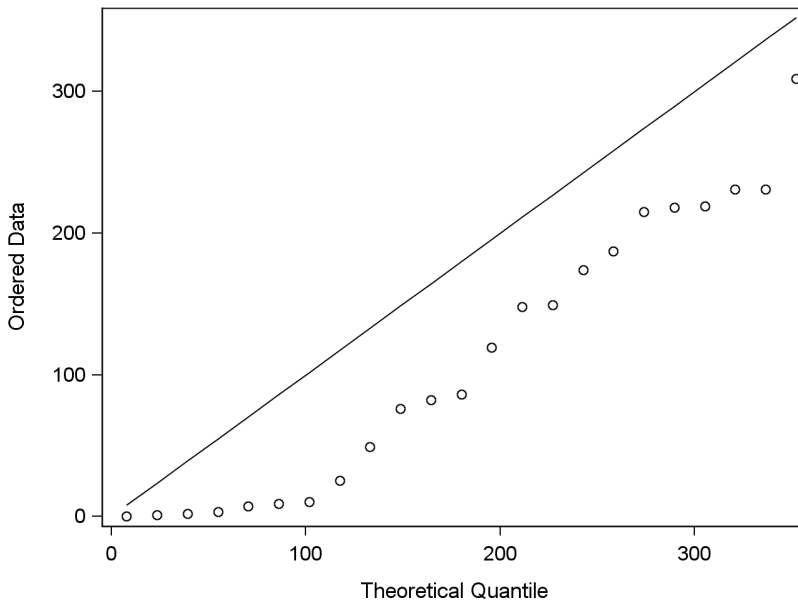
The estimated mean is 0.6158333 so the estimated exponential distribution uses $\lambda = 1/0.6158333$. Plotting the $(i - 0.5)/60$ quantiles from this distribution together with the ordered data values $y_{(i)}$ gives the following graph^x:



The exponential model appears reasonable since the data points are close to the line.

10.

- A. If the top is balanced, it should land anywhere in the 360 degree angle, with all angles equally likely.
- B. If the top is not balanced, some angles might appear more often than others.
- C. Nature: The top's composition and spinning behavior. Design: Spin it a few times. Measurement: the angle from the peak. DATA: the collection of angles that will be recorded.
- D. Here is the graph^{xi}.

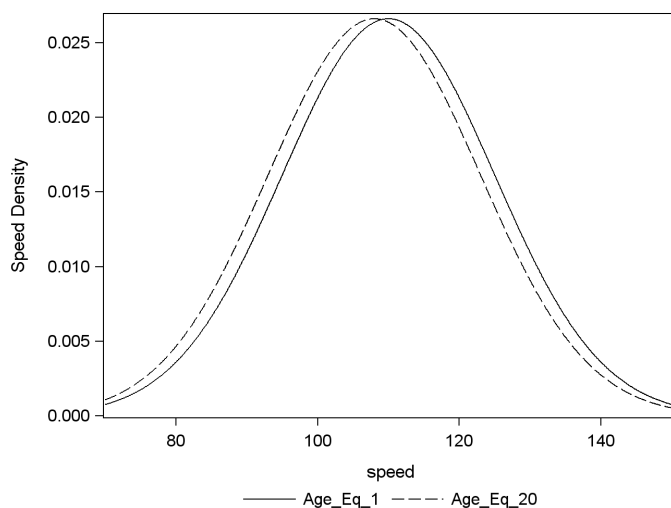


It does not appear that the $U(0, 360)$ model is plausible, since the observed and expected quantiles differ greatly, as shown in the graph.

5. Conditional Distributions and Independence

Solutions to Exercises

1. How about “effect of music on the mind?” and the paper “Effect of Music Therapy on Mood and Social Interaction Among Individuals With Acute Traumatic Brain Injury and Stroke,” in *Rehabilitation Psychology*, August 2000 Vol. 45, No. 3, 274-283.
 - A. $p(y | x)$ is the distribution of mood (Y), depending on the type of music therapy (X). There is a distribution, because different people have different moods, even when they are all exposed to the same therapy. In other words, your mood is not completely determined by whatever music you happen to be listening to, or not listening to.
 - B. If music therapy affects mood, then $p(y | X \in A_1)$ is not the same as $p(y | X \in A_2)$ for some A_1 and A_2 . In other words, the distribution of mood depends on the music therapy.
 - C. If music therapy has no effect on mood, then $p(y | X \in A_1)$ is the same as $p(y | X \in A_2)$ for all A_1 and A_2 . In other words, the distribution of mood is the same no matter what is the music therapy.
2. The graph:



Drivers of 20-year old cars might drive a little slower, with concerns about accidents related to car wear. But there should be very little difference at all.

3.
 - A. Fatalities are not that common, even when drivers are drunk. Here, $p(y | X = 0)$ is the distribution of trip outcome (alive or death) given the driver is sober.

y	$p(y X = 0)$
0, alive	0.99999999
1, death	0.00000001
Total:	1.00000000

$p(y | X = 1)$ is the distribution of trip outcome (alive or death) given the driver is drunk.

y	$p(y X = 0)$
0, alive	0.99999
1, death	0.00001
Total:	1.00000

- B. Although fatalities themselves are rare, when there is a fatality, drunk driving is frequently involved. Here, $p(x | Y = 0)$ is the distribution of driver sobriety (sober or drunk) given a successful trip outcome.

x	$p(x Y = 0)$
0, sober	0.99
1, drunk	0.01
Total:	1.00

$p(x | Y = 1)$ is the distribution of driver sobriety (sober or drunk) given a fatal trip outcome.

y	$p(y X = 1)$
0, sober	0.50
1, drunk	0.50
Total:	1.00

- C. Nature: Drunk driving behavior and fatality occurrences. Design: Collect data on drunk driving from checkpoints, collect data from accident reports. Measurement: binary outcomes X and Y . DATA: The observation you will put into your spreadsheet.

4.

A.

x	$\hat{p}(x)$
1	0.1818
2	0.0909
3	0.2121
4	0.3636
5	0.1515
Total:	1.000

- B. It is unlikely that any of the simulated data sets will show a trend as pronounced as in the original.

5. They'd all be the same graph. Y = housing expense. The definition of independence says that $p(y | X \in A_1) = p(y | X \in A_2)$ for all sets A_1 and A_2 . So when $A_1 = \{x ; x = 100\}$ and when $A_2 = \{x ; x = 110\}$, the distributions $p(y | X \in A_1) = p(y | X \in A_2)$ are identical; i.e., the distributions $p(y | X = 100) = p(y | X = 110)$ are identical. The same argument applies to $X = 100$ and $X = 120$, so all three distributions are identical under independence.

6.

- A. Hans might glue the dice together so the threes are together.

- B. Let $A_1 = \{x ; x = 3\}$ and $A_2 = \{x ; x = 4\}$. Then the distributions $p(y | X \in A_1)$ and $p(y | X \in A_2)$ are as follows:

y	$p(y X \in A_1)$
-----	--------------------

1	0.0
2	0.0
3	1.0
4	0.0
5	0.0
6	0.0
Total:	1.0

y	$p(y X \in A_2)$
1	π_1
2	π_2
3	0.0
4	π_4
5	π_5
6	π_6
Total:	1.0

These distributions are different, so the variables X and Y are dependent. You don't have to know what the π 's are.

7.
 - A. Roll the pair 1000 times, and take the subset of those 1000 rolls where the green die showed one. Suppose there are 170 such rolls. Then $\hat{p}(y | X = 1)$ is the relative frequency distribution of outcomes 1, 2, 3, 4, 5 and 6 among the red die in these 170 cases.
 - B. The true distributions $p(y | X = 1) = p(y | X = 2)$ are identical: Both are discrete uniform on the numbers 1, 2, ..., 6. But the data-based estimates will differ by randomness: On set of 1000 tosses will yield different results than another set of 1000 tosses, by chance alone.
8. They are independent. No matter what is X , Y has the Bernoulli(0.6) distribution.
9. They are dependent. The distribution of Y when $X = 0$ is different uniform distribution from the distribution of Y when $X = 1$.
10. They are dependent. The Poisson distributions of Y differ for every value of X .
11.
 - A. HHHTTHTTH
 - B. The estimated probability of heads in the next toss is the same (50%), no matter whether the previous toss was heads or tails.
 - C. No. The estimates are not the same as the true values. By chance alone, it is possible to observe a pattern like shown in Exercise 11A, even when the tosses are dependent.
12.
 - A. The distribution of the outcome of the 10,000th toss is the same, under independence, no matter what values the other 9,999 tosses take on. So the distribution is Bernoulli(0.5).
 - B. You cheat – you get heads on the first toss, then pick the coin up one millimeter and drop it.
 - C. A coin can be bent severely so that it will almost certainly land heads up.

13.

A.

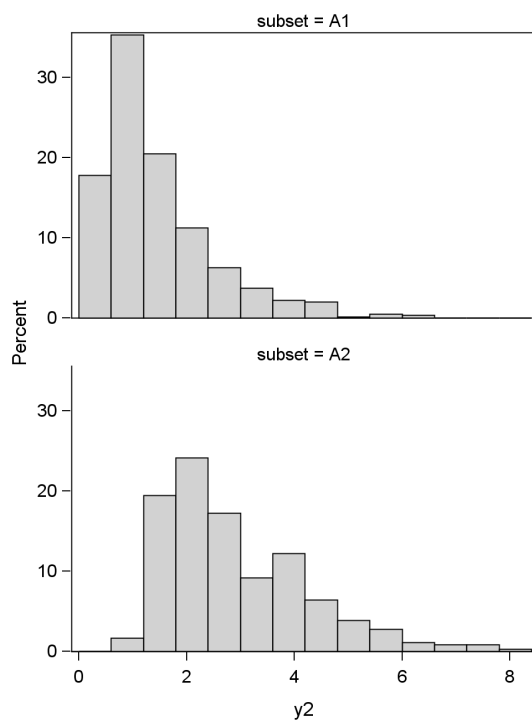
Adverse Event, y	$\hat{p}(y X = \text{Placebo})$
No	$72/80 = 0.900$
Yes	$8/80 = 0.100$
Total:	1.000

B.

Adverse Event, y	$\hat{p}(y X = \text{New Drug})$
No	$55/80 = 0.6875$
Yes	$25/80 = 0.3125$
Total:	1.0000

C. It seems that the new drug has relatively many side effects. Perhaps it would be wise to discontinue development.

D. Yes, because there is 5 or more in all four categories.

14. The graph^{xii}:

There is an appearance of morphing, suggesting that $p(y_2 | Y_1 \leq 1)$ differs from $p(y_2 | Y_1 > 1)$, and therefore that Y_1 and Y_2 are dependent.

6. Marginal Distributions, Joint Distributions, Independence, and Bayes' Theorem

Solutions to Exercises

1.

A.

Importance for your business, x	$p(x)$
Not Important	0.99999
Important	0.00001
Total:	1.00000

B.

Contains "Zia Technologies," y	$p(y X = \text{not important})$
No	0.99999
Yes	0.00001
Total:	1.00000

Contains "Zia Technologies," y	$p(y X = \text{important})$
No	0.10
Yes	0.90
Total:	1.00

C.

Importance for Business	$p(y x)$ $= p(\text{Yes} x)$	$p(x)$	$p(y x) p(x)$	$p(x y)$
Not Imp.	0.00001	0.99999	0.0000099999	0.526313296
Important	0.90	0.00001	0.0000090000	0.473686704
Totals:	-	1.000	0.0000189999	1.000000000

D. Out of all the messages flagged by this screening device, 47% are important. It seems like a useful screen, provided you are prepared to ignore slightly more than half of the flagged messages. A better screen might be preferred, one that does not admit quite as many irrelevant messages.

2.

A.

Farm, x	$p(x)$
A	0.10
B	0.90
Total:	1.00

B.

Farm, x	$p(y x)$ $= p(8 - 10 \text{ bad} x)$	$p(x)$	$p(y x) p(x)$	$p(x y)$
A	0.6	0.10	0.06	0.40

B	0.1	0.90	0.09	0.60
Totals:	-	1.000	0.15	1.00

- C. Despite the fact that farm A tends to have the poorer potatoes, a bad sample does not necessarily imply farm A since most trucks come from farm B. In fact, 60% of the poorest samples actually are truckloads from farm B.

3.

A.

Has insurance, x	Estimated pdf, $\hat{p}(x)$
Yes	0.8002
No	0.1998
Total:	1.0000

One in five do not have insurance, and the hospital must plan for this.

B.

Rating, y	Estimated pdf, $\hat{p}(y)$
1	0.1399
2	0.1994
3	0.2009
4	0.1798
5	0.2800
Total:	1.0000

This shows that people tend to be more on the satisfied side than on the dissatisfied side; still, given the large numbers of dissatisfied people, there is room for improvement.

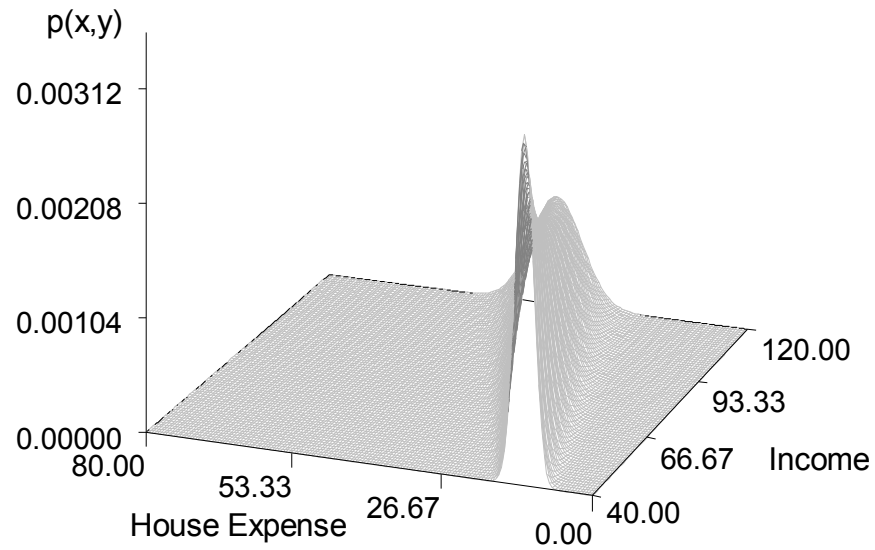
C.

Rating, y	$\hat{p}(y \text{Insurance})$	$\hat{p}(y \text{No Insurance})$
1	0.0999	0.2999
2	0.1998	0.1977
3	0.2007	0.2018
4	0.1997	0.1002
5	0.2999	0.2004
Total:	1.0000	1.0000

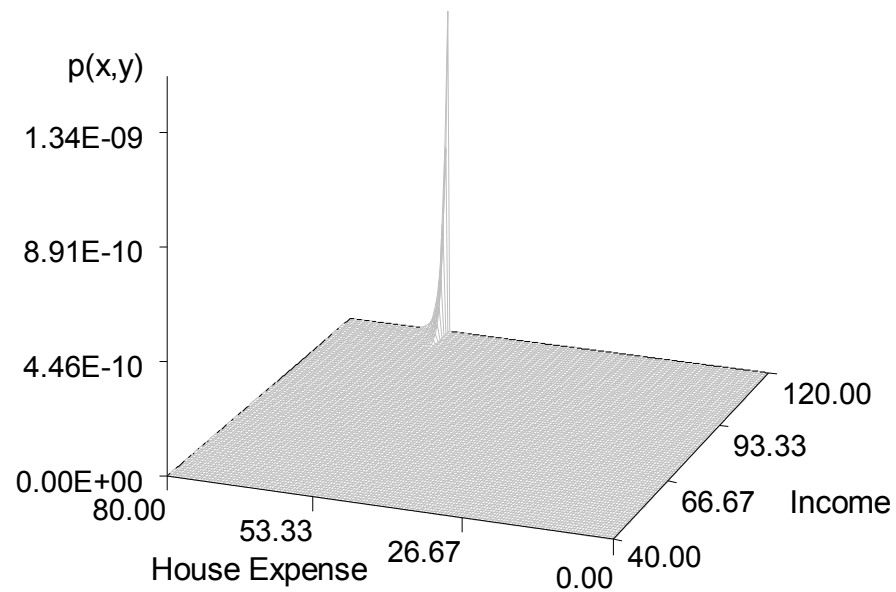
There is a pronounced tendency for people without insurance to give somewhat lower ratings—clearly a cause for concern about equal treatment.

4.

A. The graph:

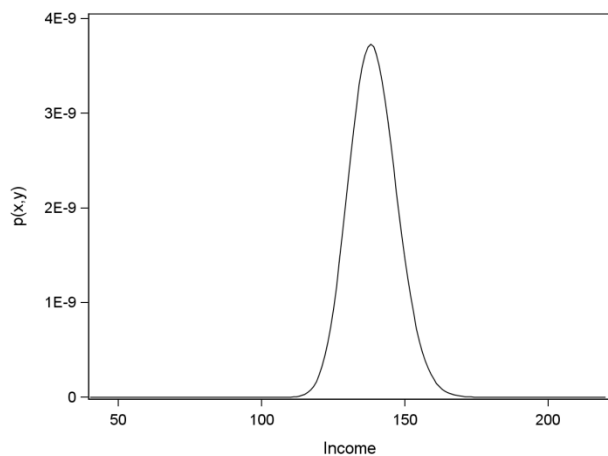


B. The graph^{xiii}:



The slice shows another view of Figure 6.8.

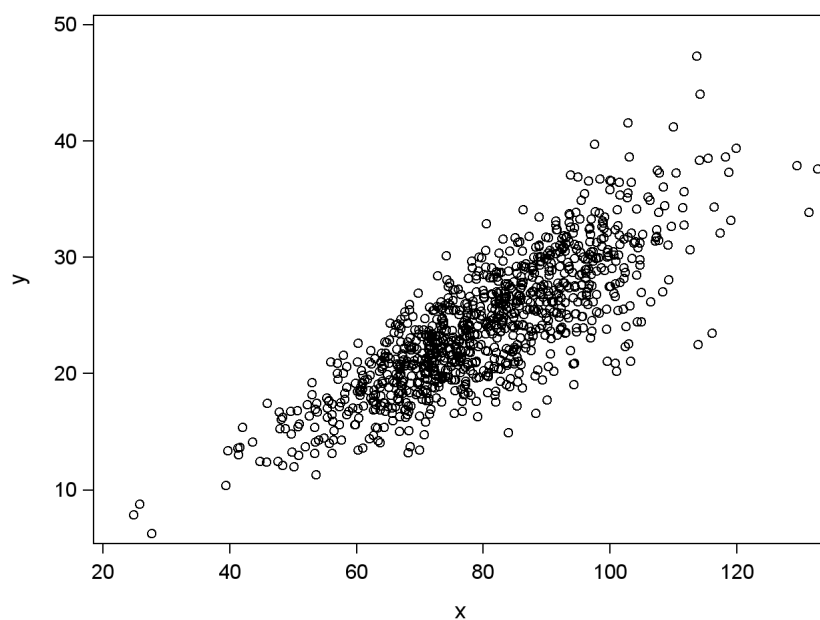
5. The graph^{xiv}:



They are different because X is no longer assumed to be bounded between 40 and 120.

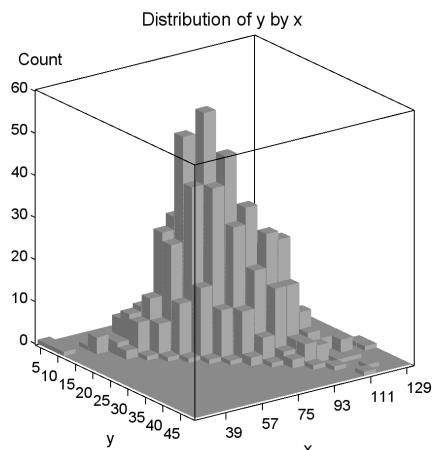
6.

A. The graph^{xv}:



Larger incomes correspond to larger house expense. There is greater variation in house expense at high income than there is at low income.

B.



The graph shows that in places in the scatterplot where there are more points, there is higher density as shown by height of the histogram. With larger X there is generally larger Y as well.

7. The sequence: T H T T T H H H T H T H H T T H T T T T H T T H T T H T H H. There is a $(.5)^{30} = 9.31 \times 10^{-10}$ probability of seeing this, or one in every set of 1,073,741,824 flips. It will take 1,073,741,824 minutes to do all these flips, or 17,895,697 hours, or 745,654 days, or 2,043 years to do it again. Have fun!

8.

A.

Stealing Behavior, x	$p(y x)$ $= p(95 x)$	$p(x)$	$p(y x) p(x)$	$p(x y)$
Stealer	0.00008727	0.99	0.00008640	0.615
Non-Stealer	0.00539910	0.01	0.00005399	0.385
Totals:	-	1.00	0.00014039	1.000

Your chance of being a non-stealer is only 0.385, in the HR manager's eyes.

B. The company thinks you are 100% a stealer no matter what you scored.

C. It's called dogmatic because no data will ever change their mind.

9. $p(\text{click} | \text{Income} = 100) \propto p(100 | \text{click})p(\text{click})$. The values of the click are yes and no, with $p(\text{yes}) = 0.01$ and $p(\text{no}) = 0.99$. From the graph, $p(100 | \text{yes})$ is about 0.0078 and $p(100 | \text{no})$ is about 0.0038. So $p(100 | \text{yes})p(\text{yes}) \cong (0.0078)(0.01) = 0.000078$ and $p(100 | \text{no})p(\text{no}) \cong (0.0038)(0.9) = 0.003762$. So $p(\text{click} | \text{Income} = 100) \cong 0.000078 / (0.000078 + 0.003762) = 0.02$; i.e., 2% of those with income 100 will click.

10.

Outcome of Car Trip, x	$p(\text{Sober} x)p(x)$	$p(x \text{Sober})$
Fatality	0.60×0.000000075	0.00000004545455
Nonfatality	0.99×0.999999925	0.99999995454545
Total:	0.989999971	Total: 1.00000000000000

11.

A. To show: $p(x|y) = \frac{p(y|x)p(x)}{p(y)}$. From (6.7), $p(x, y) = p(x|y)p(y)$. Hence

$p(x|y) = p(x, y)/p(y)$. But from (6.6), $p(y|x) = \frac{p(x, y)}{p(x)}$, implying that $p(x, y) = p(y|x)p(x)$, and the desired result then follows by substitution.

B. To show: $p(x|y) = \frac{p(y|x)p(x)}{\sum_{\text{all } x} p(y|x)p(x)}$ in the discrete case. From (6.11), $p(x|y) = \frac{p(y|x)p(x)}{p(y)}$.

Substituting (6.1), you get $p(x|y) = \frac{p(y|x)p(x)}{\sum_{\text{all } x} p(x, y)}$. Substituting (6.7), you get

$$p(x|y) = \frac{p(y|x)p(x)}{\sum_{\text{all } x} p(y|x)p(x)}.$$

C. See solution to Exercise 11B.

7. Sampling from Populations and Processes

Solutions to Exercises

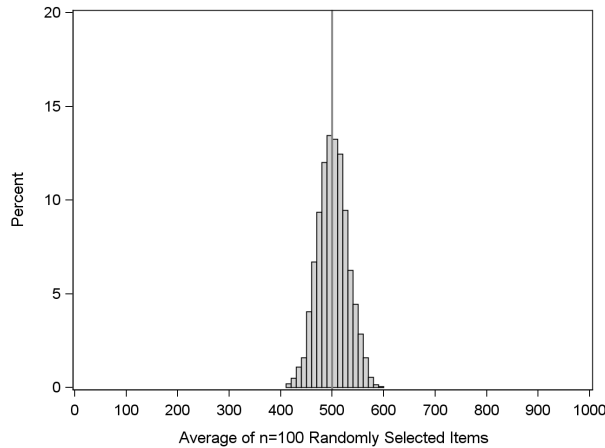
1.
 - A. In Excel, generate a column of $U(0, 1)$ random numbers next to the data column. Sort by the $U(0, 1)$ column, making sure that the data column is attached. Take the first five values of the data column following the sort.
 - B. 31.4
 - C. 24.5, different because it's calculated from different numbers.
 - D. There are many ways to do this. An inefficient, but correct way is to proceed as in Exercise 7A, but take only the first value. Then generate another set of $U(0,1)$ random numbers, sort, and take the first value. Repeat until you have $n = 5$ values. Such a sample is 37.3, 18, 56.7, 37.3, 0.7, with average 30.0. This is different from 24.5 because it is calculated from different numbers.
2.
 - A. 12
 - B.

y	$p(y)$
1	0.333
3	0.167
4	0.167
5	0.333
Total:	1.000
 - C.

y	$p(y)$
1	π_1
2	π_2
3	π_3
4	π_4
5	π_5
Total:	1.0
 - D. The population form says there is no chance of a 2. This does not generalize. Also, the population form is too specific about the probabilities in general. The actual frequencies may differ in the student online newspaper reading process
3.
 - A. The number of potatoes in the truck.
 - B. The number of potatoes in the sample.
 - C. The farmers process of putting potatoes in the truckbed; perhaps caused by variation in the field, or by systematic attempt on the farmer's part to put better potatoes on the top.
 - D. If the farmer puts the better potatoes on top, then the estimate from the scoop will tend to be too low.
4.
 - A. The number of deer out of 1,000 that have weight between 79.2 and 79.7, divided by 1,000.
 - B. The proportion of deer of this breed, spatio-temporal location, and from this design and measurement plan, whose weight will be between 79.2 and 79.7.

- C. The process interpretation is preferable because it generalizes. If there is no deer having weight between 79.2 and 79.7 in the 1,000, then the population interpretation says that this is impossible, a silly conclusion.

5.



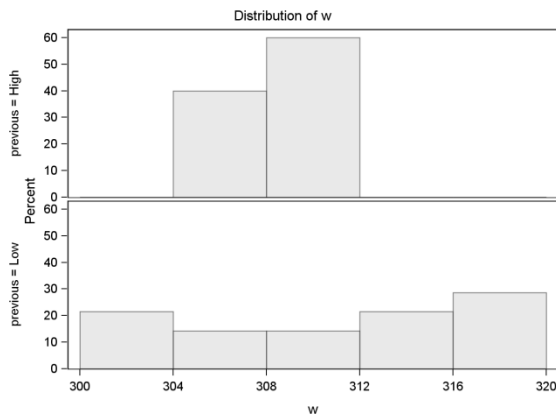
It's virtually identical to Figure 7.7; sampling fraction makes little difference.

6.

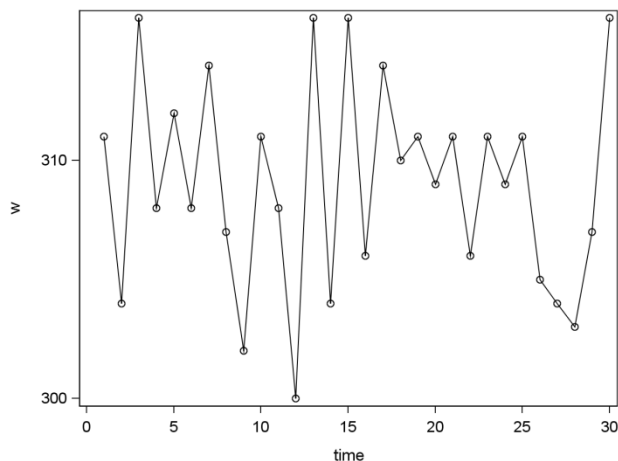
- A. Stoplights, wandering mind, stop for gas, dogs crossing the street, pedestrians, crazy drivers.
 B. What is N ? All the trips I ever took? If so is the next trip randomly sampled from the past trips?
 No, there is no population that produces the next trip data. It's from the process, pure and simple.

7.

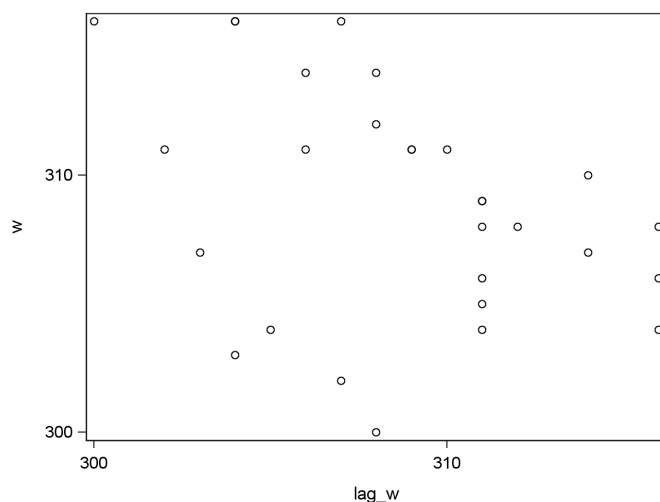
A.



Suggests that perhaps the variability after a low day is higher than the variability following a high day, suggesting non-independence; however, the sample size is small and this pattern may be explainable by chance.



There are no obvious trends, clusters, or adjacency issues to suggest non-iid behavior here.

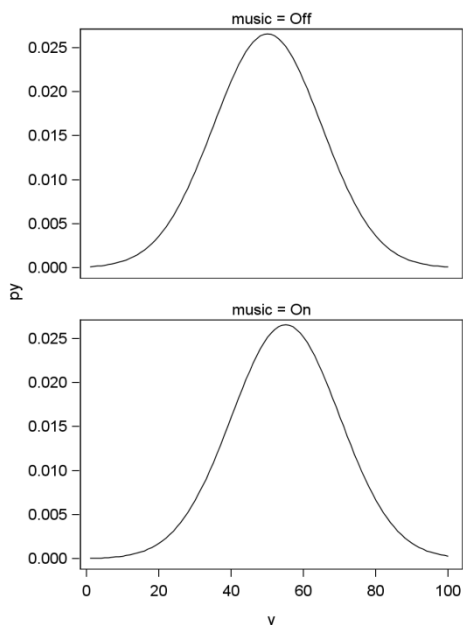


There is a hint of a negative trend, suggesting that low values tend to be followed by high values and vice versa, but this might be explainable by chance alone considering the small sample size.

- B. Generate $n = 30$ observations at random from a pdf that produces data that look like the observed data; perhaps using a rounded-off normal distribution. By design, random number generators produce iid data, so any deviations in the resulting graphs are *explained* by chance alone. Then compare the graphs of the actual data to graphs of the iid data to see whether the deviations in the actual data are *explainable* by chance alone.

8.

- A. How about “effect of music on the mind?” and the paper “Effect of Music Therapy on Mood and Social Interaction Among Individuals With Acute Traumatic Brain Injury and Stroke,” in *Rehabilitation Psychology*, August 2000 Vol. 45, No. 3, 274-283. X = music (yes or no), Y = mood. Suppose Y is measured continuously on a 0 – 100 scale. The distributions might look like this^{xvi}:



The graphs suggest a slight effect of music improving mood, but it's not much of an effect.

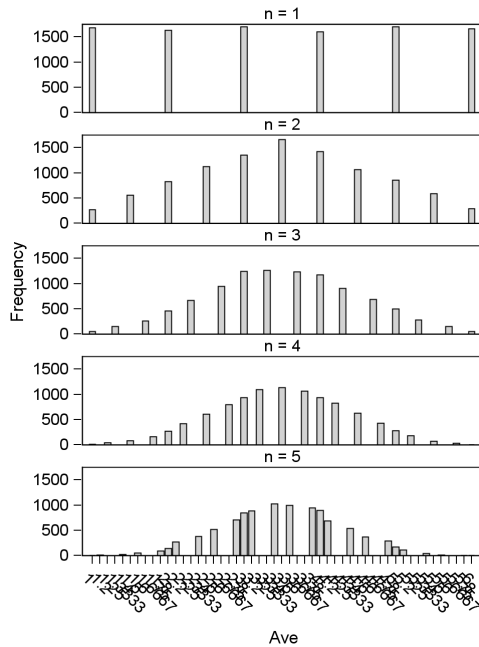
- B. The first pdf is the population distribution of mood determined from the population of N people, should they not be exposed to music. The second pdf is the population distribution of mood determined from the population of N people should they be exposed to music.
- C. The first pdf is the producer of mood for people not exposed to music. The second pdf is the producer of mood for people exposed to music.
- D. Usually, there is no definable N , unless the data are randomly sampled from a defined population of N people prior to experimentation. But even if they were so randomly sampled, there are design and measurement process elements, not part of the population, that make the population interpretation wrong.

9.

- A. v is what Bruce drinks. D is the difference between what he says he drinks and what he actually drinks.
- B. Probably usually less than 0, assuming he is embarrassed by the amount that he drinks.
- C. The numbers v and D are not selected from any population—they are just what Bruce reports at a given visit.

10.

- A. Ave_1 and Ave_2 are dependent because they share the first roll. If the first roll (Ave_1) is high, then Ave_2 also tends to be high. The same argument can be made for all adjacent averages: If one average is higher than expected, then adjacent averages also will be higher than expected.
- B. For one thing, the sample spaces are all different. For Ave_1 , $S = \{1, 2, 3, 4, 5, 6\}$; for Ave_2 , $S = \{1, 1.5, 2, 2.5, \dots, 5.5, 6\}$. For another, the variability around the mean 3.5 when there are more numbers in the average. So the distributions are all different.
- C. Here is a graph^{xvii}:



The distributions are all different.

11. Not independent because the distribution of housing expense changes as income levels change. Not identically distribution because the mean of the housing expense distribution is much lower than the mean of the income distribution.
12.
 - A. In the absence of further information, independence is a reasonable model. Knowing one BMI tells nothing about the others.
 - B. In the absence of further information, identical distributions is a reasonable model. The pdf is the process pdf that is assumed to produce BMIs for such applicants.

8. Expected Value and the Law of Large Numbers

Solutions to Exercises

1. Let Earnings = Y , and suppose the winnings are w . Hans earns $w - 1$ if he wins (the lottery does not return his 1, they keep it.) Here is the distribution of Y .

y	$p(y)$
-1	0.999
$w - 1$	0.001
Total:	1.000

Then $E(Y) = -1(0.999) + (w - 1)(0.001) = 0.001w - 1$. If Hans wants to come out even on average, then he wants $0.001w - 1 = 0$, or $w = 1,000$.

2.

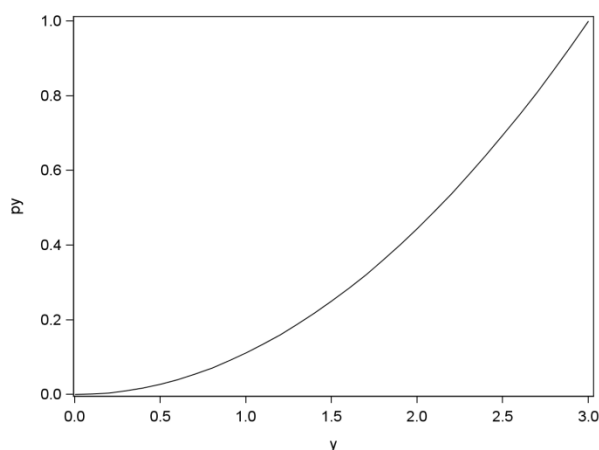
A. 2.29

B. Adequate if the past data reflects current conditions. Terribly wrong otherwise.

3.

A. $\int_0^3 (y^2/9) dy = (1/9)y^3/3 \Big|_0^3 = (1/9)3^3/3 - (1/9)0^3/3 = 1.$

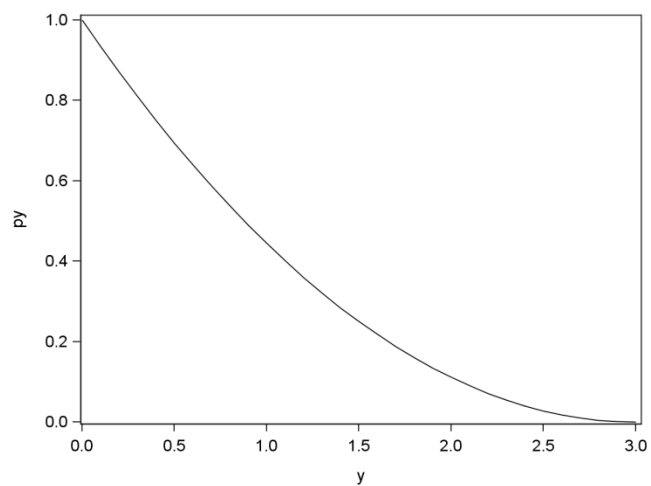
B.



C. $\int_0^3 y(y^2/9) dy = (1/9)y^4/4 \Big|_0^3 = (1/9)3^4/4 - (1/9)0^4/4 = 2.25$

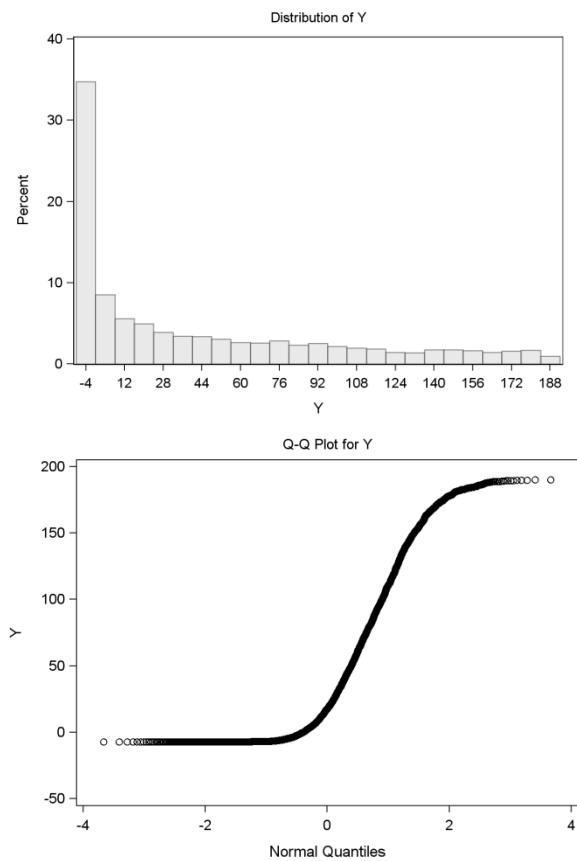
D. $\int_0^m (y^2/9) dy = (1/9)y^3/3 \Big|_0^m = (1/9)m^3/3 - (1/9)0^3/3 = m^3/27 = 0.5 \Rightarrow m = \{(.5)(27)\}^{1/3} = 2.38$

E. Median is higher. The opposite is true here:



4.

A.



The distribution is not even close to normal.

B. 0.558; the Law of Large Numbers applied to Bernoulli data.

5.

A.

y	$\hat{p}(y)$
0.7	0.066667
5.5	0.033333
5.6	0.033333
6.4	0.033333
11.6	0.033333
12.3	0.033333
13.4	0.033333
14.0	0.033333
15.2	0.033333
15.9	0.033333
16.8	0.033333
18.0	0.033333
18.1	0.033333
20.9	0.033333
21.0	0.033333
21.9	0.033333
25.9	0.066667
26.5	0.033333
32.0	0.033333
34.9	0.033333
37.3	0.033333
38.9	0.033333
39.0	0.033333
40.9	0.033333
47.9	0.033333
54.9	0.033333
56.7	0.033333
57.2	0.033333
Total: 1.0000	

B. $\hat{\mu} = \sum y\hat{p}(y) = 0.7 \times 0.066667 + 5.5 \times 0.033333 + \dots + 57.2 \times 0.033333 = 24.5333$.

C. 24.61; should be (and is) close by the LLN.

6.

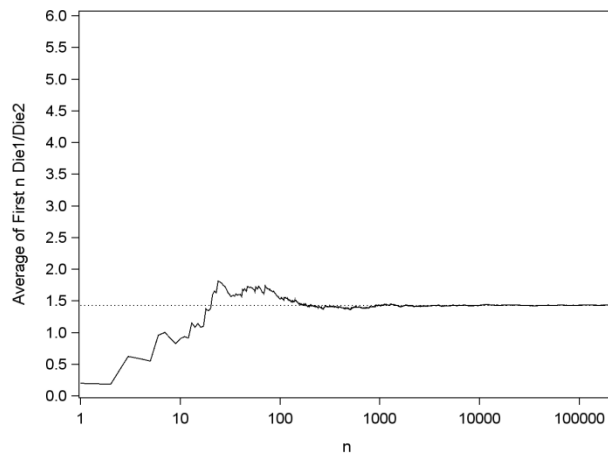
A.

y	$p(y)$
0.166667	1/36
0.200000	1/36
0.250000	1/36
0.333333	2/36
0.400000	1/36
0.500000	3/36
0.600000	1/36

0.666667	2/36
0.750000	1/36
0.800000	1/36
0.833333	1/36
1.000000	6/36
1.200000	1/36
1.250000	1/36
1.333333	1/36
1.500000	2/36
1.666667	1/36
2.000000	3/36
2.500000	1/36
3.000000	2/36
4.000000	1/36
5.000000	1/36
6.000000	1/36
Total:	1.000

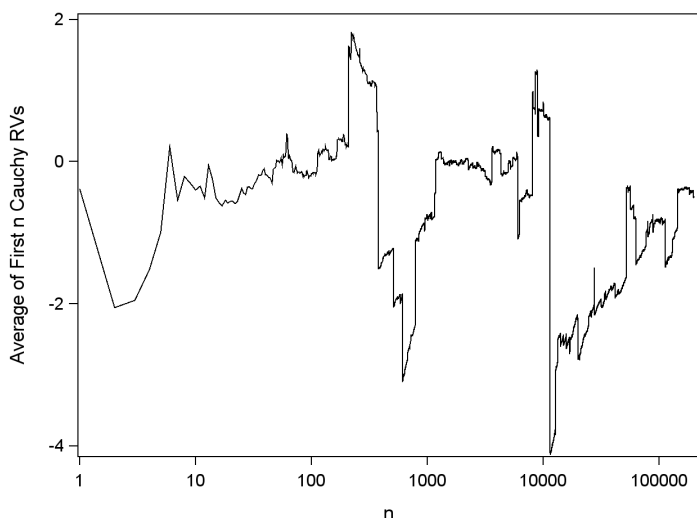
B. 1.429

C. The graph^{xviii}:



Looks like we did the probability distribution and expected value calculations correctly, because the LLN works as it is supposed to!

7.

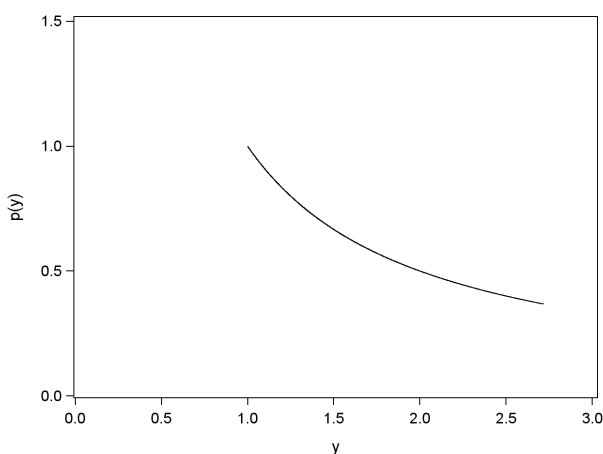


There is no defined mean for this distribution, so the LLN cannot be expected to hold.

8.

A. $\int_1^e y^{-1} dy = \ln(y)|_1^e = \ln(e) - \ln(1) = 1 - 0 = 1$, and the function is nonnegative.

B.



If this curve were a cardboard cut-out, it would balance at the mean.

C. $\int_1^e y(y^{-1}) dy = y|_1^e = e - 1 = 1.718$. That's where the curve would balance.

D. Cdf: $\int_1^y t^{-1} dt = \ln(t)|_1^y = \ln(y) - \ln(1) = \ln(y)$. Inverse cdf: $p = \ln(y) \Rightarrow y = e^p$.

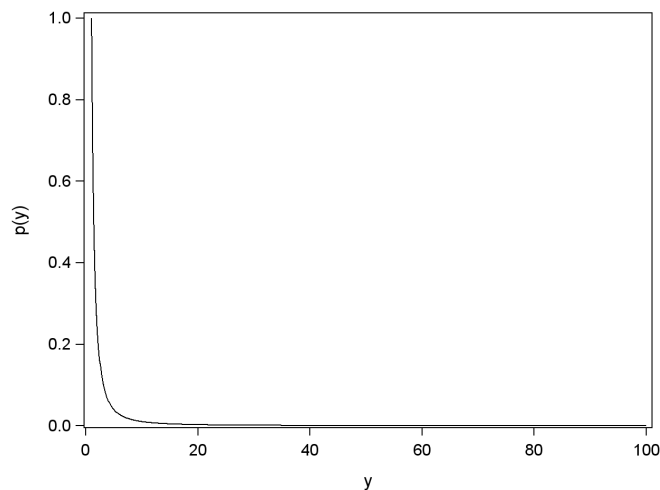
E. $p = 0.5 \Rightarrow y = e^{0.5} = 1.648$, smaller than $E(Y)$.

F. average of 10,000 = 1.704. It's different from 1.718 because it's based on random data. It's close by the LLN.

9.

A. $\int_1^\infty y^{-2} dy = -y^{-1}|_1^\infty = -\infty^{-1} - (-1^{-1}) = 0 + 1 = 1$, and the function is nonnegative.

B.

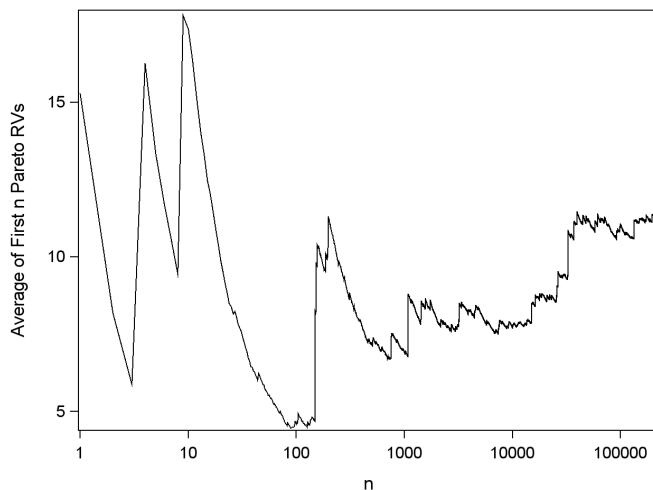


C. $\int_1^{\infty} y \times y^{-2} dy = \ln(y) \Big|_1^{\infty} = \ln(\infty) - \ln(1) = \infty$

D. cdf: $\int_1^y t^{-2} dt = -t^{-1} \Big|_1^y = -y^{-1} - (-1^{-1}) = 1 - y^{-1}$. Inverse cdf: $p = 1 - y^{-1} \Rightarrow y = (1 - p)^{-1}$.

E. $p = 0.5 \Rightarrow y = (1 - 0.5)^{-1} = 2.0$. It's smaller than the mean, which is infinity.

F.

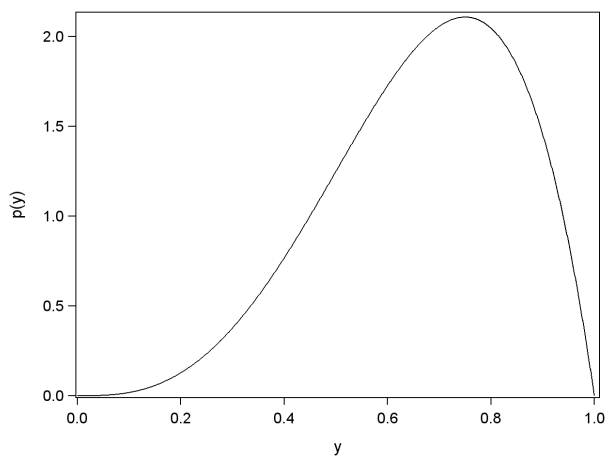


It doesn't converge because the mean is infinite.

10.

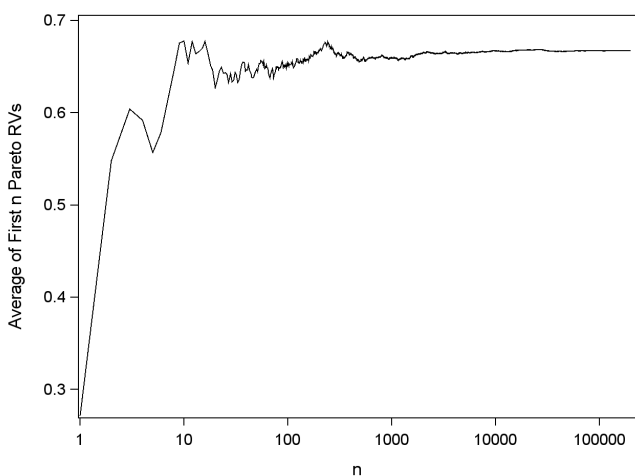
A. $c^{-1} = \int_0^1 (y^3 - y^4) dy = (y^4/4 - y^5/5) \Big|_0^1 = (1/4 - 1/5) - (0 - 0) = 1/20$. $c = 20$.

B.



C. $\int_0^1 y(20)(y^3 - y^4)dy = 20(y^5/5 - y^6/6)\Big|_0^1 = 20(1/5 - 1/6) - (0 - 0) = 2/3$. The graph, as a cardboard cut-out, would balance at $2/3$.

D.



The graph shows convergence of the running averages to $2/3$, because of the Law of Large Numbers.

9. Functions of Random Variables: Their Distributions and Expected Values

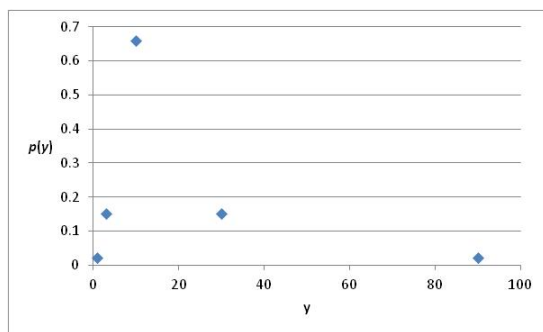
Solutions to Exercises

1.

- A. Yes, it's the additivity property applied to the bootstrap pdf.
- B. No, since $f(x) = x^2$ is a convex function, the expected value of the square (which is just the average when applied to the bootstrap pdf) is greater than the square of the expected value (or the square of the average, when applied to the bootstrap pdf).
- C. Yes, it's the linearity property of expectation, with the caveat that expectation is applied to the bootstrap pdf.
- D. No.

2.

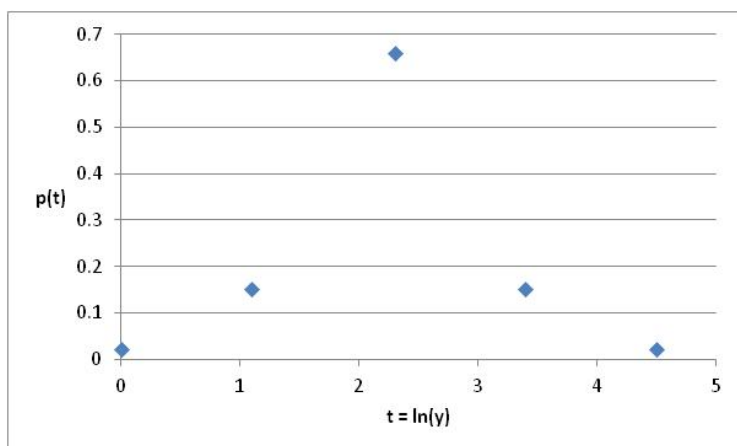
A.



This pdf will produce the numbers 1, 3, 10, 30 and 90. It will produce a lot more 3s, 10s and 30s than 1s or 90s.

- B. $E(Y) = 13.37$, point of balance. $\text{Var}(Y) = 185.6$, a measure of spread. $\text{StdDev}(Y) = 13.6$, measures deviation from mean, can be used with Chebychev's inequality; Skewness = 3.74, indicates marked asymmetry with longer right tail; Kurtosis = 17.42, indicates distribution is much more outlier-prone than the normal distribution.
- C.

$t = \ln(y)$	$p(t)$
0	0.02
1.098612	0.15
2.302585	0.66
3.401197	0.15
4.49981	0.02
Total:	1



The distribution is just as discrete as before, but is closer to symmetric and less outlier-prone.

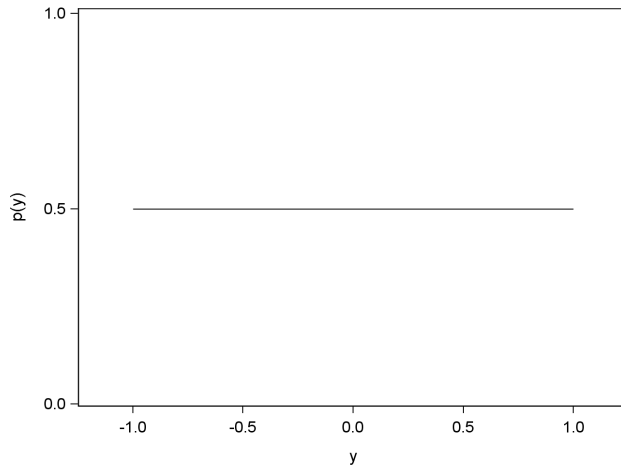
- D. The function $\ln(x)$ is concave so $E(\ln(Y)) < \ln(E(Y))$. Here, $E(T) = 2.28$, which is less than $\ln(13.37) = 2.59$, as predicted.
- E. Skewness = -0.13 , indicates near symmetry; Kurtosis = 1.31 , indicates distribution just slightly more outlier-prone than the normal distribution. The transformation made the process closer to a normal distribution, so if the methods assume normality, as they often do, then the method was indeed good.

3.

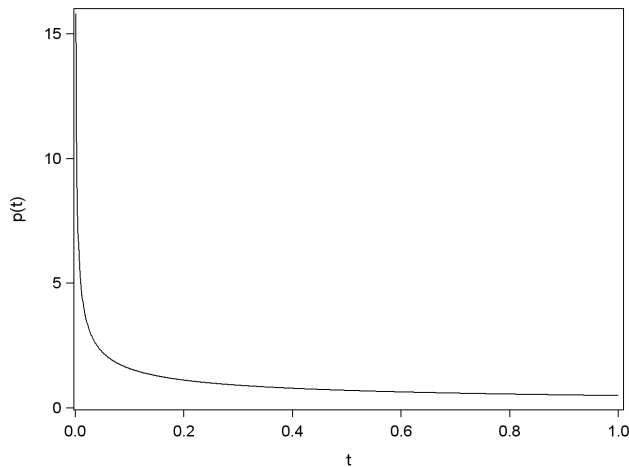
- A. $E(Y) = 3.98$, point of balance; long run average earnings. $\text{Var}(Y) = 103.02$, a measure of spread. $\text{StdDev}(Y) = 10.15$, measures deviation from mean, can be used with Chebychev's inequality; Skewness = -9.85 , indicates marked asymmetry with longer left tail; Kurtosis = 95.09 , indicates distribution is much more outlier-prone than the normal distribution.
- B. -6.169857142 to 14.12985714 ; -16.31971428 to 24.27971428 ; -26.46957143 to 34.42957143 .
- C. The only possible value within each of those ranges is 5, so the probability is 99% in all cases.
- D. $99\% > 0\%$, $99\% > 75\%$, $99\% > 88.9\%$
- E. $E(T) = E(Y_1 + Y_2 + \dots + Y_{10000})$ (by substitution)
 $= E(Y_1) + E(Y_2) + \dots + E(Y_{10000})$ (by the additivity property of expectation)
 $= 3.98 + 3.98 + \dots + 3.98$ (assuming all people are sampled from the same process, and that the pdf given is the right model for this process)
 $= 10,000(3.98)$ (by algebra)
 $= 39,800$ (by arithmetic)
- F. 39873, 40004, 41741, 39390, 37944, 38066, 38477, 40539, 39183, and 40408; average = 39562.5. With many more than 10, the average of these T s will get closer to 39,800, by the LLN.

4.

A.



B. $P(t) = \Pr(T \leq t) = \Pr(-t^{1/2} \leq Y \leq t^{1/2}) = t^{1/2}$ so $p(t) = 0.5t^{-1/2}$, for $0 < t \leq 1.0$.



Clearly, T is non-uniform. Also, the values of T are only positive, since they are squared. Finally, values of T near zero are quite likely.

C. $E(Y) = \int_{-1}^1 y(0.5)dy = (.5)(y^2/2)\big|_{-1}^1 = 1^2 - (-1)^2 = 0.0$; the graph balances at 0. $\text{Var}(Y) = E(Y^2) - \{E(Y)\}^2$; $E(Y^2) = \int_{-1}^1 y^2(0.5)dy = (.5)(y^3/3)\big|_{-1}^1 = (1/6)(1^3 - (-1)^3) = 1/3 = \text{Var}(Y)$ since $E(Y) = 0$; this a measure of spread of the graph. Skewness $= E(Y - \mu)^3/\sigma^3$; $E(Y - \mu)^3 = E(Y^3) = \int_{-1}^1 y^3(0.5)dy = (.5)(y^4/4)\big|_{-1}^1 = (1/8)(1^4 - (-1)^4) = 0$. So Skewness $= 0$, and the graph is symmetric about 0. Kurtosis $= E(Y - \mu)^4/\sigma^4 - 3$, but $E(Y - \mu)^4 = E(Y^4) = \int_{-1}^1 y^4(0.5)dy = (.5)(y^5/5)\big|_{-1}^1 = (1/10)(1^5 - (-1)^5) = 1/5$. So kurtosis $= (1/5)/(1/3)^2 - 3 = -1.2$, so the curve is less outlier-prone than the normal distribution.

D. $T = Y^2$, so $E(T) = 1/3$ as found in C, the point of balance of the graph. $\text{Var}(T) = E(T^2) - (1/3)^2 = E(Y^4) - 1/9 = 1/5 - 1/9 = 4/45$, a measure of spread of the distribution; notice that it is smaller than the spread of the untransformed distribution. $E(T - \mu)^3 = E(T^3) - 3\mu E(T^2) + 3\mu^2 E(T) - \mu^3$. But $E(T^3) = E(Y^6) = \int_{-1}^1 y^6(0.5)dy = (.5)(y^7/7)\big|_{-1}^1 = (1/14)(1^7 - (-1)^7) = 1/7$. So $E(T - \mu)^3 = 1/7 - 3(1/3)(1/5) + 3(1/3)^3 - (1/3)^3 = 1/7 - 1/5 + 2/27 = 0.016931217$, so skewness $= E(T - \mu)^3/\sigma^3 = 0.016931217/(4/45)^{3/2} = 0.6389$, indicating a curve with longer right tail than left. $E(Y - \mu)^4 = E(T^2) -$

$4\mu E(T^3) + 6\mu^2 E(T^2) - 4\mu^3 E(T) + \mu^4$. But $E(Y^8) =$

$\int_{-1}^1 y^8 (0.5) dy = (.5)(y^9/9) \Big|_{-1}^1 = (1/18)(1^9 - (-1)^9) = 1/9$. So $E(Y - \mu)^4 = (1/9) - 4(1/3)(1/7) + 6(1/3)^2(1/5) - 4(1/3)^3(1/3) + (1/3)^4 = 0.016931$ and Kurtosis $= E(Y - \mu)^4 / \sigma^4 - 3 = (0.016931 / (4/45)^2) - 3 = -0.857$. Thus the distribution is less outlier-prone than the normal distribution.

E. Based on 20000 simulations, the average T is 0.333, close to the true value $1/3$, the estimated variance is 0.08888, close to the true value $4/45$, the estimated skewness is 0.650, close to the true value 0.6389, and the estimated kurtosis is -0.844, close to the true value -0.857.

5. 7.98; the result is less than 10 because $E|Y - 10|^2 > \{E|Y - 10|\}^2$; i.e., $\sigma^2 > \text{MAD}^2$, or $\sigma > \text{MAD}$.
6. A. Depends on what “reasonable” means. If it means 95% sure, then the range is -5% to 15%, meaning Hans might lose up to 5%.
B. Because the distribution is bell-shaped, like a normal distribution, therefore the 68-95-99.7 rule is more precise.
C. Precisely, no. Nothing is perfect. Approximately, yes, although returns are known to come from distributions with heavier tails than the normal distribution.
7. $\sum (y_i - \bar{y}) = (y_1 - \bar{y}) + (y_2 - \bar{y}) + \dots + (y_n - \bar{y}) = y_1 + y_2 + \dots + y_n - \bar{y} - \bar{y} - \dots - \bar{y} = \sum y_i - n\bar{y}$. Now, $\bar{y} = (1/n) \sum y_i$, so $n\bar{y} = \sum y_i$, implying $\sum (y_i - \bar{y}) = 0$.
For any distribution, $E(Y - \mu) = E(Y) - \mu = \mu - \mu = 0$. For the bootstrap distribution, $E(Y - \mu) = (1/n) \sum (y_i - \bar{y})$, so $(1/n) \sum (y_i - \bar{y}) = 0$, implying $\sum (y_i - \bar{y}) = 0$.
8. $E(aX + bY) = E(aX) + E(bY)$ (by the additivity property of expectation)
 $= aE(X) + bE(Y)$ (by the linearity property of expectation).
9. A.

x	$p(x)$
1	0.2
2	0.2
3	0.4
5	0.2

B. 2.8, 1.76, 1.327, 0.370, -0.783

10. -2

11.

A. Skewness $= \frac{2(\theta_2 - \theta_1)(\theta_1 + \theta_2 + 1)^{1/2}}{(\theta_1 + \theta_2 + 2)(\theta_1 \theta_2)^{1/2}}$; kurtosis is more complicated.

B. Skewness $= \frac{2(2 - 4)(4 + 2 + 1)^{1/2}}{(4 + 2 + 2)(4 \times 2)^{1/2}} = \frac{-4(7)^{1/2}}{(8)(8)^{1/2}} = -0.468$.

C. Estimated skewness = -0.469.

D. The estimate is based on random data and changes every time you change the random number seed.

E. LLN

10. Distributions of Totals

Solutions to Exercises

1.

A. They are values that tell you how often the individual responses will occur for this process. For example, in the long run of observations from this customer process, 5% of the observed values will be 1.

B. 3.9, 1.69

C. In the long run of observations from this customer process, the sample average of the observed values will tend towards 3.9.

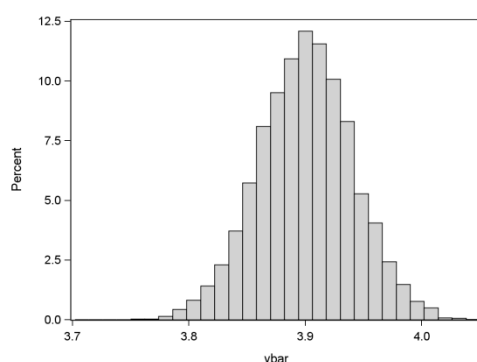
D. $E(\bar{Y}) = E\{(1/1000)(Y_1 + Y_2 + \dots + Y_{1000})\}$ (by substitution)
 $= (1/1000) E(Y_1 + Y_2 + \dots + Y_{1000})$ (by the linearity property of expectation)
 $= (1/1000) \{E(Y_1) + E(Y_2) + \dots + E(Y_{1000})\}$ (by the additivity property of expectation)
 $= (1/1000)(3.9 + 3.9 + \dots + 3.9)$ (since all Y s are sampled from the same identical distribution, whose mean is 3.9)
 $= (1/1000)(1000(3.9)) = 3.9$ (by algebra and arithmetic). Independence is not needed.

E. $\text{Var}(\bar{Y}) = \text{Var}\{(1/1000)(Y_1 + Y_2 + \dots + Y_{1000})\}$ (by substitution)
 $= (1/1000)^2 \text{Var}(Y_1 + Y_2 + \dots + Y_{1000})$ (by the linearity property of variance)
 $= (1/1000)^2 \{\text{Var}(Y_1) + \text{Var}(Y_2) + \dots + \text{Var}(Y_{1000})\}$ (by the additivity property of variance when the variables are independent)
 $= (1/1000)^2(1.69 + 1.69 + \dots + 1.69)$ (since all Y s are sampled from the same identical distribution, whose variance is 1.69)
 $= (1/1000)^2(1000(1.69)) = 1.69/1000 = 0.00169$ (by algebra and arithmetic).
 Independence *is* needed.

F. 0.0411 . $3.9 \pm 3(0.0411)$ gives 3.777 to 4.023. At least 88.9% of the samples of size $n = 1,000$ from this process will give a $\bar{Y}$ that is between 3.777 and 4.023.

G. 9,982 out of 10,000 samples produced a $\bar{Y}$ between 3.777 and 4.023, or 99.82%. Since 99.82% > 88.9%, Chebychev's inequality appears to be working correctly^{xix}.

H.



The graph continuous, symmetric and bell-shaped, so the CLT seems to be working very well.

I. 69.92%, 95.49%, and 99.82%, very close to 68-95-99.7.

J. The 68-95-99.7 rule because it is more accurate.

2.

$$\text{Var}(V) = E\{V - E(V)\}^2 \quad (\text{by definition})$$

$$= E\{aX + bY - E(aX + bY)\}^2 \quad (\text{by substitution})$$

$= E[aX + bY - \{aE(X) + bE(Y)\}]^2$	(by the linearity and additivity properties of expectation)
$= E\{aX + bY - (a\mu_X + b\mu_Y)\}^2$	(by definition)
$= E\{a(X - \mu_X) + b(Y - \mu_Y)\}^2$	(by algebra)
$= E\{a^2(X - \mu_X)^2 + b^2(Y - \mu_Y)^2 + 2ab(X - \mu_X)(Y - \mu_Y)\}$	(by algebra)
$= a^2E\{(X - \mu_X)^2\} + b^2E\{(Y - \mu_Y)^2\} + 2abE\{(X - \mu_X)(Y - \mu_Y)\}$	(by the linearity and additivity properties of expectation)
$= a^2\text{Var}(X) + b^2\text{Var}(Y) + 2abE\{(X - \mu_X)(Y - \mu_Y)\}$	(by definition of variance)
$= a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$	(by definition of covariance)

3.

A. Strategy 1: $Y_1 = 20R_1$. Strategy 2: $Y_2 = 12R_1 + 8R_2$.B. $E(Y_1) = E(20R_1) = 20E(R_1) = 20(0.05)$ (by the linearity property of expectation). So $E(Y_1) = 1.0$. If you made this investment repeatedly, under identical conditions, your long-run average earnings would be 1.

$E(Y_2) = E(12R_1 + 8R_2) = 12E(R_1) + 8E(R_2) = 12(0.05)$ (by the linearity and additivity properties of expectation). So $E(Y_2) = 1.0$. If you made this investment repeatedly, under identical conditions, your long-run average earnings would be 1. Neither strategy is preferred in terms of expected value.

C. $\text{Var}(Y_1) = \text{Var}(20R_1) = 20^2\text{Var}(R_1) = 20^2(0.04)^2$ (by the linearity property of variance). So $\text{StdDev}(Y_1) = 20(0.04) = 0.8$. If you made this investment repeatedly, under identical conditions, your earnings would be between $1 - 2(0.8)$ and $1 + 2(0.8)$ at least 75% of the time.

$\text{Var}(Y_2) = \text{Var}(12R_1 + 8R_2) = 12^2\text{Var}(R_1) + 8^2\text{Var}(R_2) = 12^2(0.04)^2 + 8^2(0.04)^2$ (by the linearity and additivity properties of variance when the random variables are independent). So $\text{StdDev}(Y_2) = \{12^2(0.04)^2 + 8^2(0.04)^2\}^{1/2} = 0.577$. If you made this investment repeatedly, under identical conditions, your earnings would be between $1 - 2(0.577)$ and $1 + 2(0.577)$ at least 75% of the time. If you want a chance at higher earnings, then you like strategy 1. If you want less of a chance of a loss, then you like strategy 2.

D. Strategy 1 is unaffected. For strategy 2, $\text{Var}(12R_1 + 8R_2) = 12^2\text{Var}(R_1) + 8^2\text{Var}(R_2) + 2(12)(8)\text{Cov}(R_1, R_2) = 12^2(0.04)^2 + 8^2(0.04)^2 + 2(12)(8)(.9)(.04)(.04) = 0.6093$, and $\text{StdDev}(Y_2) = 0.781$. The two strategies are virtually identical in this case.E. Strategy 1 is unaffected. For strategy 2, $\text{Var}(12R_1 + 8R_2) = 12^2\text{Var}(R_1) + 8^2\text{Var}(R_2) - 2(12)(8)\text{Cov}(R_1, R_2) = 12^2(0.04)^2 + 8^2(0.04)^2 - 2(12)(8)(.9)(.04)(.04) = 0.0563$, and $\text{StdDev}(Y_2) = 0.237$. Now, your earnings would be between $1 - 2(0.237)$ and $1 + 2(0.237)$ at least 75% of the time. You are unlikely to lose money under strategy 2 in this case.

- F. The smaller the correlation (in the number line sense, where -1 is the smallest possible value), the greater the effect of diversification on reducing risk.

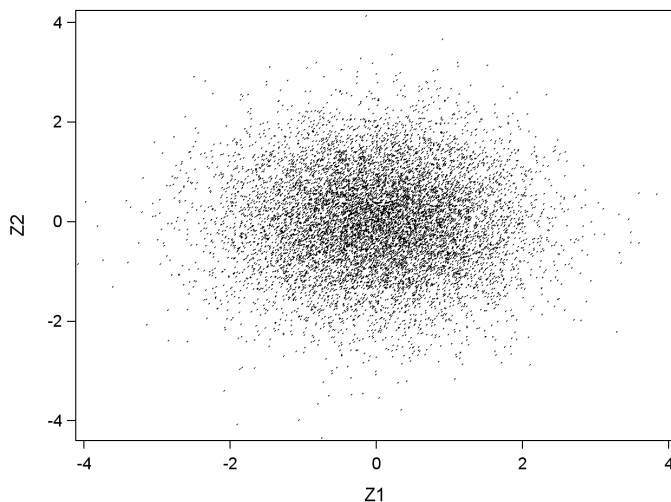
4.

- A. $\text{Var}(Y) = \text{Var}(\beta_0 + \beta_1 X + D) = (\beta_1)^2 \sigma_x^2 + \sigma^2$, by the linearity and additivity properties of variance under independence.
- B. $(\beta_1)^2 \sigma_x^2 / \{(\beta_1)^2 \sigma_x^2 + \sigma^2\}$
- C. $\text{Cov}(Y, X) = E(Y - \mu_Y)(X - \mu_X) = E\{(\beta_0 + \beta_1 X + D - (\beta_0 + \beta_1 \mu_X))(X - \mu_X)\} = E[\{\beta_1(X - \mu_X) + D\}(X - \mu_X)] = \beta_1 E(X - \mu_X)^2 + E\{D(X - \mu_X)\} = \beta_1 \sigma_x^2 + \text{Cov}(D, X) = \beta_1 \sigma_x^2$.
- D. $\{\text{Corr}(Y, X)\}^2 = \{\text{Cov}(Y, X)\}^2 / \{\text{Var}(Y)\text{Var}(X)\} = (\beta_1)^2 \sigma_x^4 / \{((\beta_1)^2 \sigma_x^2 + \sigma^2) \sigma_x^2\} = (\beta_1)^2 \sigma_x^2 / \{(\beta_1)^2 \sigma_x^2 + \sigma^2\}$

5. Since $\text{Var}(X) = 1$, $\text{Var}(Y) = 1$ implies that $(\beta_1)^2 + \sigma^2 = 1$. Further, $\text{Corr}(Y, X) = 0.9$ implies $0.9 = \beta_1$, and hence $\sigma^2 = 1 - 0.9^2$ and $\sigma = 0.4359$. So generate $X \sim N(0, 1)$, $D \sim N(0, 0.4359^2)$, and $Y = 0.9X + D$. The plug-in estimate of variance of T is 3.747, close the true value $1 + 1 + 2(.9)(1)(1) = 3.8$.
6. From the solution to Exercise 4, the squared correlation is $a^2 \sigma_x^2 / \{a^2 \sigma_x^2 + \sigma^2\}$. Here, the D term is absent, so you can assume it is always 0.0, in which case $\text{Var}(D) = 0$. Then $\rho^2 = a^2 \sigma_x^2 / \{a^2 \sigma_x^2 + 0\}$, implying $|\rho| = 1.0$.

7.

- A. Here is the graph^{xx}:

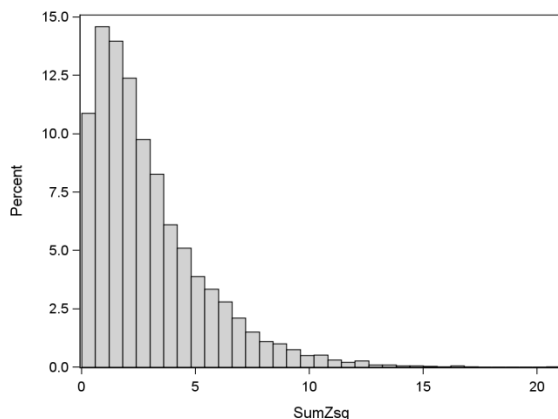


This shows little, if any relationship between Z_1 and Z_2 .

- B. -0.00597, this suggests a possible negative relationship between Z_1 and Z_2 , but the number is so close to zero that the relationship is very weak, if there is any relationship at all.
- C. For independent variables the covariance is zero, hence the correlation is also zero.
- D. The estimate is based on a random sample. A different random sample will give a different number altogether.

8.

A. Here is the graph^{xxi}:



This is a right-skew-distribution. The mean appears to be around 3.

- B. Less and less skewed. The means are higher and higher.
- C. With larger number in the sum, the distribution is closer to a normal distribution, because of the CLT. This is seen in the simulations.

9.

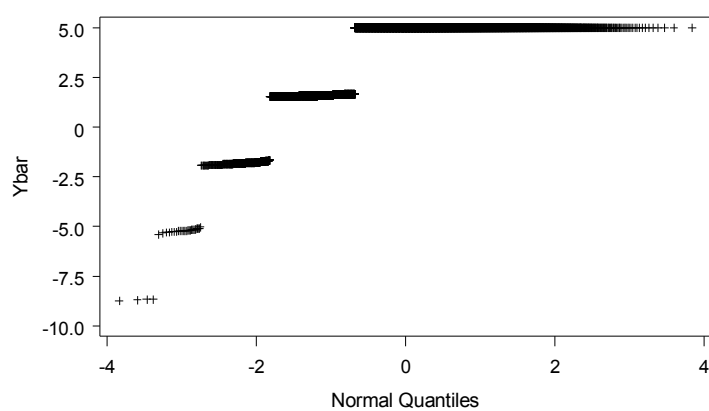
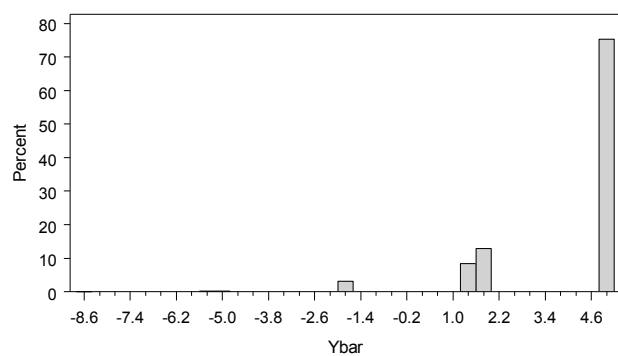
- A. Discrete uniform on 1, 2, ..., 6. Discrete uniform on 1, 2, ..., 6.
- B. CLT assumes independent. These are not independent: if you know Y_1 , you know Y_2 and all the rest of the Y 's.

10. From the marginal bootstrap distributions X and Y we know that the bootstrap means of X and Y are $\bar{x}, \bar{y}$, respectively. From Table 10.3, the bootstrap joint probabilities are $p(x_i, y_j) = 1/n$ when $i = j$ and $=0$ otherwise. So $\sum_i \sum_j (x_i - \mu_x)(y_j - \mu_y) p(x_i, y_j) = \sum_i (x_i - \mu_x)(y_i - \mu_y) p(x_i, y_i) + \sum \sum_{i \neq j} (x_i - \mu_x)(y_j - \mu_y) p(x_i, y_j)$
 $p(x_i, y_j) = \sum_i (x_i - \bar{x})(y_i - \bar{y}) (1/n) + \sum \sum_{i \neq j} (x_i - \mu_x)(y_j - \mu_y)(0) = (1/n) \sum_i (x_i - \bar{x})(y_i - \bar{y})$.

11. $-8259 < 20T - 100,000 < -2267 \Leftrightarrow (100,000 - 8259)/20 < T < (100,000 - 2267)/20 \Leftrightarrow 4587 < T \leq 4886$. The probability is $P(4886) - P(4586)$, where $P(t)$ is the binomial cdf. This gives 0.9973. The approximation is good.

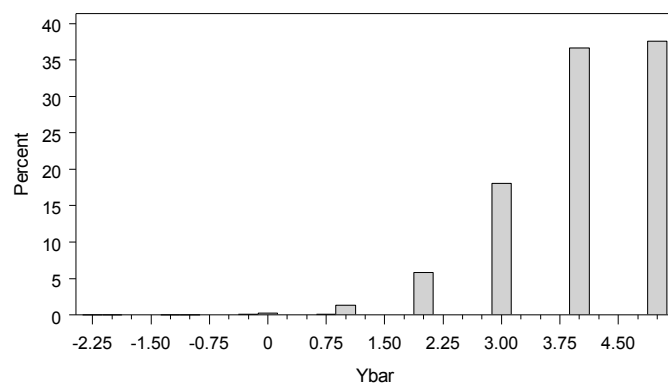
12.

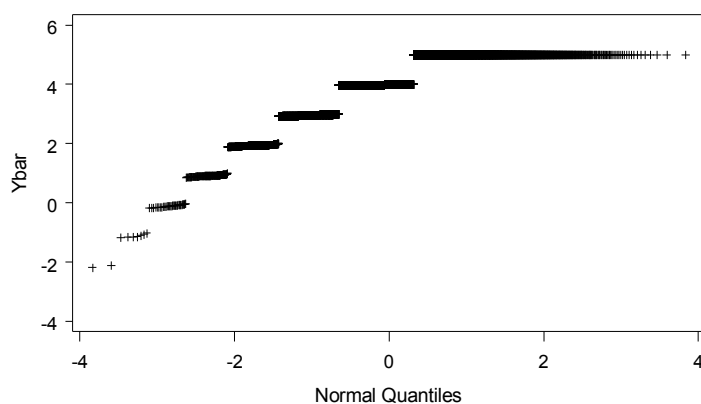
- A. 5.0



Not close to normal at all.

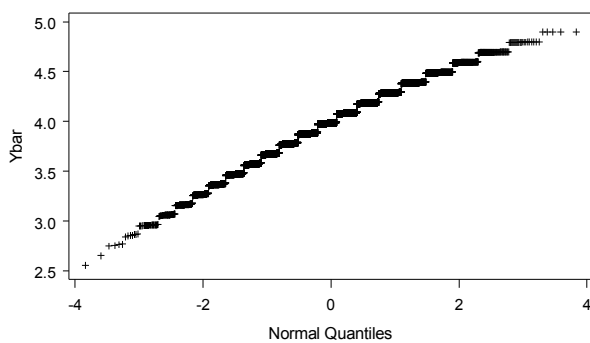
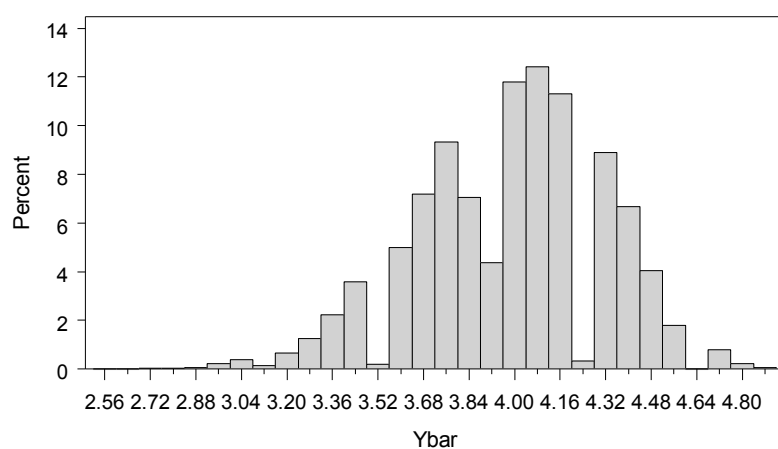
B.





Still not close to normal with $n = 100$.

C. $n = 1000$ is perfect but approximately normal:



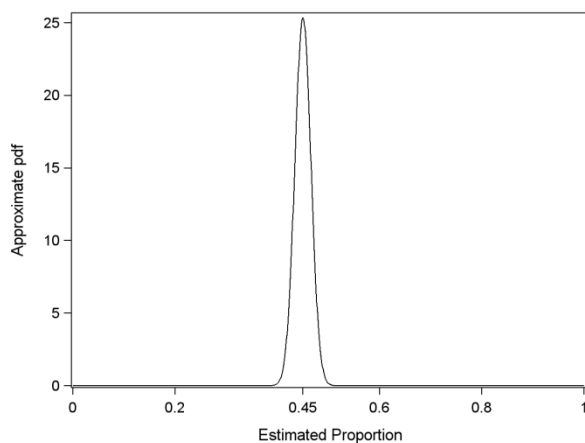
D. No difference in the conclusions with total versus average—the graphs look the same as regards normal appearance or lack thereof.

13.

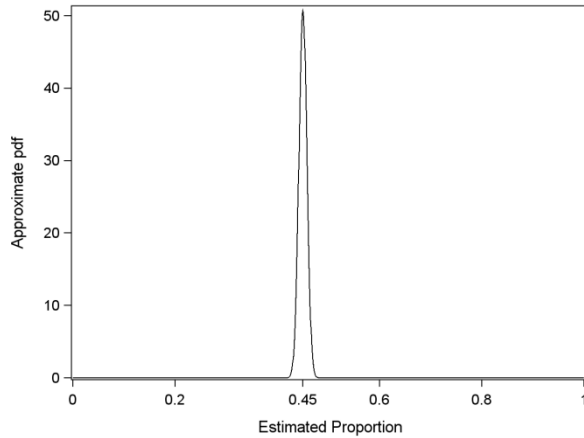
$\text{Cov}(X, Y) = E\{(X - \mu_X)(Y - \mu_Y)\}$	(by definition)
$= E(XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y)$	(by algebra)
$= E(XY) - \mu_X E(Y) - \mu_Y E(X) + \mu_X \mu_Y$	(by linearity and additivity properties of expectation, noting that the μ s are constants)
$= E(XY) - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y$	(since $E(X) = \mu_X$ and $E(Y) = \mu_Y$ by definition)
$= E(XY) - \mu_X \mu_Y$	(by algebra).

14.

- A. $Y \sim \text{Bernoulli}(\pi)$
 B. $E(Y) = 0 \times (1 - \pi) + 1 \times \pi = \pi$. $\text{Var}(Y) = E(Y^2) - \{E(Y)\}^2 = \pi - \pi^2 = \pi(1 - \pi)$
 C. It is an estimate of π
 D. $E(\bar{Y}) = E(Y) = \pi$. $\text{Var}(\bar{Y}) = \text{Var}(Y) / n = \pi(1 - \pi) / n$.
 E. $\bar{Y} \sim N(\pi, \pi(1 - \pi) / n)$
 F. Here is the graph^{xxii}.



G.



- H. In 68%, 95%, and 99.7% of samples of $n = 1,000$ from this process, respectively, the estimated proportion will be within ± 0.0157 , ± 0.0315 and ± 0.0472 of the true process proportion. In 68%, 95%, and 99.7% of samples of $n = 4,000$ from this process, respectively, the estimated proportion will be within ± 0.0079 , ± 0.0157 and ± 0.0236 of the true process proportion.
- I. With a larger sample size, the estimated proportion is closer to the true proportion.
15. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$; if $X = Y$, this formula gives $\text{Var}(X + X) = \text{Var}(X) + \text{Var}(X) + 2\text{Cov}(X, X)$. But by definition, $\text{Cov}(X, X) = E\{(X - \mu_X)(X - \mu_X)\} = \text{Var}(X)$, so the formula gives $4\text{Var}(X)$. This is correct by the linearity property of variance: $\text{Var}(X + X) = \text{Var}(2X) = 2^2\text{Var}(X) = 4\text{Var}(X)$.

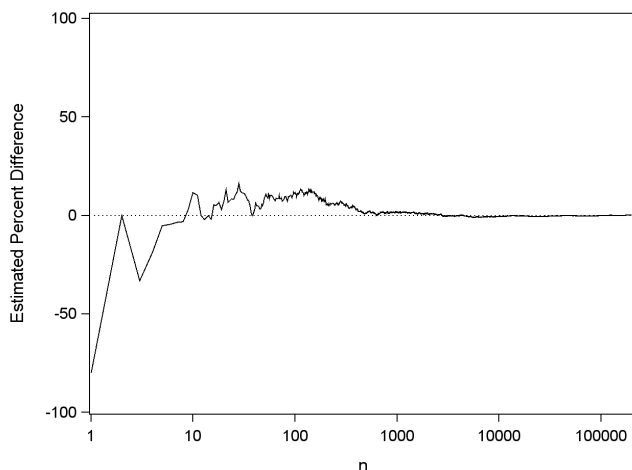
11. Estimation: Unbiasedness, Consistency, and Efficiency

Solutions to Exercises

1.

- A. $E(Y) = 0 \times (1 - \pi) + 1 \times \pi = \pi$, the probability that the material passes. Therefore Y is an unbiased estimator.
- B. That Y is produced by the Bernoulli(π) process, where π is the probability that the material passes the test.

2. The graph^{xxiii}:



3.

- A. Average = 3.75, Stdev = 4.18. The Stdev is closer to the true value 4.0, so the Stdev is a better estimate in this sample.
- B. Average of averages^{xxiv} = 3.996; Average of std devs = 3.830.
- C. Bias of average is estimated to be $3.996 - 4.0 = -0.004$. Bias of stdev is estimated to be $3.830 - 4.0 = -0.170$. The usual standard deviation estimator is biased low (Jensen's inequality), the average appears unbiased and so is preferred from that standpoint. Variance of average is estimated to be 0.816. Variance of stdev is estimated to be 1.323. The average appears to be preferred from the standpoint of variance. ESD of average is estimated to be $(-0.004)^2 + 0.816 = 0.816$. ESD of stdev is estimated to be $(-0.170)^2 + 1.323 = 1.352$. The average appears to be preferred from the standpoint of ESD.

4.

- A. Cdf is $P(y) =$

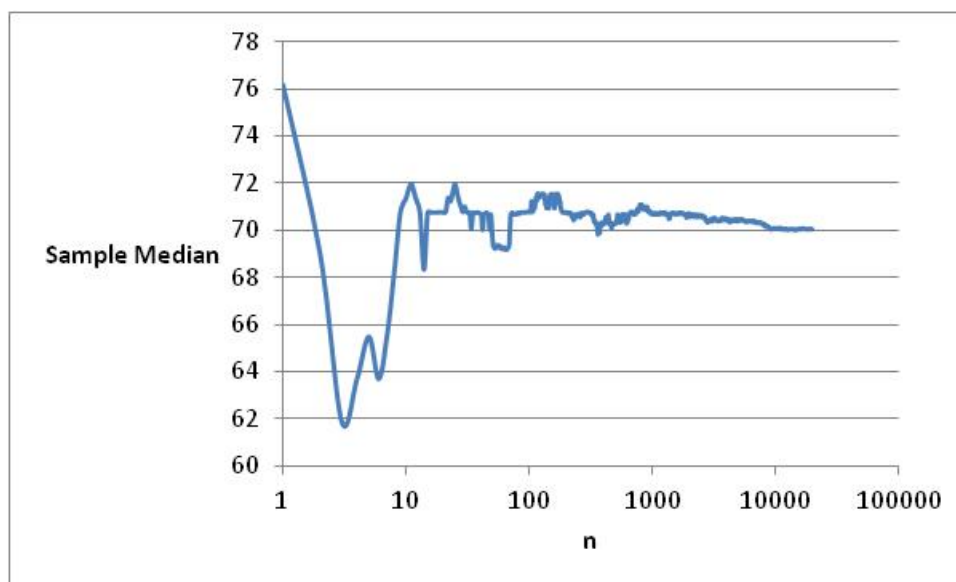
$$\int_0^y 0.5 \exp(-0.5t) dt = 0.5 \exp(-0.5t) / (-0.5) \Big|_0^y = -\exp(-0.5y) - (-\exp(-0.5(0))) = 1 - \exp(-0.5y).$$

Setting $P(y) = 0.5$ and solving gives $1 - \exp(-0.5y) = 0.5 \Rightarrow \exp(-0.5y) = 0.5 \Rightarrow y = -2\ln(0.5) = 1.386$.

- B. Since $\bar{Y}$ is an unbiased estimator of the mean, $E(\bar{Y}) = 2.0$ and hence $E(\bar{Y}) \neq 1.386$.
- C. From Chapter 10 we know that $\text{Var}(\bar{Y}) = \sigma^2/n = 2^2/10 = 0.4$. We also know that $E(\bar{Y}) = 2.0$, hence $\text{bias}(\bar{Y})$ as an estimate of 1.386 is $2.0 - 1.386 = 0.614$ and $\text{ESD}(\bar{Y}) = 0.4 + 0.614^2 = 0.777$.
- D. $\text{Bias} = 1.492 - 1.386 = 0.106$. $\text{Variance} = 0.3842$. $\text{ESD} = 0.3842 + 0.106^2 = 0.395$.
- E. Median.
5. Consider the following table, showing the variance for various combinations. For example, there are 8 combinations leading to 1.0, for a probability of $8/36 = 0.2222$. The rest of the probabilities are obtained similarly.

	1	2	3	4	5	6
1	0	0.25	1	2.25	4	6.25
2	0.25	0	0.25	1	2.25	4
3	1	0.25	0	0.25	1	2.25
4	2.25	1	0.25	0	0.25	1
5	4	2.25	1	0.25	0	0.25
6	6.25	4	2.25	1	0.25	0

6. 288.386315, 288.5307609. Not much difference for large n . ($n = 999$ here is large).
- 7.
- A. Yes: $E\{(\hat{\theta}_1 + \hat{\theta}_2)/2\} = (1/2)E\{(\hat{\theta}_1 + \hat{\theta}_2)\} = (1/2)\{E(\hat{\theta}_1) + E(\hat{\theta}_2)\}$ (by linearity and additivity) $= (1/2)(\theta + \theta)$ (by unbiasedness), $= \theta$.
- B. Yes; follow the method of solution in 7A.
- C. Yes; follow the method of solution in 7A.
- D. No; follow the method of solution in 7A and get 1.2θ , not θ .
- E. No, it's biased low by Jensen's inequality since $x^{1/2}$ is concave.
- F. No, it's biased high by Jensen's inequality since x^2 is convex.
- 8.
- A. Follow the method of solution in 7A to find that $c_1 + c_2 = 1.0$.
- B. The variance is $f(c_1) = 100(c_1)^2 + (1 - c_1)^2$. So $f'(c_1) = 200c_1 - 2(1 - c_1) = 0$ implies $c_1 = 1/101$, and hence $c_2 = 100/101$.
9. Here is the graph, done in excel using random number generation, and then =MEDIAN(A\$1:An) to calculate the running medians. The estimated median appears to settle on the true median, 70.



10. No, as shown in Figure 8.10, the average is not consistent because the random variable has infinite variance.

11. B is Bernoulli(π), where π is the probability that Y is less than 200. By the LLN as applied to Bernoullis shown in Section 8.5, the average of the B is a consistent estimator of π .

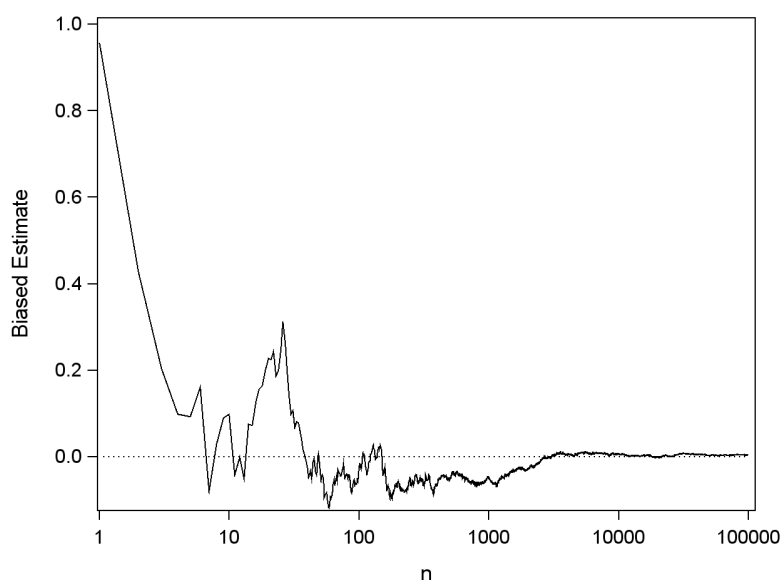
12.

A. $E(\hat{\theta}) = E\{(1/n)(Y_1 + Y_2 + \dots + Y_n) + 1/n\} = E\{(1/n)(Y_1 + Y_2 + \dots + Y_n)\} + 1/n = \mu + 1/n \neq \mu$

B. $(1/n)(Y_1 + Y_2 + \dots + Y_n)$ is a consistent estimator of μ and $(1/n)$ tends towards 0.

C. Simulate 100,000 values of $(1/4)(Y_1 + Y_2 + Y_3 + Y_4) + 1/4$, where each Y is $N(0, 1)$, and average them. You get something like 0.2521, not very close to 0, but much closer to 0.25.

D. Here is a graph^{xxv}:



13. By definition, $\text{Cov}(Y_i, Y_i) = E[\{Y_i - E(Y_i)\}\{Y_i - E(Y_i)\}] = \text{Var}(Y_i)$.

14.

- A. Since $E(Y_i) = \pi$ for all i , $E(\bar{Y}) = \pi$, by the linearity and additivity properties of expectation.
- B. Since $E(Y_i) = \pi$ for all i , $E(Y_1) = \pi$ in particular.
- C. $\text{Var}(\bar{Y}) = \sigma^2/n$, while $\text{Var}(Y_1) = \sigma^2$. Since both are unbiased, ESD = Variance, and $\bar{Y}$ is much more efficient, having $\text{ESD} = \sigma^2/n$.
- D. Here, $\sigma^2 = \pi(1 - \pi)$ because $E(Y) = E(Y^2) - \{E(Y)\}^2 = \pi - \pi^2 = \pi(1 - \pi)$. Also $(1/n) \sum (Y_i - \bar{Y})^2$ is the variance of the bootstrap distribution, so the computing formula $\text{Var}(Y) = E(Y^2) - \{E(Y)\}^2$ applies, giving $(1/n) \sum (Y_i - \bar{Y})^2 = (1/n) \sum (Y_i)^2 - \bar{Y}^2 = \bar{Y} - \bar{Y}^2 = \bar{Y}(1 - \bar{Y})$ since $Y^2 = Y$. But the n formula for variance is biased by the factor $(n - 1)/n$, so that $E(\bar{Y}(1 - \bar{Y})) = ((n - 1)/n) \pi(1 - \pi)$.
- E. $(\bar{Y}(1 - \bar{Y}))(n/(n - 1))$.

15.

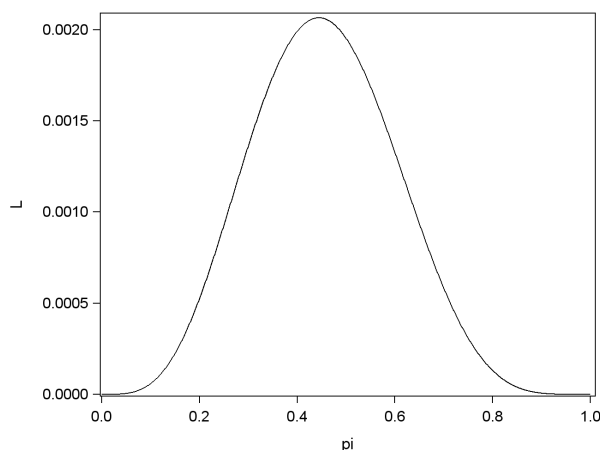
- A. $E\{(1/n) \sum (X_i - \mu_X)(Y_i - \mu_Y)\} = (1/n) \sum E\{(X_i - \mu_X)(Y_i - \mu_Y)\}$, by linearity and additivity properties of expectation. Since the pairs are iid from the same joint pdf, $E\{(X_i - \mu_X)(Y_i - \mu_Y)\} = \sigma_{XY}$ for all $i = 1, 2, \dots, n$; hence $(1/n) \sum E\{(X_i - \mu_X)(Y_i - \mu_Y)\} = (1/n)(n\sigma_{XY}) = \sigma_{XY}$.
- B. No, because the result is an estimate not an estimator.
- C. Because you don't know μ_X and you don't know μ_Y .

12. Likelihood Function and Maximum Likelihood Estimates

Solutions to Exercises

1.

A. $L(\pi) = \pi^4(1 - \pi)^5$; here is the graph^{xxvi}:



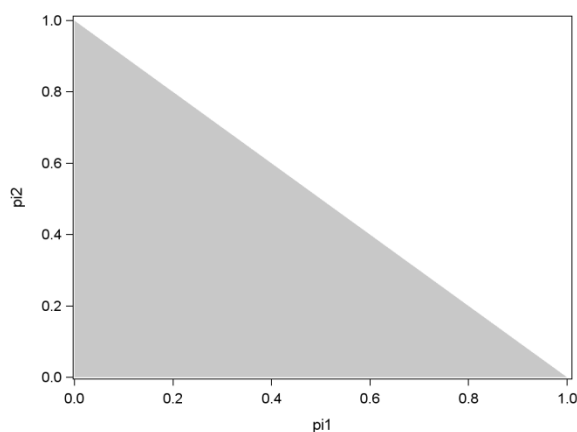
B. Using increments $\pi = 0.000, 0.001, \dots, 0.999, 1.000$, inspection shows that the maximum of L occurs when $\pi = 0.444$.

C. $LL(\pi) = 4\ln(\pi) + 5\ln(1 - \pi)$. $LL'(\pi) = 4/\pi - 5/(1 - \pi) = 0 \Rightarrow 4(1 - \pi) - 5\pi = 0 \Rightarrow \hat{\pi}_{MLE} = 4/9$.

D. Using EXCEL's solver with 0.5 as an initial value gives $\hat{\pi}_{MLE} = 0.444444446293876$.

E. $LL''(\pi) = -4/\pi^2 - 5/(1 - \pi)^2$, so $\hat{\sigma}^2 = (4/(4/9)^2 + 5/(5/9)^2)^{-1} = 0.0274$ and $\hat{\sigma} = 0.166$. The Wald interval is 0.113 to 0.776, this captures roughly 95% of the area under the likelihood function shown in 1A.

2. The graph^{xxvii}:

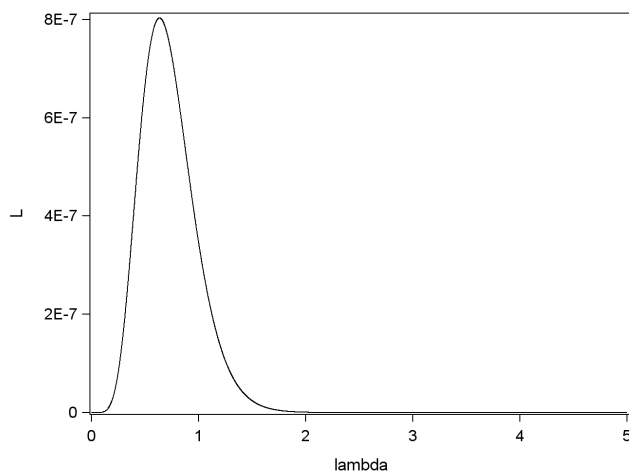


3. The graph of this sample space fills all of the two-dimensional space. You can't show the whole thing!

4. The likelihood for an individual Scot's response y_i is $L_i = (\pi_1)^{I(y_i = \text{support})} (\pi_2)^{I(y_i = \text{do not support})} (1 - \pi_1 - \pi_2)^{I(y_i = \text{no opinion})}$. So the joint likelihood is $(\pi_1)^{\sum I(y_i = \text{support})} (\pi_2)^{\sum I(y_i = \text{do not support})} (1 - \pi_1 - \pi_2)^{\sum I(y_i = \text{no opinion})} = \pi_1^{392} \pi_2^{401} (1 - \pi_1 - \pi_2)^{209}$.

5.

A. $L(\lambda) = \prod_i (\lambda^{y_i} e^{-\lambda} / y_i!) = \lambda^{\sum y_i} e^{-n\lambda} / \prod y_i! = \lambda^7 e^{-11\lambda} / 48$; here is the graph^{xxviii}.



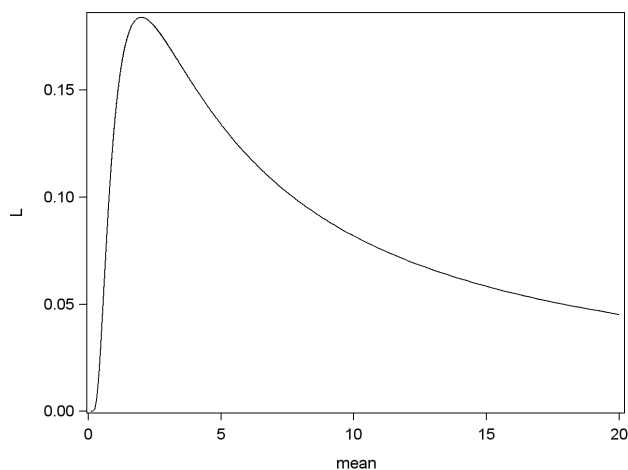
B. Using increments $\lambda = 0.000, 0.001, \dots, 4.999, 5.000$, inspection shows that the maximum of L occurs when $\lambda = 0.636$.

C. $LL(\lambda) = 7\ln(\lambda) - 11\lambda - \ln(48)$; $LL'(\lambda) = 7/\lambda - 11 = 0 \Rightarrow \hat{\lambda} = 7/11$.

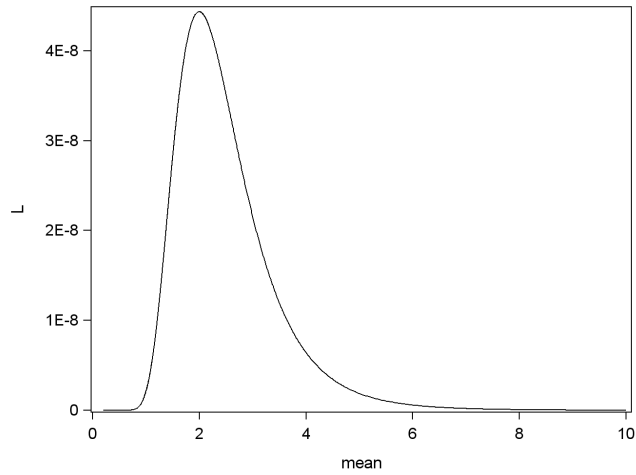
D. Using EXCEL's solver with 1.0 as an initial value gives $\hat{\lambda}_{MLE} = 0.63636363671665$.

1. $LL''(\lambda) = -7/\lambda^2$, so $\hat{\sigma}^2 = (7/(7/11)^2)^{-1} = 0.05785$ and $\hat{\sigma} = 0.2405$. The Wald interval is 0.155 to 1.117, this captures roughly 95% of the area under the likelihood function shown in 1A.

6. The graphs^{xxix}:



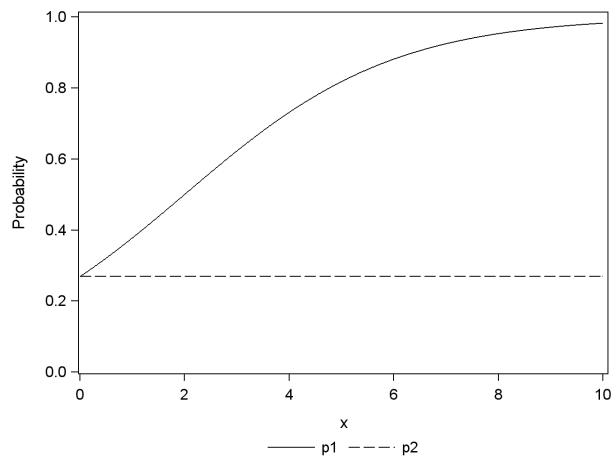
Based on the data, the expected waiting time can be high, but not low.



Based on the data, the expected waiting time appears to be between 1 and 5.

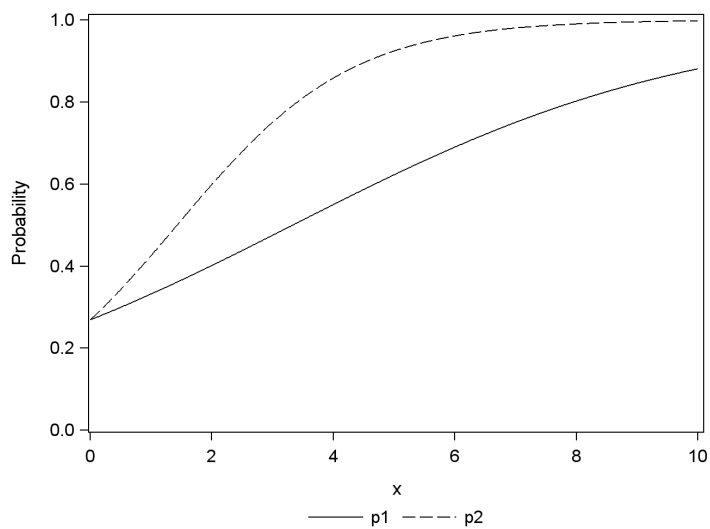
7.

A. The graph^{xxx}:



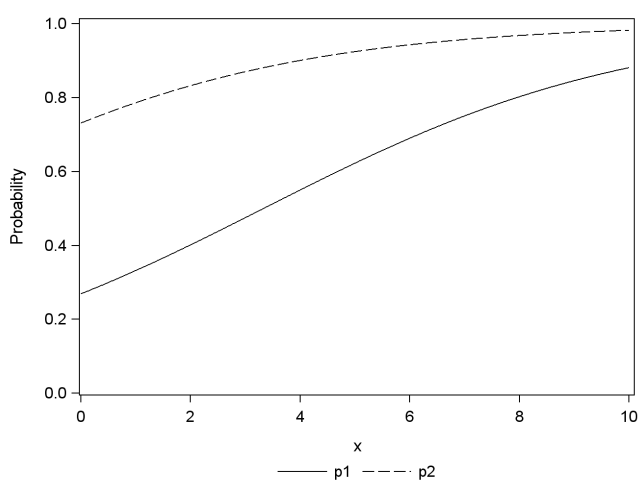
$\beta_1 = 0$ causes the probability function to be flat.

B.



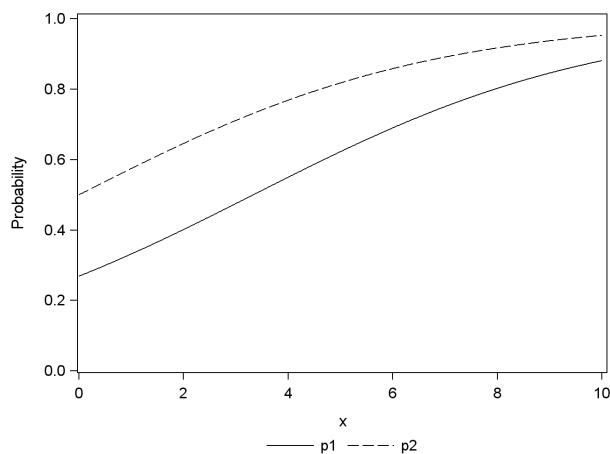
Larger β_1 causes the probability function to rise more steeply.

C.



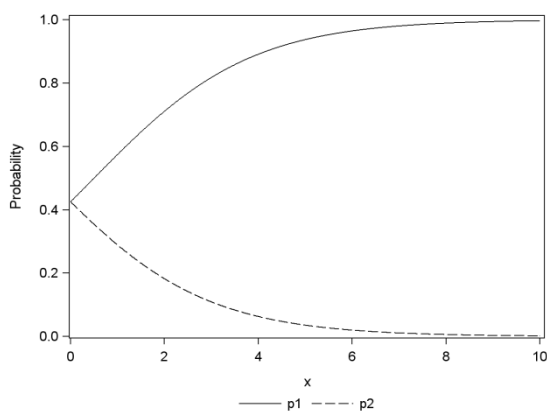
Larger β_0 causes larger probabilities overall.

D.



$\beta_0 = 0$ causes the probability to be 0 when $x = 0$.

E.



$\beta_1 < 0$ causes the probabilities to decrease for larger x .

8.

A. $LL(\beta_0, \beta_1) = \ln \{ \exp(\beta_0 + \beta_1(2)) / (1 + \exp(\beta_0 + \beta_1(2))) \} + \ln \{ 1 / (1 + \exp(\beta_0 + \beta_1(0.5))) \} + \dots + \ln \{ \exp(\beta_0 + \beta_1(3.2)) / (1 + \exp(\beta_0 + \beta_1(3.2))) \}$

B. $\hat{\beta}_0 = -3.1533$; $\hat{\beta}_1 = 1.8725$

C. $0.0565 \leq \beta_1 \leq 3.6885$. The interval tells us that β_1 is most likely positive, implying that larger x correspond to larger probabilities.

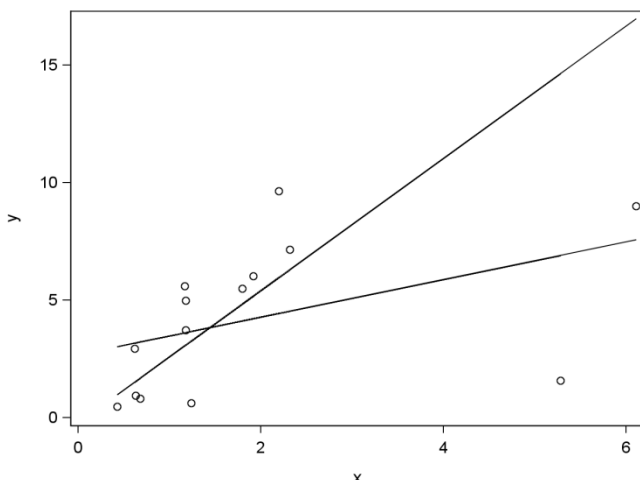
9.

A. Specify the normal pdfs $N(\beta_0 + \beta_1 x, \sigma^2)$ for each y observation, take the logs of each, add them up, and let the computer choose $(\beta_0, \beta_1, \sigma)$ to maximize the sum. Get $\hat{\beta}_0 = 2.683$, $\hat{\beta}_1 = 0.799$, $\hat{\sigma} = 2.698$.

B. Specify the normal pdfs $N(\beta_0 + \beta_1 x, x^2 \sigma^2)$ for each y observation, take the logs of each, add them up, and let the computer choose $(\beta_0, \beta_1, \sigma)$ to maximize the sum. Get

$$\hat{\beta}_0 = -0.224, \hat{\beta}_1 = 2.815, \hat{\sigma} = 1.519.$$

C. Here is the graph^{xxxii}; larger slope corresponds to the heteroscedastic model.



Under the constant variance assumption, the estimation procedure treats all data points equally in the fit. Under the heteroscedastic model, data points with smaller variance are given more weight, hence the heteroscedastic line comes closer to the data points near 0 in the graph than does the ordinary regression line.

10.

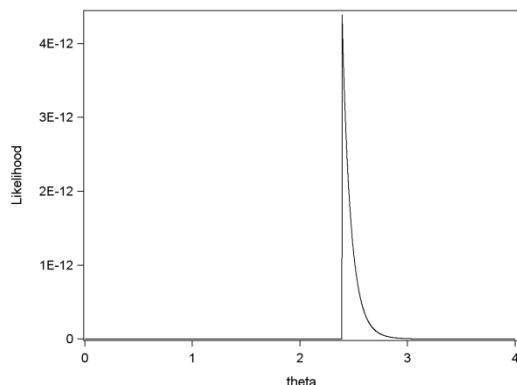
A. $\hat{\mu} = 14.29, \hat{\delta} = 5.95, \hat{\theta} = 1.32$; can't tell because you don't know whether a data value is from the treatment group or the control group.

B. $5.12 \leq \delta \leq 6.79$; yes, there is a difference between treatment and control because $\delta \neq 0$. Again, there is no telling whether the treatment makes the values higher or lower.

11.

A. For the first observation, $L(\theta) = 1/\theta$, for $1.26 < \theta$; $L(\theta) = 0$ otherwise. For the second observation, $L(\theta) = 1/\theta$, for $2.22 < \theta$; $L(\theta) = 0$ otherwise. To get the likelihood for the first two observations, you have to multiply those two functions. But the second function is 0 when $\theta < 2.22$, so the product is $L(\theta) = 1/\theta^2$, for $2.22 < \theta$; $L(\theta) = 0$ otherwise. Extending that logic to all 30 observations, the likelihood is 0 whenever θ is less than any observation, hence it is zero when θ is less than the maximum, 2.39. Thus $L(\theta) = 1/\theta^{30}$, for $2.39 < \theta$; $L(\theta) = 0$ otherwise.

B. Here is the graph^{xxxiii}.



The MLE is 2.39.

C. The plausible range of θ is from 2.39 to about 2.8.

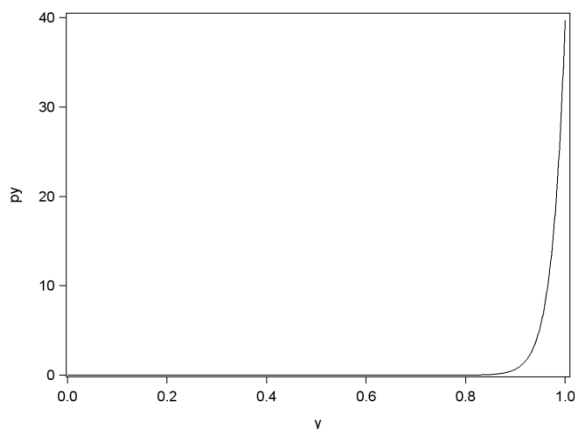
D. The derivative is not zero: The curve is downward sloping at $\theta = 2.39$.

12.

A. $\int_0^1 \theta y^{\theta-1} dy = \theta(y^\theta / \theta) \Big|_0^1 = 1^\theta - 0^\theta = 1.0$. Also the function is non-negative over the range 0 to 1, so it's a valid pdf.

B. $L(\theta) = \theta(1 - 0.012)^{\theta-1} \times \theta(1 - 0.001)^{\theta-1} \times \theta(1 - 0.043)^{\theta-1} \times \theta(1 - 0.008)^{\theta-1} \times \theta(1 - 0.059)^{\theta-1}$
 $= \theta^5 \{(1 - 0.012) \times (1 - 0.001) \times (1 - 0.043) \times (1 - 0.008) \times (1 - 0.059)\}^{\theta-1} = \theta^5 (0.88173)^{\theta-1}$. So $LL(\theta) = 5\ln(\theta) + (\theta - 1)\ln(0.88173)$, and $LL'(\theta) = 5/\theta + \ln(0.88173) = 0 \Rightarrow \hat{\theta} = 5/\{-\ln(0.88173)\} = 39.72$.

C. The graph^{xxxiii}:

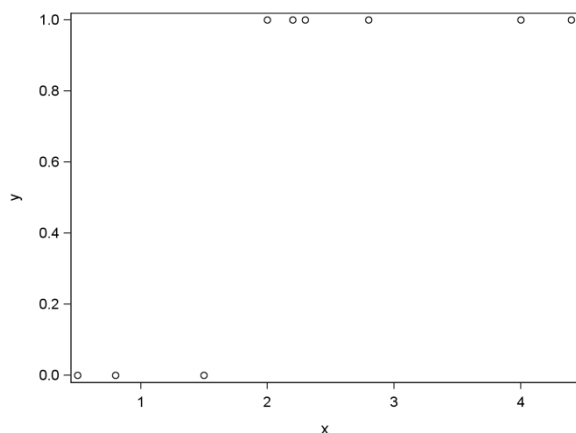


The p -value is greater than 0.10 if and only if $Y < 0.9$. The probability is

$\int_0^{0.90} \theta y^{\theta-1} dy = \theta(y^\theta / \theta) \Big|_0^{0.90} = 0.90^\theta - 0^\theta = 0.90^\theta$. This probability may be estimated as $0.90^{39.72} = 0.015$.

13.

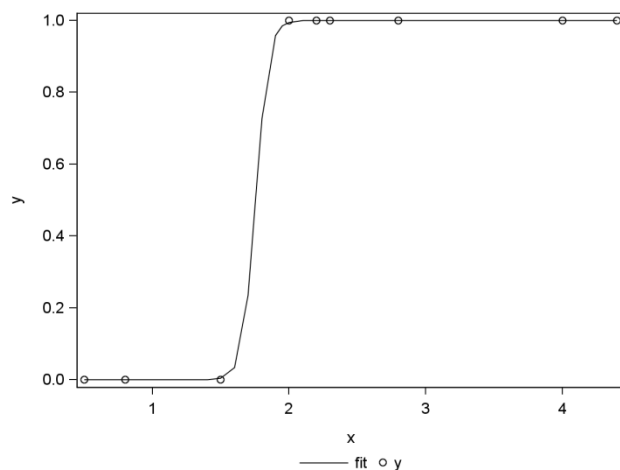
A. The graph:



The y data are perfectly separated: whenever $x < 1.8$, $y = 0$; otherwise $y = 1$.

B. $\exp(\beta_0 + \beta_1(2)) / (1 + \exp(\beta_0 + \beta_1(2))) \times 1 / (1 + \exp(\beta_0 + \beta_1(0.5))) \times \dots \times 1 / (1 + \exp(\beta_0 + \beta_1(0.8)))$

C. The graph:



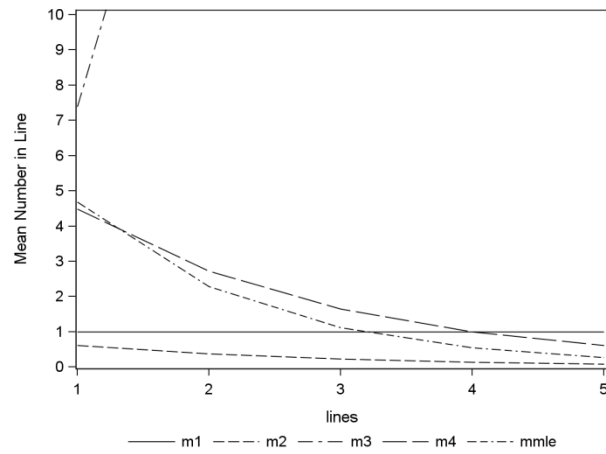
It seems that the model is trying to make the probabilities all 0's for the low values and all 1s for the high values.

14.

A. For (0, 0): $LL = -41.9$. For (0, -0.5): $LL = -58.2$. For (1, 1): $LL = -546.9$. For (2, -0.5): $LL = -22.6$. The setting (2, -0.5) is best supported by the data.

B. 2.259136484, -0.71626629; $LL = -22.00$. The log-likelihoods in A are all smaller because the MLE maximizes the log-likelihood function.

C. Here is the graph^{xxxiv}:



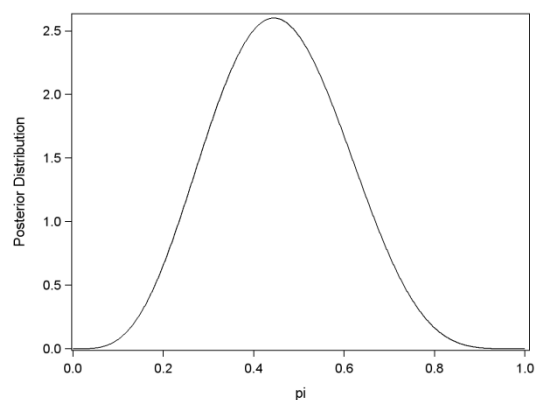
The graph labeled mmle is best supported by the data.

13. Bayesian Statistics

Solutions to Exercises

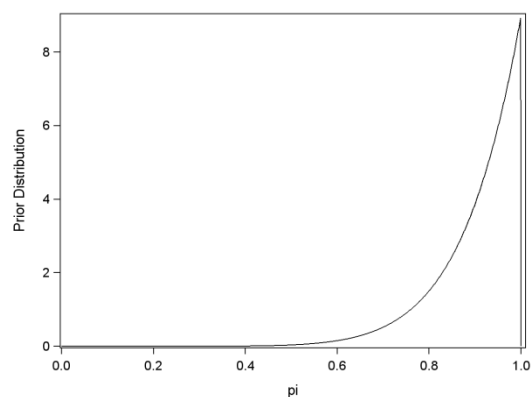
1.

A.



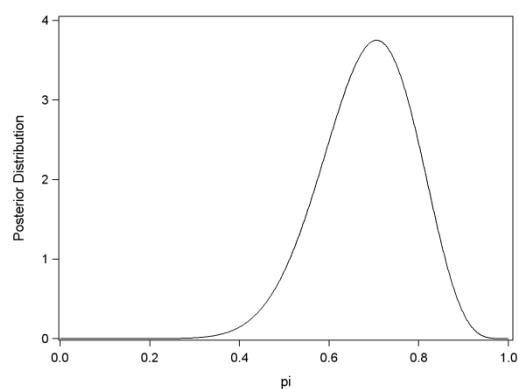
This is exactly the same as the likelihood function, but the vertical axis is scaled so that the area under the curve is 1.0.

B.



You would use this prior if you think the probability π is likely to be close to 1.0.

C.

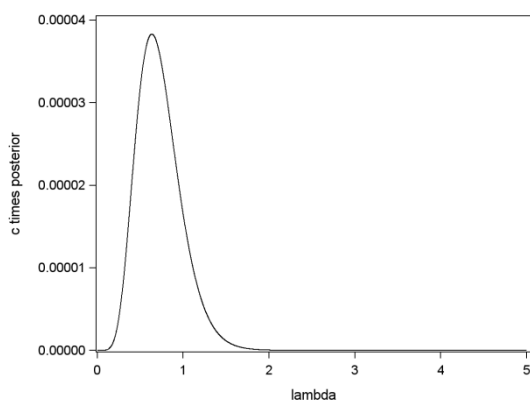


If you have a prior that expresses a strong opinion that the parameter is near 1.0, it has the effect of pulling the posterior towards 1.

2.

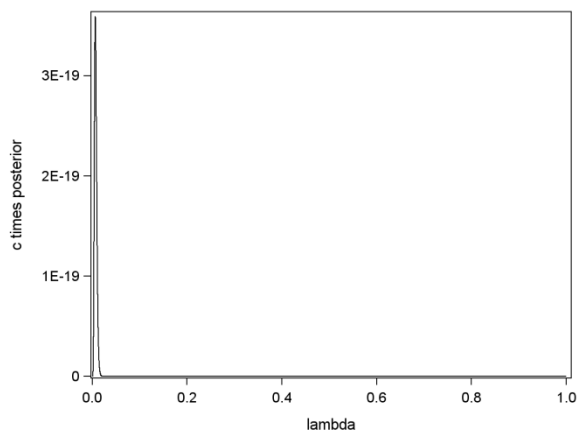
A. From Exercise 5 in Chapter 12, the likelihood function is $L(\lambda) = \lambda^7 e^{-11\lambda}/48$. So the posterior is $p(\lambda | \text{data}) \propto (\lambda^7 e^{-11\lambda}/48) \times 0.01 \exp(-0.01\lambda) \propto (\lambda^7 e^{-11.01\lambda})$

B. C



This posterior looks nearly identical to the likelihood function graphed in Exercise 12.5, so the prior has had little effect.

C. The posterior is $p(\lambda | \text{data}) \propto (\lambda^7 e^{-11\lambda}/48) \times 1000 \exp(-1000\lambda) \propto (\lambda^7 e^{-1111\lambda})$; the graph is



This prior changes the posterior from the likelihood a great deal and so has a great effect on the analysis. This prior is highly informative, while the previous one was vague.

3. Let θ_i be the probability for company i , where i is the company name. The likelihood function for θ is as follows:

Company, θ	Likelihood×Prior	Posterior
BankTen	$(0.196)^8(0.200)(0.088)(0.2)$ $= 7.6664\text{E-}09$	0.225642229
DoggyTreats	$(0.025)^8(0.239)(0.391)(0.2)$ $= 2.85184\text{E-}15$	8.39371E-08

CraftyCrafts	$(0.085)^8(0.327)(0.334)(0.2)$ $= 5.95217\text{E-}11$	0.001751882
AgBus	$(0.238)^8(0.209)(0.061)(0.2)$ $= 2.62495\text{E-}08$	0.772593268
InternetSavvy	$(0.045)^8(0.245)(0.517)(0.2) =$ $4.25978\text{E-}13$	1.25376E-05
Total:	3.39759E-08	1.0000

Most likely they came from AgBus.

4.
 - A. Because the the area under the curve is infinity, not 1.0. Therefore it is not a proper pdf.
 - B. $p(\theta | y) \propto p(y | \theta) p(\theta) = (1/(2\pi)^{1/2})\exp(-0.5(y - \theta)^2)(1.0)$. As a function of θ this is the $N(y, 1)$ pdf, so $\theta | Y = y \sim N(y, 1)$.
5.
 - A. $p(\pi_1, \pi_2 | \text{data}) \propto \pi_1^{392} \pi_2^{401} (1 - \pi_1 - \pi_2)^{209}$
 - B. It looks the same when graphed, but the z axis (that determines height when the function is viewed as a contour map) is scaled so that the volume under the graph is 1.0.
6. The maximum of the function $c \times f(x)$ occurs at the same x that maximizes $f(x)$. The likelihood function and the posterior distribution with a uniform prior differ only by a constant of proportionality, so the θ that maximizes one also maximizes the other.
7.
 - A. $L(\theta | \text{data}) = 1/\theta$, for $\theta > 0.6$; $= 0$ otherwise. So $p(\theta | \text{data}) \propto (1/\theta)I(\theta > 0.6) \times (1/5)I(0 < \theta < 5) \propto (1/\theta)I(0.6 < \theta < 5)$. To get the constant, note that $\int_{0.6}^5 \theta^{-1} d\theta = \ln(\theta)|_{0.6}^5 = \ln(5) - \ln(0.6) = 2.120$. So the posterior distribution is $p(\theta | \text{data}) = 1/(2.120\theta)$, for $0.6 < \theta < 5$.
 - B. The posterior cdf is $P(\theta | \text{data}) = (2.120)^{-1} \int_{0.6}^{\theta} \tau^{-1} d\tau = (2.120)^{-1} \ln(\tau)|_{0.6}^{\theta} = (2.120)^{-1} (\ln(\theta) - \ln(0.6))$. Setting $P(\theta | \text{data}) = 0.025$ yields $\ln(\theta) - \ln(0.6) = 2.120(0.025)$, hence $\theta = \exp(\ln(0.6) + 2.120(0.025)) = 0.633$. Setting $P(\theta | \text{data}) = 0.975$ yields $\ln(\theta) - \ln(0.6) = 2.120(0.975)$, hence $\theta = \exp(\ln(0.6) + 2.120(0.975)) = 4.741$. The equal tail interval is 0.633 to 4.741.
 - C. For the HPD interval, the left endpoint must be 0.6 since the peak of the curve is at the left boundary. So you just need the 0.95 quantile for the upper boundary, which is $\exp(\ln(0.6) + 2.120(0.95)) = 4.500$, and the HPD interval is 0.6 to 4.500.
 - D. Similar to the solution for Exercise 11A of Chapter 12, the likelihood function here is $L(\theta | \text{data}) = 1/\theta^7$ for $0.8 < \theta$, using the $U(0, 5)$ prior for θ , the posterior is $p(\theta | \text{data}) \propto 1/\theta^7$ for $0.8 < \theta < 5$. To get the constant, note that $\int_{0.8}^5 \theta^{-7} d\theta = -\theta^{-6}/6|_{0.8}^5 = -5^{-6}/6 - (-0.8^{-6}/6) = 0.636$. So the posterior distribution is $p(\theta | \text{data}) = 1/(0.636\theta^7)$, for $0.8 < \theta < 5$. The rest follows the previous solutions. You should notice that with more data the intervals are narrower.

- 8.
- A. $\mu | \text{data} \sim N(0.000200189, 0.00008151^2)$.
 - B. Using the normal cdf, get 0.007.
 - C. Because it says there is a chance the persistent mean return might actually be negative, in which case investments will lose money in the long run.

- 9.
- A. 9983 out of 10000 posterior samples showed $\beta_1 > 0$, hence the estimated posterior probability that $\beta_1 > 0$ is 0.9983. So, yes, there is about a 99.83% chance that larger X really do correspond to larger probabilities of observing $Y=1$.
 - B. 0.9224 to 1.0000

- 10.
- A. 0.9986
 - B. 28.00
 - C. 15.37 to 40.85
 - D. 0.9865^{xxxv}

- 11.
- A. Recall $L(\theta) = 1/\theta^{30}$, for $2.39 < \theta$; $L(\theta) = 0$ otherwise so $p(\theta | \text{data}) = c/\theta^{30}$, for $2.39 < \theta$; $= 0$ otherwise. The constant c is found as follows:

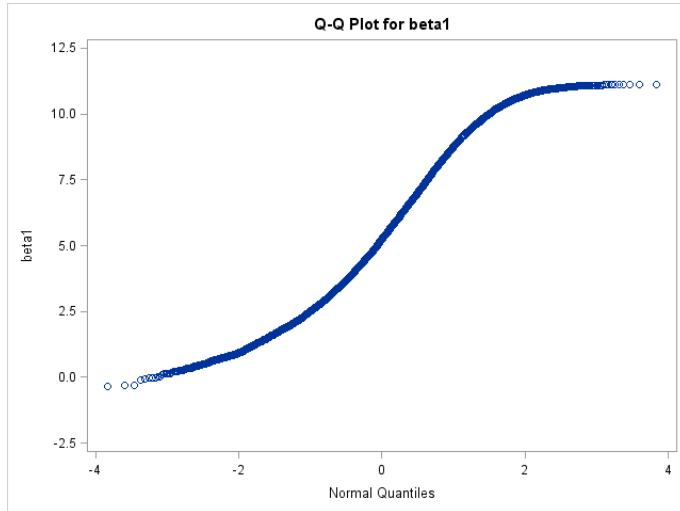
$$\int_{2.39}^{\infty} \theta^{-30} d\theta = -\theta^{-29} / 29 \Big|_{2.39}^{\infty} = -\infty^{-29} / 29 - (-2.39^{-29} / 29) = 3.66491\text{E}-13. \text{ So } p(\theta | \text{data}) = 2.72858\text{E}+12/\theta^{30}, 2.39 < \theta. \text{ The mean is}$$

$$\int_{2.39}^{\infty} \theta(2.72858\text{E}+12\theta^{-30}) d\theta = -2.72858\text{E}+12\theta^{-28} / 28 \Big|_{2.39}^{\infty} = 2.72858\text{E}+12 \times 2.39^{-28} / 28 = 2.475357$$

The median m satisfies $\int_{2.39}^m (2.72858\text{E}+12\theta^{-30}) d\theta = 0.5$; solving you get

$$-\theta^{-29} / 29 \Big|_{2.39}^m = -m^{-29} / 29 - (-2.39^{-29} / 29) = (.5)3.66491\text{E}-13, \text{ or } m = 2.448.$$

- B. The MLE is the maximum observation, but you know θ is more than any data value, being the theoretical maximum. So the estimates in A are more sensible.
12. The likelihood function is proportional to the posterior in the case of a uniform prior. See the solution to 12C in Chapter 12.
13. Using PROC GENMOD of SAS, the uniform prior produced an error: ERROR: Error in Bayesian Markov Chain Monte Carlo computation. Bayesian analysis can not be completed. Using a jeffreys prior gave results. Here is a q-q plot of the simulated β_1 values.



It seems that the quasi-separation case can be solved using a prior, but the results are quite sensitive to the prior that you pick.

14.

- A. Estimated^{xxxvi} to be 0.0 (0/10,000 samples from the posterior yielded $\beta_1 > 0$).
- B. $E(Y | X = 2) = \exp(\beta_0 + \beta_1(2))$, calculating this for each of 10,000 simulated pairs (β_0, β_1) and picking the 5th and 95th percentiles gives the range 1.56 to 3.19. The mean max number in line when there are two lines is between 1.56 and 3.19 with 90% posterior probability.
- C. Here is the summary from 10,000 simulations:

Ystar	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	1032	10.32	1032	10.32
1	2260	22.60	3292	32.92
2	2507	25.07	5799	57.99
3	1946	19.46	7745	77.45
4	1200	12.00	8945	89.45
5	623	6.23	9568	95.68
6	261	2.61	9829	98.29
7	118	1.18	9947	99.47
8	33	0.33	9980	99.80
9	13	0.13	9993	99.93
10	4	0.04	9997	99.97
11	1	0.01	9998	99.98

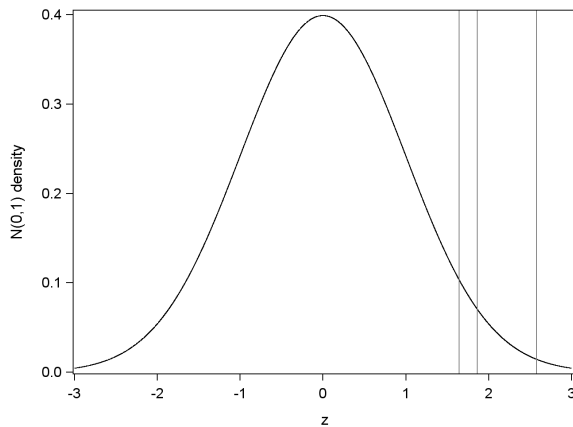
Ystar	Frequency	Percent	Cumulative Frequency	Cumulative Percent
12	1	0.01	9999	99.99
13	1	0.01	10000	100.00

When there are two lines, most of the time ($\cong 95.68\%$ of the time) the maximum number in line will be 5 or fewer. Occasionally, there can be much larger numbers in line.

14. Frequentist Statistical Methods

Solutions to Exercises

1.
 - A. I am 16.7% confident that the die is showing a 3.
 - B. There is a 16.7% probability that the die is showing a 3.
2.
 - A. I am $100\pi\%$ confident that the die is showing a 3.
 - B. There is a 16.7% probability that the die is showing a 3. (There is a lot more to this particular answer, but this will do, assuming vagueness on the Bayesian's part.)
3. The graph^{xxxvii}:



The area between the + and – critical values should be equal to the confidence level. So with 95% confidence, you need 2.5% in each tail, and hence the quantile is the 0.975 quantile, which includes the 95% in the middle as well as the 2.5% in the left tail.

4.
 - A. $1.16 \leq \mu_X \leq 2.67$
 - B. In approximately 90% of repeated samples from the process, similarly constructed intervals (which will all be different than the one in A) will contain the true process mean μ_X .
 - C. $2.06 \leq \mu_Y \leq 6.36$
 - D. In approximately 99% of repeated samples from the process, similarly constructed intervals (which will all be different than the one in C) will contain the true process mean μ_Y . This type of interval is correct more often than the one in A, so I have more confidence that it contains the parameter.
 - E. For A, 8498 out of 10,000 are correct, so the true confidence level is approximately 84.98%. For C, 9706 out of 10,000 are correct, so the true confidence level is approximately 97.06%.

5.

A.

y	$p(y)$
0	$1 - \pi$
1	π
Total:	1.00

$$\text{Mean} = \pi, \text{ Variance} = \pi(1 - \pi)$$

B.

y	$\hat{p}(y)$
0	0.35
1	0.65
Total:	1.00

Mean = 0.65, Variance = $0.65(0.35) = 0.2275$. The distribution in A is correct because the model produces the data. The data do not produce the data. Or in other words, Nature is not changed by the 13/20 1's – whatever the natural probability π was before data collection is the same as that after. Also, π is unknown (*model has unknown parameters*); you can't assume it is any specific number like 0.65.

C. $0.65 \pm 1.96(0.2275)^{1/2}/(20)^{1/2}$ or $0.441 \leq \pi \leq 0.859$.

D. Just plug in the numbers.

E. In approximately 95% of repeated samples from the process, similarly constructed intervals (which will all be different than the one in C) will contain the true process mean π .

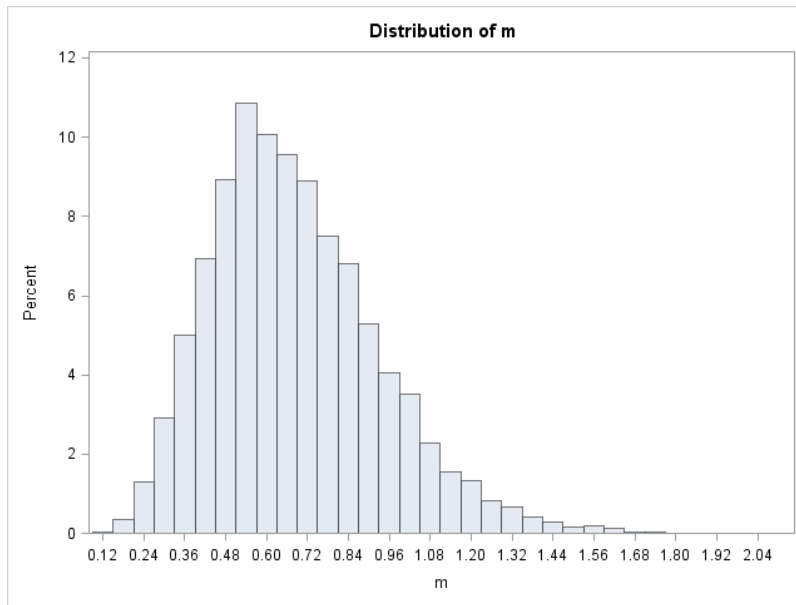
F. When $\pi = 0.60$, $9249/10000 = 92.49\%$ of the simulated intervals contained π . This is reasonably close to 95%. When $\pi = 0.01$, $1835/10000 = 18.35\%$ of the simulated intervals contained π . This is *not* reasonably close to 95%. The ugly rule of thumb requires expected frequencies > 5 in most categories. When $\pi = 0.60$, the expected frequencies are 8 and 12 for 0 and 1, respectively, meeting the ugly rule of thumb. When $\pi = 0.01$, the expected frequencies are 19.8 and 0.2 for 0 and 1, respectively, not meeting the ugly rule of thumb.

6. $0.000200189 \pm 1.96(0.00008151)$ or $0.00004 \leq \mu \leq 0.00036$. This interval assumes approximate normality of the average, whereas the other did not. Assuming approximate normality, you get a tighter range than when you use Chebychev's inequality, which does not assume any distribution.

7.

A. $-0.12 \leq \mu \leq 1.40$

B. Here is the graph^{xxxviii}:



C. $0.29 \leq \mu \leq 1.25$. The asymmetric interval is preferred because the symmetric interval admits negative values, which are impossible for the mean of a Poisson distribution.

8.

A. $0.745 \leq \mu \leq 1.246$

B. $1.49 \leq \theta \leq 2.49$

C. That interval was 2.39 to 2.8, and was estimated from the likelihood, which is asymmetric, rather than from the assumption of a normal distribution for the average, which is a symmetric distribution. In particular, θ can't be less than 2.39, but the standard interval $1.49 \leq \theta \leq 2.49$ does not respect this fact.

15. Are Your Results Explainable by Chance Alone?

Solutions to Exercises

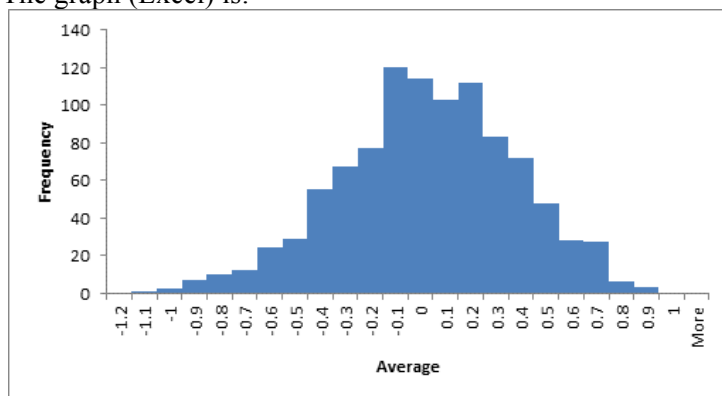
1.

- A. $Y_1, Y_2, \dots, Y_{100} \sim_{\text{iid}} \text{Bernoulli}(0.5)$
- B. Under the null model, $E(Y_i) = 0.5$ and the $\text{Var}(Y_i) = 0.5(1 - 0.5) = 0.25$. Hence $E(T) = 100(0.5) = 50$ and $\text{Var}(T) = 100(0.25) = 25$. Applying the CLT, $T \sim N(50, 5^2)$.
- C. $pv = 2\Pr(T \leq 45) \approx 2\Pr(V \leq 45)$, where $V \sim N(50, 5^2)$. Using software, you get $pv \approx 0.3173$. Hans' data are easily explainable by chance alone.

2.

- A. The CLT states that $\bar{D}$ is approximately normally distributed. Since under the null model, $E(D_i) = 0$, $E(\bar{D}) = 0$ as well under the null model. You know that $\text{Var}(\bar{D}) = \text{Var}(D_i)/n = \sigma^2/10$. The parameter $\sigma^2 = E(D - \mu_D)^2$ is unknown, but is equal to $E(D^2)$ under the null model, and can be estimated via the plug-in estimator $(1/n)\Sigma(D_i)^2$, giving $\hat{\sigma}^2 = 1.143$. Thus an approximate null distribution of $\bar{D}$ is $N(0, 1.143/10)$.
- B. The observed average is $\bar{d} = 0.089$. Thus $pv \approx 2\Pr(V \geq 0.089)$, where $V \sim N(0, 1.143/10)$. Using software, you get $pv \approx 0.792$. The results are easily explainable by chance alone.
- C. The bootstrap centered data are $0.22 - 0.089 = 0.131, -1.79 - 0.089 = -1.879, \dots, -0.28 - 0.089 = -0.369$.

The graph (Excel) is:



The observed value 0.089 is near the center.

- D. Using the 1000 values graphed in C, there are 802 that are either ≥ 0.089 or ≤ -0.089 . So the two-sided p -value is estimated to be 0.802, agreeing nicely with the normal approximation p -value 0.792.

3.

- A. $Y_1, Y_2, \dots, Y_{1250} \sim_{\text{iid}} p(y)$, where $p(y)$ is given as follows:

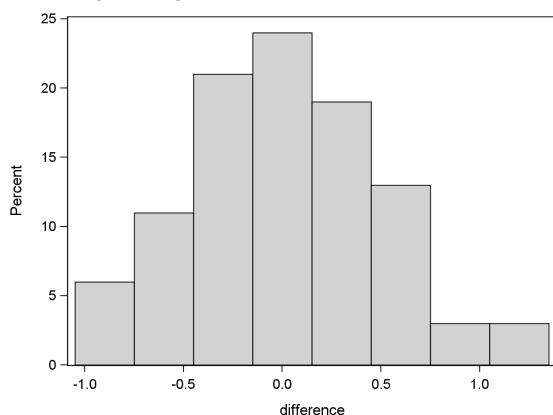
y	$p(y)$
1	π_1
2	π_2
3	π_3
4	π_4
5	π_5
Total:	1.0

The parameters are as follows: π_1 = the probability that a person answers 1, π_2 = the probability that a person answers 2, etc.

- B. $\bar{Y}_1 - \bar{Y}_2 \sim N(0, \sigma^2(1/1000 + 1/250))$. The parameter σ is unknown but 1.4 is an estimate; hence $\bar{Y}_1 - \bar{Y}_2 \sim N(0, (1.4)^2(1/1000 + 1/250))$.
- C. The observed difference is $3.50 - 2.94 = 0.56$. The two-sided p -value is thus $pv \cong 2 \times \Pr(V \geq 0.56)$, where $V \sim N(0, (1.4)^2(1/1000 + 1/250))$. Using software that evaluates the normal cdf, $pv \cong 1.5 \times 10^{-8}$. The difference between average satisfaction of newer and existing customers is larger than can be explained by chance alone.
- D. The discrete uniform model on 1, 2, 3, 4, 5 seems reasonable, having mean 3.0 and standard deviation 1.4. Simulating 100 sets of 1250 numbers, and taking first 250 of each set to be simulated newer customers and the remaining 1000 to be simulated existing customers, and calculating the difference between averages, the 100 differences ranged from -0.27 to 0.398 . Thus the estimate of the p -value using these simulated data is $0/100 = 0.0$, and the normality assuming estimate $pv \cong 1.5 \times 10^{-8}$ appears reasonable.

4.

- A. $Y_{ij} \sim_{\text{iid}} p(y)$, where $i = 1, 2$ (routes) and $j = 1, 2, \dots, n_i$ (driving excursions within route).
- B. Maybe it takes on average 15 minutes, with a standard deviation of 1 minute. Then $p(y)$ could be specified as the $N(15, 1^2)$ distribution for the purposes of a simulation model. Simulating 100 trials of 10 trips using each route, the difference between average times looks as shown in the following histogram:



- C. Based on that histogram, a difference in averages of 0.15 would be easily explainable by chance alone, so I would not conclude that the new route was faster.

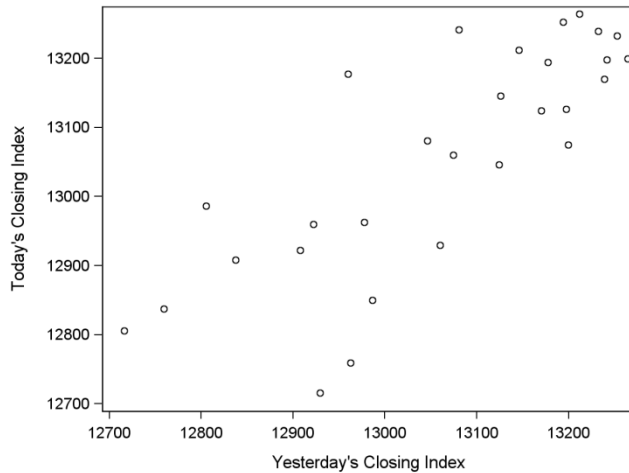
5. 0.0013, 0.0228, 0.1587, 0.5000, 0.8413, 0.9772, 0.9987

6.

- A. Because it's just an estimate based on a sample of $n = 18,834$ observations. A different sample would give a different estimate.
- B. Under the bootstrap model, $\Pr(Y \leq 0)$ is the proportion of the 18,834 observations that are less than or equal to zero. This should be very close to the true proportion, assuming the 18,834 are iid from $p_0(y)$, by the LLN as discussed in Section 8.5.

7.

- A. $Y_1, Y_2, \dots, Y_{30} \sim_{\text{iid}} p(y)$
- B.



C. 0.7928

D. Using 100 iid bootstrap samples, each having size $n=30$, and then finding the lag correlation for each sample, the values ranged from -0.53 to 0.40 . Thus the probability of observing a lag correlation of 0.7928 or higher under the iid sampling model is estimated to be $0/100 = 0.0$. (The true probability is not really zero, but close.) The lag correlation is too large to be explained by the iid model.

8.

A. -0.145

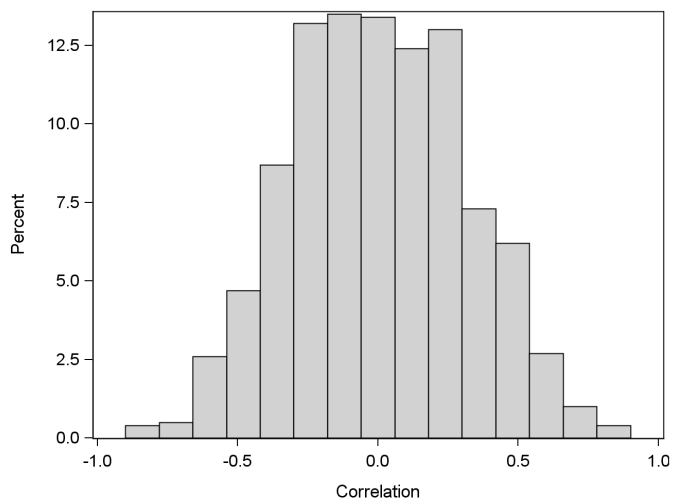
B.

x	$\hat{p}(x)$
1	4/12
3	2/12
4	2/12
5	4/12
Total:	1.00

C.

y	$\hat{p}(y)$
1	4/12
2	4/12
3	3/12
5	1/12
Total:	1.00

D. Here is the histogram^{xxxix}.



E. 68%; this is the estimated p -value.

9.

A. 0.74613

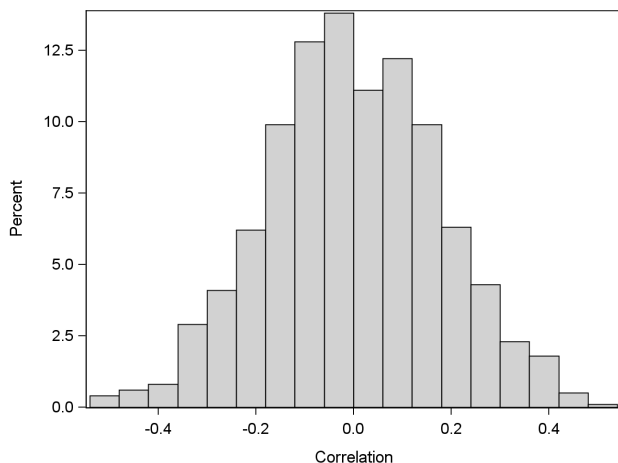
B.

x	$\hat{p}(x)$
1	6/33
2	3/33
3	7/33
4	12/33
5	5/33
Total:	1.00

C.

y	$\hat{p}(y)$
1	6/33
2	5/33
3	7/33
4	10/33
5	5/33
Total:	1.00

D. Here is the histogram^{x1}.

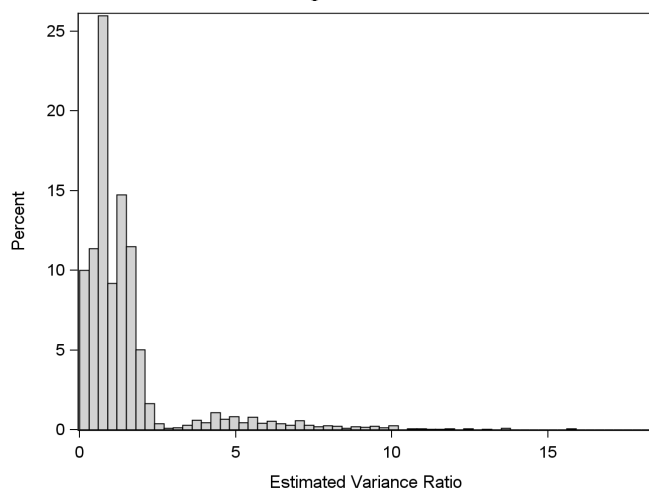


E. 0%; this is the estimated p -value.

10.

A. 0.6615

B. Based on 100,000 random permutations^{xli}:



C. 0.569

11.

A. Either $U(0,1)$, or approximately $U(0,1)$.

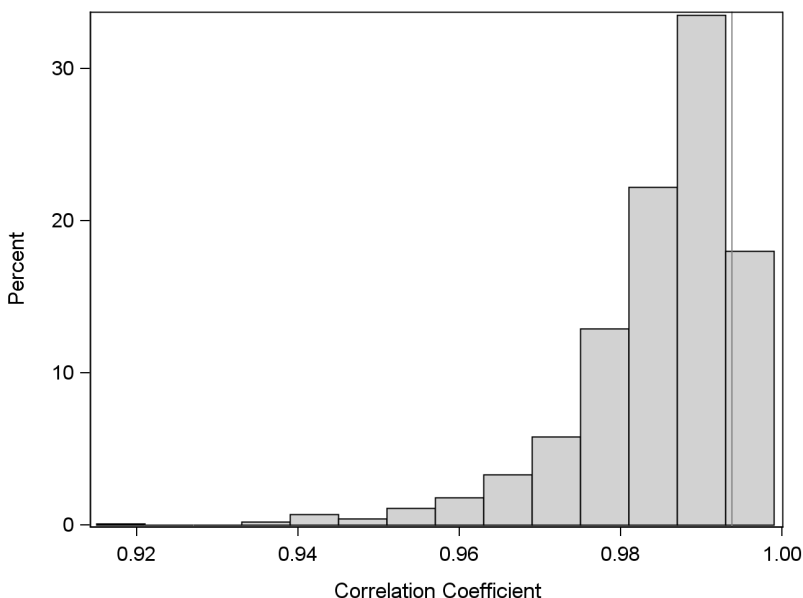
B. 652 out of 1000, or 0.652. He has about a 65% chance of finding a false positive.

C. 994 out of 1000, or 0.994. He has about a 99.4% chance of finding a false positive.

D. 52 out of 1000 with 20 tests and 45 out of 1000 with 100 tests. In either case, he has about a 5% chance of finding a false positive if he uses the Bonferroni method. The Bonferroni method is effective at screening out false positives.

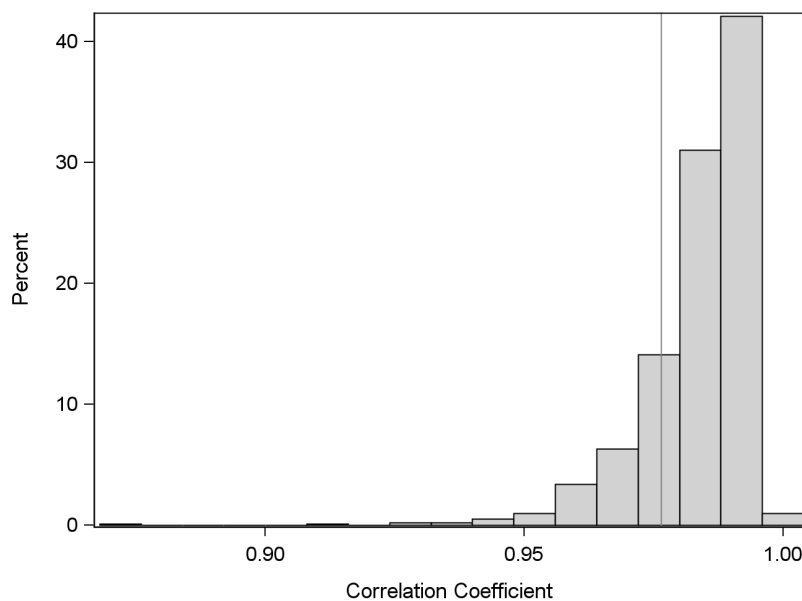
12.

The correlation coefficient for the exponential q-q plot is 0.99373. Simulating 1000 data sets from the exponential distribution with $\lambda = 1$ gives the following distribution^{xlii} of similarly calculated correlation coefficients:



The p -value is the proportion of correlations less than or equal to 0.99373 (the vertical line in the graph), and is estimated to be $858/1000 = 0.858$. The deviation of the original q-q plot from a line is easily explainable by chance alone if the data come from an exponential distribution.

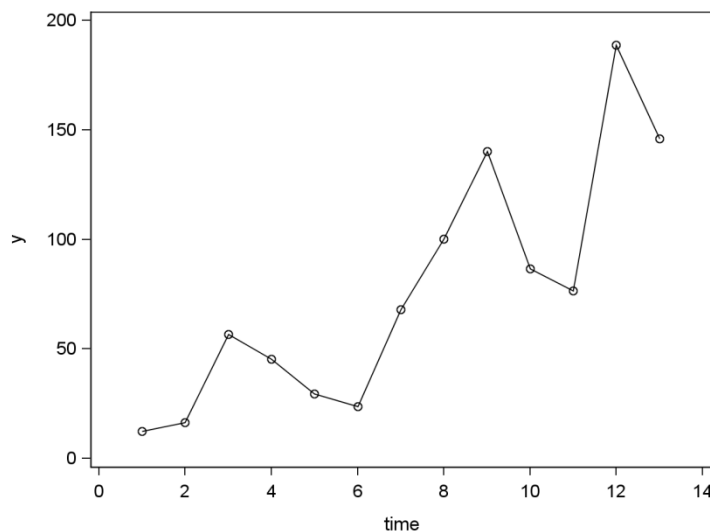
13. The correlation coefficient for the uniform q-q plot is 0.97649. Simulating 1000 data sets from the $U(0, 360)$ distribution gives the following distribution of similarly calculated correlation coefficients:



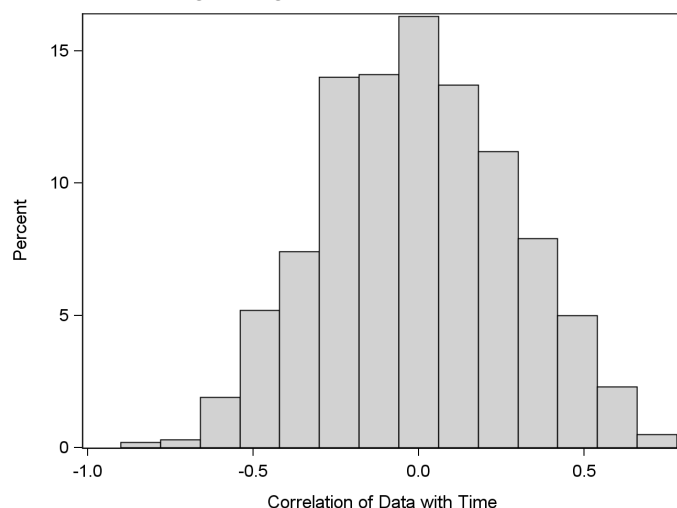
The p -value is the proportion of correlations less than or equal to 0.97649 (the vertical line in the graph), and is estimated to be $192/1000 = 0.192$. The deviation of the original q-q plot from a line is easily explainable by chance alone if the data come from the $U(0, 360)$ distribution.

(Comments: The correlation coefficient measures deviation from the line of best fit, not the deviation from the expected line. A statistic that measures deviation from the expected line will show different results; see the *Kolmogorov-Smirnov test*. Also, the sample size is small here so randomness has a large effect.)

14.

A. The graph is:^{xliii}

- B. 0.842. There is a strong relationship between the data value and the time that it is observed.
- C. Data are random. They rarely if ever are equal to their expected values. (Never, in the continuous case).
- D. Simulating 1000 iid samples of size $n = 13$ each from the bootstrap distribution, and calculating the correlation between the observation and the time order for each sample produces the following histogram of correlation coefficients.



Out of 1000 simulations under the iid model, the smallest correlation was -0.837 and the largest was 0.750 . None were more than 0.842 or less than -0.842 so the p -value is estimated to be $0/1000$, or 0.0 . In other words, the probability of seeing a correlation as extreme as 0.842 between the data and the time order is nearly zero, under the iid model. Thus the time trend in the original data does not appear to be explainable by chance alone.

16. Chi-Squared, Student's t , and F -Distributions, with Applications

Solutions to Exercises

1. Let $T = aY + b$, where $a < 0$. Then:

$P(t) = \Pr(T \leq t)$	(by definition of cdf)
$= \Pr(aY + b \leq t)$	(by substitution)
$= \Pr\{Y \geq (t - b)/a\}$	(by algebra, since $a < 0$)
$= 1 - P_Y\{(t - b)/a\}$	(since $\Pr(Y = (t - b)/a) = 0$, and by definition of the cdf of Y).

Now, $p(t) = P'(t)$, since the derivative of the cdf is the pdf (see Chapter 2). So

$P'(t) = (\partial/\partial t)P(t)$	(by definition of derivative)
$= (\partial/\partial t) [1 - P_Y\{(t - b)/a\}]$	(by substitution from above)
$= -P'_Y\{(t - b)/a\} (\partial/\partial t) \{(t - b)/a\}$	(by the chain rule, property D9 given in Section 2.5 of Chapter 2, and by linearity property of derivative)
$= -P'_Y\{(t - b)/a\} \times 1/a$	(by the linearity and additivity properties of derivatives).

Now, $P'_Y(y) = p_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(y - \mu)^2}{2\sigma^2}\right\}$, by the fact that the derivative of the cdf is equal to the pdf, and by the assumption that $Y \sim N(\mu, \sigma^2)$. So, by substitution and algebra:

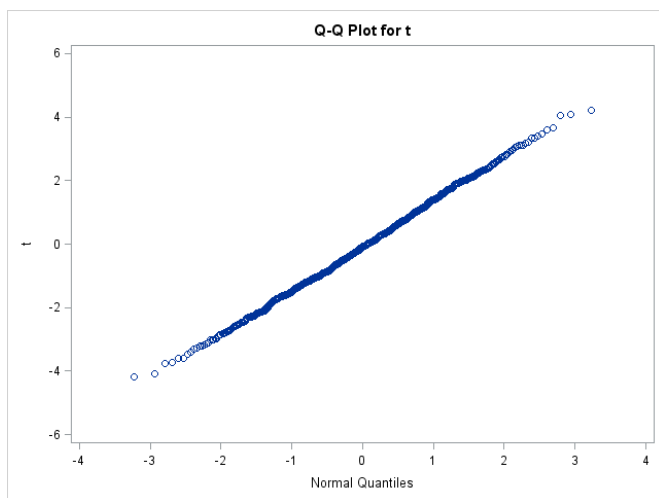
$$\begin{aligned}
 -P'_Y\{(t - b)/a\} \times 1/a &= p_Y\{(t - b)/a\} \times 1/a \\
 &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{\{(t - b)/a - \mu\}^2}{2\sigma^2}\right] \times 1/(-a) \\
 &= \frac{1}{\sqrt{2\pi} |a| \sigma} \exp\left[-\frac{\{t - (a\mu + b)\}^2}{2a^2\sigma^2}\right]
 \end{aligned}$$

This form you can recognize as the $N(a\mu + b, a^2\sigma^2)$ pdf for the random variable T .

2. Let home time = Y_1 and driving time = Y_2 . Then $Y_1 \sim N(65, 10^2)$ and $Y_2 \sim N(30, 3^2)$. By the additivity property of the normal distribution under independence, her total time $T = Y_1 + Y_2 \sim N(95, 109)$. The probability that she is late is $\Pr(T > 105) = 0.169$. She is late around one out of six class days.

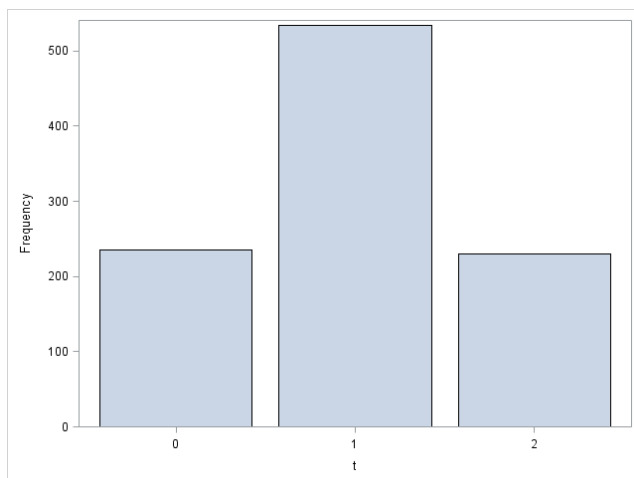
3.

- A. Generate 1000 Y_1 and Y_2 , both distributed as $N(0,1)$, for example. Add them and draw a q-q plot of the resulting 1000 sums $Y_1 + Y_2$. Here's how it may look:



It looks like these T could have come from a normal distribution.

- B. Generate 1000 Y_1 and Y_2 , both distributed as $\text{Bernoulli}(0.5)$, for example. Add them and draw a bar chart showing frequencies of the resulting 1000 sums $Y_1 + Y_2$. Here's how it may look^{xliv}:



These T are *not* from a Bernoulli distribution because there are three values of T , while there are only two values for a Bernoulli RV.

4.

- A. $N(0, 0.5)$ B. $N(0, 1)$ C. T_1 since the denominator is $\{(Z_3)^2/1\}^{1/2}$, the square root of a chi-squared random variable divided by its degrees of freedom (1 in this case).

5.

- A. The standard deviations are larger than the means. Therefore, if the purchase amounts were normally distributed, then there would have to be a lot of negative purchase amounts. So the normality assumption of the two-sample t -test is clearly violated.

- B. The t -statistic is $T = \hat{\delta} / s.e.(\hat{\delta})$. Here, $\hat{\delta} = 49.21 - 50.24 = -1.03$, and $s.e.(\hat{\delta}) = \hat{\sigma}_{pooled} \sqrt{1/n_1 + 1/n_2}$, where $\hat{\sigma}_{pooled}^2 = \frac{(n_1 - 1)\hat{\sigma}_1^2 + (n_2 - 1)\hat{\sigma}_2^2}{(n_1 - 1) + (n_2 - 1)} = \{(123410 - 1)(56.10)^2 + (156770 - 1)(69.04)^2\} / \{(123410 - 1) + (156770 - 1)\} = 63.665^2$. So $t = -1.03 / \{63.665(1/123410 + 1/156770)^{1/2}\} = -4.25$. The probability of seeing a t -statistic this far from 0 by chance alone is $p_v = 2 \times \Pr(T_{(123410-1) + (156770-1)} \leq -4.25) = 0.00002$. The difference is very unlikely due to chance alone.
- C. The confidence interval is $-1.03 \pm c \{63.665(1/123410 + 1/156770)^{1/2}\}$, where $c = t_{0.975, (123410-1) + (156770-1)} = 1.96$. So with 95% confidence, $-1.50 \leq \delta \leq -0.56$. The interval excludes the possibility that $\delta = 0$, which corroborates the analysis of B that states that the results are not explainable by chance alone. The chance-only model supposes that $\delta = 0$.
- D. The difference in average purchase after clicks on ad 1 is somewhere between 1.50 and 0.56 less than the difference in average purchase amounts after clicks on ad 2. It's not a very big difference, but money is money, and so yes, the difference does seem practically significant. All else being equal, the advertiser would be wise to use ad 2.
- 6.
- A. The pairs of observations on the same person might be correlated because they share that person's unique characteristics. Hans should know better.
- B. The differences d_i are 3, 4, 0, -1, 2, 0, -1, 2, -2, 4, -2, and 3. Since each difference is from a different person, they are reasonably assumed independent, unless there is some collusion, where people jointly decide how to answer the survey. Hidden grouping variables such as individuals who are married could also induce dependence.
- C. The differences are discrete, taken only values -4, -3, ..., 4, and so are obviously not produced by a normal distribution.
- D. The exact 95% confidence interval for a sample from $N(\mu, \sigma^2)$ is $\bar{Y} - c \times s.e.(\bar{Y}) \leq \mu \leq \bar{Y} + c \times s.e.(\bar{Y})$. Here the Y 's are the D 's, $c = t_{12-1, 0.975} = 2.201$, the mean of the d values is 1.0, the standard deviation of the d values is 2.2563, and the standard error of the average d is $2.2563/12^{1/2} = 0.6513$. Hence the interval for μ_d is $1.0 \pm 2.201 \times 0.6513$, or $-0.43 \leq \mu_d \leq 2.43$. The process mean difference between online newspaper preference and print newspaper preference is between -0.43 and 2.43, with 95% confidence.
- E. Yes, because the interval admits that μ_d could be zero, in which case the difference between the sample averages would be explained by chance alone.
- 7.
- A. The estimated standard deviation of the differences is 2.2563. The critical values from Equations (16.4) and (16.5) are $c_L = 0.6449$ and $c_U = 1.3374$. So the interval is $2.2563/1.3374 \leq \sigma_d \leq 2.2563/0.6449$, or $1.69 \leq \sigma_d \leq 3.50$.
- B. The normality assumption is obviously violated. The independence assumption is reasonable, barring collusion or clustered samples. The identical distribution assumption is reasonable.
8. The interval is $\hat{\delta} \pm c \times s.e.(\hat{\delta})$; here $\hat{\delta} = -0.33 - 0.07 = -0.40$. The pooled variance estimate is $\{(27 - 1)1.07^2 + (32 - 1)1.70^2\} / \{(27 - 1) + (32 - 1)\} = 2.0940$ and the pooled standard deviation is $2.0940^{1/2} = 1.447$. The critical value is $c = t_{(27-1) + (32-1), 0.975} = 2.002$, so the interval is $-0.40 \pm 2.002 \times 1.447(1/27 + 1/32)^{1/2}$, or $-1.16 \leq \delta \leq 0.36$. The difference between the mean returns for the pre-9/11 process and the post-9/11 process is between -1.16 and 0.36, with 95% confidence. In particular, it is not clear that 9/11 affected mean returns, since the difference may be positive or negative.

9.

- A. The interval is $\hat{\theta} / F_{n_1-1, n_2-1, 1-\alpha/2} \leq \theta \leq \hat{\theta} / F_{n_1-1, n_2-1, \alpha/2}$. Here, $\hat{\theta} = 1.07^2 / 1.70^2 = 0.396$, and the critical values are $F_{27-1, 32-1, 1-0.05/2} = 2.095$ and $F_{27-1, 32-1, 0.05/2} = 0.465$, so the interval is $0.396/2.095 \leq \theta \leq 0.396/0.465$, or $0.189 \leq \theta \leq 0.852$. There is evidence that the process variance (called volatility) prior to 9/11 is less than the process variance after 9/11.
- B. The p -value is $2 \times \Pr(F_{26,31} \leq 0.396) = 0.0092$. The difference between the observed ratio and 1.0 is not easily explained by chance alone, under the model where the process variances are equal. This corroborates 9A which also ruled out equality (or $\theta = 1$).

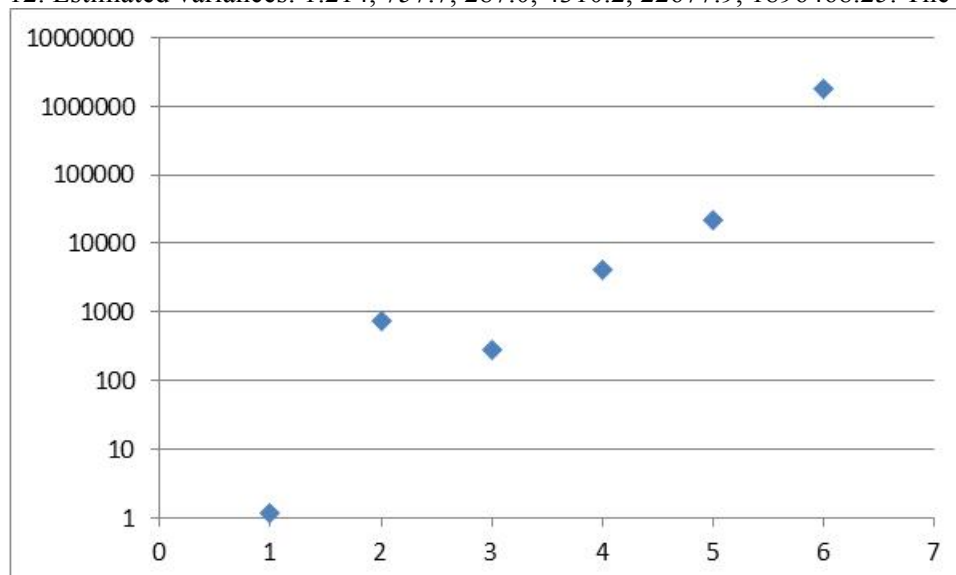
10.

- A. $Y_{ij} \sim \text{independent } N(\mu_i, \sigma^2)$, where $i = 1, 2, 3$; and $j = 1, 2, \dots, 20$.
- B. Wait times can't be negative and there are occasional long wait times when call volume is heavy, so the distribution will likely be right skewed rather than normal.
- C. Independence may be reasonable, although adjacency could cause dependencies. In periods of time where call volume is heavy, long wait times will tend to follow other long wait times.
- D. Homoscedasticity may be reasonable, although if one call center is understaffed, the variability in wait times may be higher, say ranging from 0 to 5 minutes, whereas in a well-staffed call center, the wait times might all be between 0 and 1 minute.
- E. $F_{2,57} = 3.47$, $pv = 0.0380$. There is only a 0.0380 chance of seeing an F -statistic as extreme as 3.47 when the process mean wait times are the same for all three centers. The difference between the sample means of the three shifts is therefore not easily explained by chance alone.

11.

- A. $F_{3,396} = 0.51$, $pv = 0.673$. The difference between the proportions (recall that a proportion is a mean of $\{0, 1\}$ data) is easily explainable by chance alone, under a model where the true proportions are really equal. Hans, it's not surprising that you got different results every time.
- B. All assumptions are reasonable—and even true unless Hans is a trickster—except for the normality assumption.
- C. $\chi^2_3 = 1.550$, $pv = 0.671$. The p -value is nearly identical to that of the F -test and the conclusion is identical.

12. Estimated variances: 1.214, 757.7, 287.0, 4310.2, 22677.9, 1896468.25. The graph:



Hans, here is the solution to your dilemma. Even though the data are all sampled from the same distribution, the sample variance just gets bigger and bigger; it never converges. The true variance is infinity, because that's where the sample variance goes as you collect more data.

17. Likelihood Ratio Tests

Solutions to Exercises

1.

$$\begin{aligned}
 & L(\sigma^2 | y_1, y_2, \dots, y_n) \\
 &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \frac{-(y_1 - m_0)^2}{2\sigma^2} \times \\
 & \frac{1}{\sqrt{2\pi\sigma^2}} \exp \frac{-(y_2 - m_0)^2}{2\sigma^2} \times \dots \times \\
 & \frac{1}{\sqrt{2\pi\sigma^2}} \exp \frac{-(y_n - m_0)^2}{2\sigma^2} \quad \text{(by substitution)} \\
 &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \\
 & \times \exp \left\{ \frac{-(y_1 - m_0)^2 - (y_2 - m_0)^2 - \dots - (y_n - m_0)^2}{2\sigma^2} \right\} \quad \text{(by algebra and properties of exponents)} \\
 &= \left(\frac{1}{\sqrt{2\pi}} \right)^n \left(\frac{1}{\sqrt{\sigma^2}} \right)^n \exp \left\{ \frac{-\sum_i (y_i - m_0)^2}{2\sigma^2} \right\} \quad \text{(by properties of exponents and definition of summation } \Sigma) \\
 &= (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp \left\{ \frac{-\sum_i (y_i - m_0)^2}{2\sigma^2} \right\} \quad \text{(by properties of exponents)}
 \end{aligned}$$

Hence $LL(\sigma^2 | y_1, y_2, \dots, y_n) = -(n/2)\ln(2\pi) - (n/2)\ln(\sigma^2) - \Sigma(y_i - m_0)^2/(2\sigma^2)$, implying that $(\partial/\partial\sigma^2)LL = -(n/2)(1/\sigma^2) + \Sigma(y_i - m_0)^2/(2\sigma^4)$. Setting this derivative to zero at $\sigma^2 = \hat{\sigma}_0^2$ and solving gives the result that $\hat{\sigma}_0^2 = \{(y_1 - m_0)^2 + \{(y_2 - m_0)^2 + \dots + \{(y_n - m_0)^2\}/n$.

2. In (17.3), $\chi^2 = n \times \ln \{1 + (\bar{y} - m_0)^2 / \hat{\sigma}_1^2\}$. But the t -statistic for testing $H_0 : \mu = m_0$ is

$t = (\bar{y} - m_0)/(\hat{\sigma}/n^{1/2})$, implying $t^2 = (\bar{y} - m_0)^2/(\hat{\sigma}^2/n)$. Since $\hat{\sigma}_1^2 = \frac{n-1}{n}\hat{\sigma}^2$, it follows that $t^2 = (\bar{y} - m_0)^2/(\hat{\sigma}_1^2/(n-1)) = (n-1)(\bar{y} - m_0)^2/\hat{\sigma}_1^2$, or $(\bar{y} - m_0)^2/\hat{\sigma}_1^2 = t^2/(n-1)$. Substituting this latter expression in (17.3) gives you (17.4).

3.

A. $\chi^2 = 2(LL_1 - LL_0) = 2n \ln(m_0/\bar{y}) - 2n(1 - \bar{y}/m_0) = -2n \ln(\bar{y}/m_0) - 2n(1 - \bar{y}/m_0)$, because of the fact that $\ln(m_0/\bar{y}) = -\ln(\bar{y}/m_0)$, by property of logarithms.

- B. Write $\bar{y}/m_0 = (\bar{y} - m_0 + m_0)/m_0 = 1 + (\bar{y} - m_0)/m_0$. Then $\ln(\bar{y}/m_0) \cong (\bar{y} - m_0)/m_0 - (1/2)\{(\bar{y} - m_0)/m_0\}^2$. By algebra and cancelling terms, this implies that $\chi^2 \cong -2n[(\bar{y} - m_0)/m_0 - (1/2)\{(\bar{y} - m_0)/m_0\}^2] - 2n(1 - \bar{y}/m_0) = n\{(\bar{y} - m_0)/m_0\}^2$.
- C. Viewed as a function of random data, the chi-squared statistic is approximately equal to $n\{(\bar{Y} - m_0)/m_0\}^2$. By the CLT, under H_0 , $\bar{Y} \sim N(m_0, m_0^2/n)$, by property of exponential distributions, implying that $(\bar{Y} - m_0)/(m_0/n^{1/2}) \sim N(0, 1)$ by the linearity and additivity properties of the normal distribution. Squaring, you get $n\{(\bar{Y} - m_0)/m_0\}^2 \sim \chi_1^2$ by definition of the chi-squared distribution.
4. Here, $\sum_i \sum_j (y_{ij} - \mu_i)^2 = (36 - \mu_1)^2 + (23 - \mu_1)^2 + \dots + (23 - \mu_1)^2 + (30 - \mu_2)^2 + (25 - \mu_2)^2 + \dots + (36 - \mu_2)^2$. Taking the derivative with respect to μ_1 , all the terms involving μ_2 disappear, so the problem is just like finding the minimizer of $(36 - \mu_1)^2 + (23 - \mu_1)^2 + \dots + (23 - \mu_1)^2$, which is simply the average of the values 36, 23, ..., 23, as shown in Example 2.9. A similar argument applies for μ_2 .
5. Consider the function $f(x) = n \times \ln\{1/(1-x)\} = -n \times \ln(1-x)$. Then $f'(x) = -n/(1-x)(-1) = n/(1-x) > 0$ for all $0 < x < 1$. So the function is monotonically increasing.
- 6.
- | If R^2 is... | Then... |
|-----------------------|---|
| equal to 1.0 | there is a perfect linear relationship between X and Y . |
| between 0.49 and 1.0 | there is a strong linear relationship between X and Y . |
| between 0.09 and 0.49 | there is a moderate linear relationship between X and Y . |
| between 0.0 and 0.09 | there is little to no linear relationship between X and Y . |
| equal to 0.0 | there is no linear relationship between X and Y . |
7. The LR χ^2 is $n \ln \left\{ 1 + \frac{\sum_i (\hat{y}_i - \bar{y})^2 / n}{\hat{\sigma}_1^2} \right\}$ and the model F -statistic is $\frac{\sum_i (\hat{y}_i - \bar{y})^2 / k}{n \hat{\sigma}_1^2 / (n - k - 1)}$. Algebraically, $\chi^2 = n \ln[1 + \{k/(n - k - 1)\}f]$. The result then follows because $c_1 \times \ln(1 + c_2 x)$ is monotonically increasing as shown in the chapter.
8. Here, the x variables do not affect the likelihood. So you can find the MLE for π exactly as shown in Example 12.3.
9. Following substitution of π_6 for $1 - \pi_1 - \dots - \pi_5$, the equations $f_j/\pi_j - f_6/(1 - \pi_1 - \dots - \pi_5) = 0$, for $j = 1, 2, \dots, 5$, unpack to $f_1/\pi_1 - f_6/\pi_6 = 0$; $f_2/\pi_2 - f_6/\pi_6 = 0$; ...; $f_5/\pi_5 - f_6/\pi_6 = 0$. Equivalently, $\pi_6 f_1 = \pi_1 f_6$; $\pi_6 f_2 = \pi_2 f_6$; ...; $\pi_6 f_5 = \pi_5 f_6$. Summing, and using $n = f_1 + f_2 + \dots + f_6$, you get $\pi_6(n$

$-f_6) = (1 - \pi_6)f_6$, or $\pi_6 n - \pi_6 f_6 = f_6 - \pi_6 f_6$, or $\pi_6 n = f_6$. This gives the result for $\hat{\pi}_6$; the rest are obtained by back-substitution.

10.

A. The data value Y_i is the answer of student i , either A, B, C, D, or E.

B. In the restricted model I assume that $Y_1, Y_2, \dots, Y_{670} \sim_{\text{iid}} p_0(y)$, where $p_0(y)$ is as follows:

Answer, y	$p_0(y)$
A	1/5
B	1/5
C	1/5
D	1/5
E	1/5
Total:	1.00

C. In the unrestricted model I assume that $Y_1, Y_2, \dots, Y_{670} \sim_{\text{iid}} p_1(y)$, where $p_1(y)$ is as follows:

Answer, y	$p_1(y)$
A	π_1
B	π_2
C	π_3
D	π_4
E	π_5
Total:	1.00

D. The expected values are $e_j = 670(1/5) = 134$. So the LR χ^2 is $\chi^2 = 2\sum f_i \ln(f_i/e_i) = 2\{120\ln(120/134) + 130\ln(130/134) + 150\ln(150/134) + 160\ln(160/134) + 110\ln(110/134)\} = 12.803$. The p -value is $pv \equiv \Pr(\chi^2_4 \geq 12.803) = 0.012$. The Pearson χ^2 is $\chi^2 = \sum (f_i - e_i)^2/e_i = (120 - 134)^2/134 + \dots + (110 - 134)^2/134 = 12.836$, with p -value also 0.012. Either way, the frequency pattern is not easily explained by random selection of answers.

11. The parameter λ of the Poisson distribution is estimated by $\hat{\lambda} = \{0(35) + 1(34) + 2(24) + 3(6) + 4(1)\}/100 = 1.04$. The calculations are as follows:

Pupfish	Water Scoops	Poisson Probability	Expected Value
0	35	0.353454682	35.35
1	34	0.367592869	36.76
2	24	0.191148292	19.11
3	6	0.066264741	6.63
≥ 4	1	0.021539416	2.15
Total:	100	1.00	100

Then the LR chi-squared statistic is $2\{35\ln(35/35.35) + \dots + 1\ln(1/2.15)\} = 2.203$ and the Pearson chi-squared statistic is $(35 - 35.35)^2/35.35 + \dots + (1 - 2.15)^2/2.15 = 2.137$. The degrees of freedom in either case would be $df = 4$ (four parameters in the full model, none in the restricted), but because you had to estimate a parameter, you lose a degree of freedom, giving three parameters. So the p -values are 0.531 and 0.544 respectively. In either case, the difference between the observed and Poisson expected frequencies is easily explainable by chance, using a model where the data are produced by a Poisson distribution.

12. Using $LL_1 = \sum_{i=1}^5 \sum_{j=1}^5 f_{ij} \ln(f_{ij}/f_{i.})$ and $LL_0 = \sum_{i=1}^5 \sum_{j=1}^5 f_{ij} \ln(f_{ij}/n)$, you get $LL_1 - LL_0 =$

$$\sum_{i=1}^5 \sum_{j=1}^5 f_{ij} \{\ln(f_{ij}/f_{i.}) - \ln(f_{ij}/n)\} = \sum_{i=1}^5 \sum_{j=1}^5 f_{ij} \{\ln[(f_{ij}/f_{i.})/(f_{ij}/n)]\} =$$

$$\sum_{i=1}^5 \sum_{j=1}^5 f_{ij} \{\ln[f_{ij}/\{f_{i.}f_{.j}/n\}]\} = \sum_{i=1}^5 \sum_{j=1}^5 f_{ij} \{\ln[f_{ij}/e_{ij}]\}.$$

13.

- A. $LL_0 = \ln\{(1/3)^4(2/3)^5\} = 4\ln(1/3) + 5\ln(2/3) = -6.421774695$. The MLE for π is $4/9$, so $LL_1 = \ln\{(4/9)^4(5/9)^5\} = 4\ln(4/9) + 5\ln(5/9) = -6.182654189$.
- B. $2(LL_1 - LL_0) = 2\{-6.182654189 - (-6.421774695)\} = 0.4782$. This is called the likelihood ratio chi-squared statistic.
- C. The expected values are $9(1/3) = 3$ and $9(2/3) = 6$. Applying (17.8) gives $2\{4\ln(4/3) + 5\ln(5/6)\} = 0.478$.
- D. $pv = 0.489$. There is one parameter (π) in the unrestricted model but none in the restricted model. So $df = 1 - 0 = 1$.
- E. It is approximate because the distribution of the chi-squared statistic is only *approximately* a chi-squared distribution. The approximation gets better as the sample size gets larger, but the sample size is pretty darn small here ($n = 9$).
- F. $\chi^2 = (4 - 3)^2/3 + (5 - 6)^2/6 = 0.50$, $pv = 0.480$.

14. The MLE for λ is the sample average, 1.04. Plugging this into the Poisson pdf gives the following:

Pupfish	Water Scoops	Poisson Probability	Expected Value
0	35	0.353454682	35.35
1	34	0.367592869	36.76
2	24	0.191148292	19.11
3	6	0.066264741	6.63
≥ 4	1	0.021539416	2.15
Total:	100	1.00	100

So the sum of the log likelihoods for the 100 sampled observations is $35\ln(0.353454682) + 34\ln(0.367592869) + 24\ln(0.191148292) + 6\ln(0.066264741) + 1\ln(0.021539416) = -130.262$, so $AIC_1 = -2(-130.262) + 2(1) = 262.52$. On the other hand, the multinomial log likelihood is $35\ln(0.35) + 34\ln(0.34) + 24\ln(0.24) + 6\ln(0.06) + 1\ln(0.01) = -129.1597299$, so $AIC_2 = -2(-129.160) + 2(4) = 266.32$. Despite its higher log likelihood, the multinomial model is judged worse by the AIC statistic because it has more parameters.

15. Since $LL_1 = (-n/2)\ln(2\pi) - (n/2)\ln(\hat{\sigma}_1^2) - n/2$, $AIC_1 = n\ln(2\pi) + n\ln(\hat{\sigma}_1^2) + n + 2(k + 2)$, where k is the number of X variables in the model. In the restricted model, $AIC_0 = n\ln(2\pi) + n\ln(\hat{\sigma}_0^2) + n + 2(2)$, since there only 2 parameter, β_0 and σ^2 , in the restricted model. $AIC_0 < AIC_1$ if and only if $n\ln(\hat{\sigma}_0^2) < n\ln(\hat{\sigma}_1^2) + 2k$.

16. The sum of squares function to be minimized is $\sum_i \{y_i - (\beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik})\}^2$, which in this example is $(8.99 - (\beta_0 + \beta_1(6.11)))^2 + (5.50 - (\beta_0 + \beta_1(1.80)))^2 + \dots + (4.97 - (\beta_0 + \beta_1(1.18)))^2$. As

with the joint likelihood function, when the derivatives are zero, an extreme of the function is found. Taking the derivative with respect to β_0 and setting to zero gives

$$\begin{aligned}
 &2(8.99 - (\beta_0 + \beta_1(6.11))(-1) + 2(5.50 - (\beta_0 + \beta_1(1.80))(-1) + \dots + 2(4.97 - (\beta_0 + \beta_1(1.18))(-1) = 0, \\
 &\text{or} \\
 &(8.99 - (\beta_0 + \beta_1(6.11))) + (5.50 - (\beta_0 + \beta_1(1.80))) + \dots + (4.97 - (\beta_0 + \beta_1(1.18))) = 0, \\
 &\text{or} \\
 &8.99 + 5.50 + \dots + 4.97 = \beta_0 + \beta_0 + \dots + \beta_0 + (6.11 + 1.80 + \dots + 1.18)\beta_1, \\
 &\text{or} \\
 &58.93 = 14\beta_0 + 26.76\beta_1.
 \end{aligned}$$

Taking the derivative with respect to β_1 and setting to zero gives

$$\begin{aligned}
 &2(8.99 - (\beta_0 + \beta_1(6.11))(-6.11) + 2(5.50 - (\beta_0 + \beta_1(1.80))(-1.80) + \dots + 2(4.97 - (\beta_0 + \beta_1(1.18))(-1.18) = 0, \text{ or} \\
 &(8.99 - (\beta_0 + \beta_1(6.11)))(6.11) + (5.50 - (\beta_0 + \beta_1(1.80)))(1.80) + \dots + (4.97 - (\beta_0 + \beta_1(1.18)))(1.18) = 0, \text{ or} \\
 &8.99(6.11) + 5.50(1.80) + \dots + 4.97(1.18) = (6.11 + 1.80 + \dots + 1.18)\beta_0 + (6.11^2 + 1.80^2 + \dots + 1.18^2)\beta_1 \text{ or} \\
 &143.2522 = 26.76\beta_0 + 89.4792\beta_1.
 \end{aligned}$$

Solving these two equations simultaneously gives the least squares estimates 2.6827252 and 0.798649, which are identical to the MLEs.

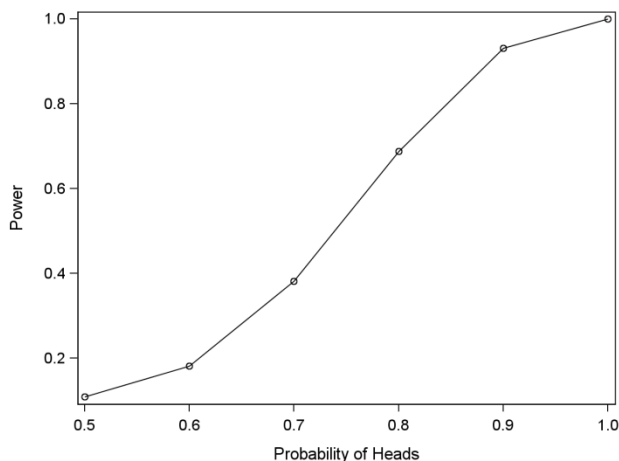
18. Sample Size and Power

Solutions to Exercises

1.

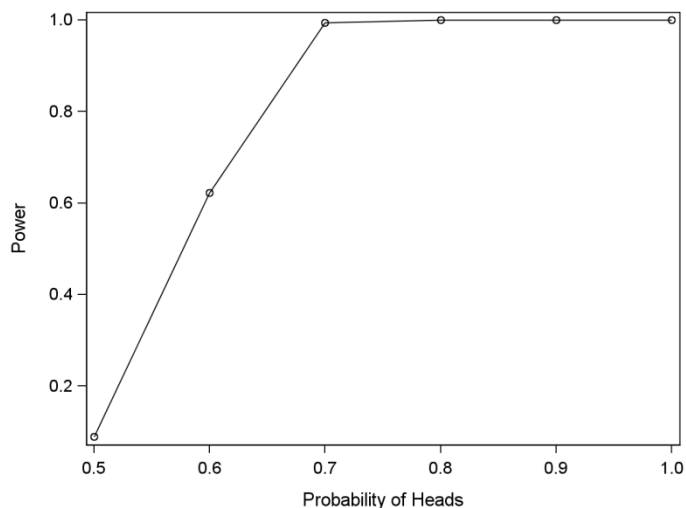
A. $1089/10000 = 0.1089$

B. For $\pi = 0.6$ there are $1812/10000 = 0.1812$ rejections. For $\pi = 0.7$ there are $3815/10000 = 0.3815$ rejections. For $\pi = 0.8$ there are $6877/10000 = 0.6877$ rejections. For $\pi = 0.9$ there are $9305/10000 = 0.9305$ rejections. For $\pi = 1.0$ there are $10000/10000 = 1.0000$ rejections. Here is the graph^{xlv}:



C. With 100 flips, $\hat{\pi} \sim N(0.50, 0.5(1-0.5)/100)$ in the null case, or $\hat{\pi} \sim N(0.50, 0.05^2)$. The type I error rate in the original study was 0.1089, let's call it 10%. So the critical values for $\hat{\pi}$ are the 0.05 and 0.95 quantiles of the $N(0.50, 0.05^2)$ distribution, or 0.418 and 0.582. So Hans should reject H_0 if the number of heads in 100 tosses is either 41 or less, or 59 or more.

D. For $\pi = 0.5$ there are $887/10000 = 0.0887$ rejections. For $\pi = 0.6$ there are $6226/10000 = 0.6226$ rejections. For $\pi = 0.7$ there are $9941/10000 = 0.9941$ rejections. For $\pi = 0.8, 0.9$, and 1.0 there are $10000/10000 = 1.0000$ rejections. Here is the graph^{xlvi}:



Power is much higher with $n = 100$ than it is with $n = 10$.

2.

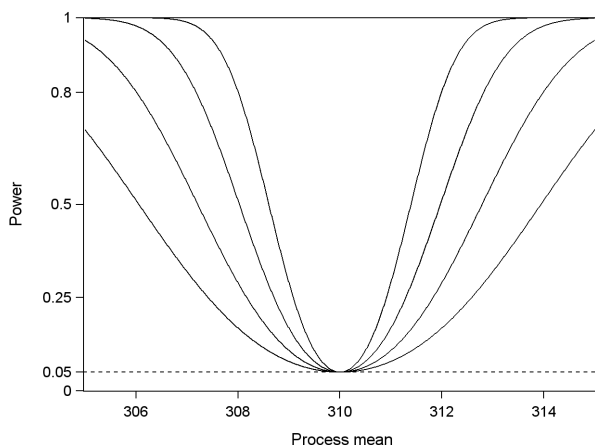
- A. $a.m. = 0.03$ sounds good. $a.m. = 0.50$ is too high since it means you would get within ± 0.50 . So if the estimate was 0.7, the true proportion might be anywhere from 0.20 to 1.20, or 20% to 100%.
- B. $\alpha = 0.05$ is standard. $\alpha = 0.80$ is too high, because it means there is an 80% chance that the true proportion will be more than $a.m.$ away from your estimate.
- C. Maybe 0.50, or 50%. I really am not sure, so I'll pick the middle. But I really doubt that 99% of students own cars. Many of the cars students drive are on loan from their parents. Students don't have much money.
- D. $n = 1.96^2 (0.5)(1 - 0.5)/0.03^2 = 1,067$ students. Changing $a.m.$ to 0.04, you get 600. Changing α to 0.10, but keeping $a.m. = 0.03$ you get 752. The sample size is very sensitive to the initial guesses.

3.

- A. $a.m. = 1.0$ sounds good. $a.m. = 10$ is too high since it means you would get within ± 10 . So if the estimate was 6.0, the true average might be anywhere from -4.0 to 16.0 cups per week.
- B. $\alpha = 0.05$ is standard. $\alpha = 0.80$ is too high, because it means there is an 80% chance that the true average will be more than $a.m.$ away from your estimate.
- C. Maybe 7.0. That's one cup a day. Some drink more some drink less. But I really doubt that it is 70.0 – that would be 10 cups a day.
- D. The square root of the mean is equal to the standard deviation of a Poisson distribution. Assuming the number of cups of coffee drunk is Poisson-like, the guess of σ is therefore $(7.0)^{1/2} = 2.65$.
- E. $n = 1.96^2 (2.65)^2 / 1^2 = 27$ students. Changing $a.m.$ to 0.5, you get 107. Changing α to 0.10, but keeping $a.m. = 1$ you get 19. The sample size is very sensitive to the initial guesses.

4. $E(T_\delta) = E\{N(\delta, 1)\} \times E\{1/\sqrt{\chi_{df}^2/df}\}$, by the product rule for expectation under independence. By Jensen's inequality $E(T_\delta) > \delta$.

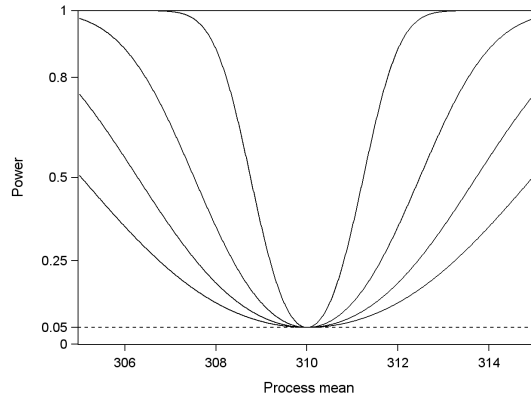
5. Here is Figure 18.5.



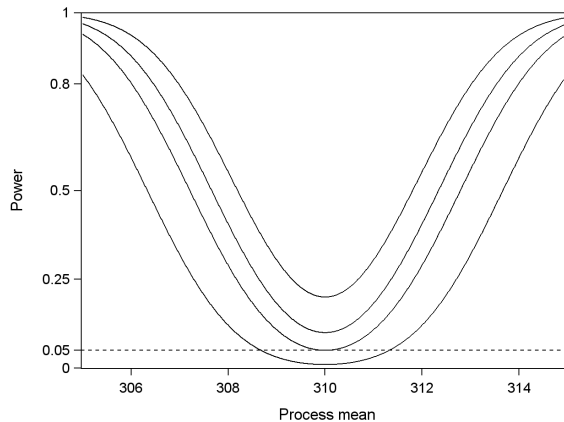
- A. For larger n the power goes to 1 even more quickly as the process mean deviates from the null value (310 in this case).
- B. When the true mean is 310.001 the power will go to 1 for large n .

6.

- A. Standard deviations are 2, 4, 6 and 8; highest power curve is $\sigma = 2^{xlvii}$.



B. α values are 0.20, 0.10, 0.05 and 0.01; highest power is for $\alpha=0.20$ ^{xlviii}.



7.

- A. With $n = 100$ per group, 284 out of 1000 studies yielded $p_v < 0.05$; estimated power is 0.284.
- B. With $n = 200$ per group, 442 out of 1000 studies yielded $p_v < 0.05$; estimated power is 0.442.
- C. With $n = 400$ per group, 770 out of 1000 studies yielded $p_v < 0.05$; estimated power is 0.770.
- D. The plant scientist should plant at least 400 in each group in order to be able to detect that there is a difference between the groups with reasonably high probability, when in fact there is a 10% increase.

8. I'd like the average to be estimated to within an $a.m. = 0.25$ on the five-point scale; $a.m. = 1.0$ seems way too large, and even $a.m. = 0.50$ seems large in that the interval range covers a whole point on scale that spans 4 points total (one to five). The standard deviation can be guessed using a discrete uniform distribution on the numbers 1, 2, ..., 5; this gives $s = 1.4$. The usual $\alpha = 0.05$ seems acceptably low, leading to $n = 1.96^2 (1.4)^2 / 0.25^2 = 123$. Assuming a 10% response rate on the emails, I'd send out about 1,230, hoping for 123 responses.

9.

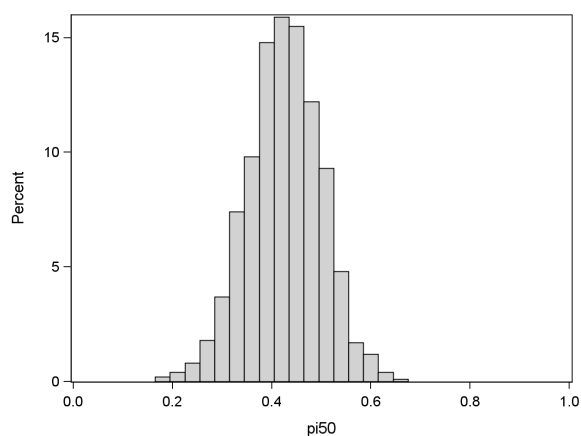
- A. If $\theta = 2.0$, then $\Pr(Y < 0.20) = 0.10$.
- B. If $\theta = 1.0$, then $\Pr(Y \geq 0.20) = 0.80$.
- C. If $\theta = 1.0$, then $\Pr(Y < 0.20) = 0.20$.
- D. The mean of a $U(0,2)$ is 1.0 and the variance is $1/3$. So when $n = 10$, $\bar{Y} \sim N(1.0, (1/3)/10)$, and the 0.10 quantile is 0.766.
- E. If $\theta = 1.0$, then $\bar{Y} \sim N(0.5, (1/12)/10)$, and $\Pr(\bar{Y} < 0.766) \cong 0.998$.

10.

- A. $\chi^2 = 4.933$, .05 critical value ($df=2$) is 5.99, $pv = 0.0849$.
- B. In this example H_0 is false, and we failed to reject H_0 ($pv > 0.05$) so this is a Type II error.
- C. Out of 100 simulated samples of 20 in each group, there were 43 rejections, so the estimated power is 0.43.
- D. Now I get $68/100 = 0.68$. Better.

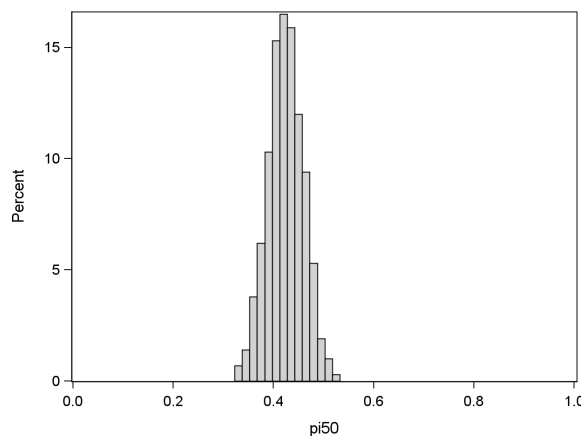
11.

- A. $\hat{\beta}_0 = -5.5859$, $\hat{\beta}_1 = 0.0876$,
 $\hat{\pi}(50) = \exp(-5.5859 + (0.0876)(50)) / \{1 + \exp(-5.5859 + (0.0876)(50))\} = 0.23$.
- B. Based on 1000 simulations, here is the histogram^{xlix} of the estimates of the probability.



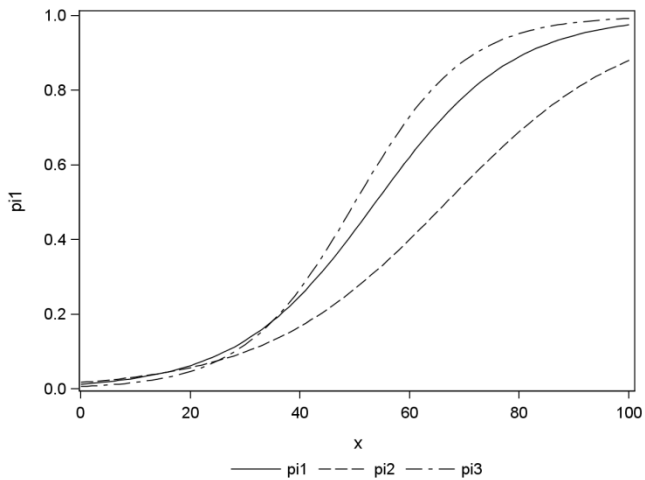
Seems like too much variability in the estimates.

- C. Using $n = 400$ you get the following.

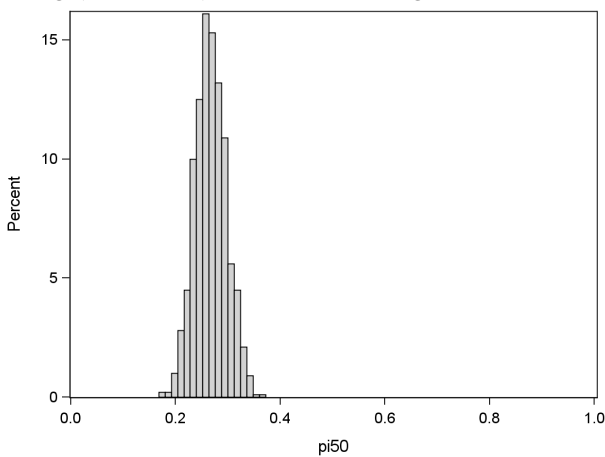


This shows substantially more accuracy and seems adequate. But if the *a.m.* is ± 0.03 you'll need more because it still appears too wide.

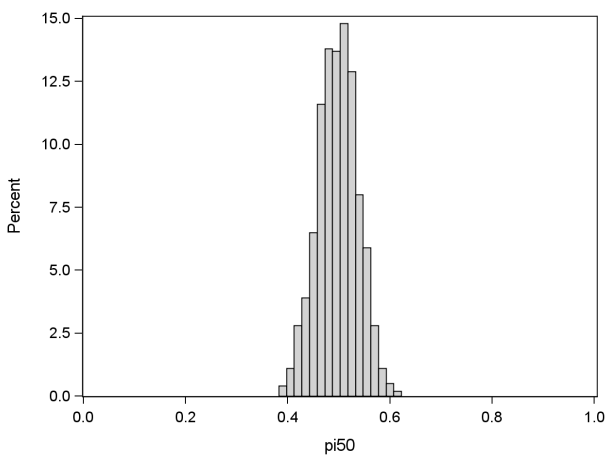
- D. Here are some graphs: pi1 is the original setting $(\beta_0, \beta_1) = (-4.3, 0.08)$, pi2 uses $(-4.0, 0.06)$, and pi3 uses $(-5.0, 0.10)$.



Using $n = 400$ simulations with these two new parameter setting yields the following histograms: For setting $(-4.0, 0.06)$, here is the histogram of estimates of $\pi(50)$:

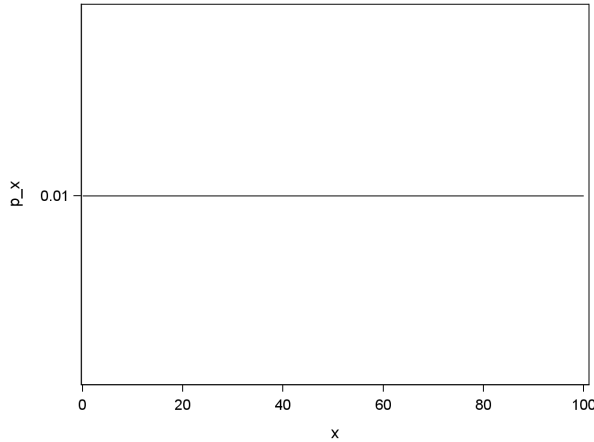


For setting $(-5.0, 0.1)$, here is the histogram of estimates of $\pi(50)$:



All results seem similar, so the accuracy of the estimates seems relatively stable across parameter settings.

E. Here is the graph:



This model would be reasonable if the likelihood of a person with $x_1\%$ burn to appear at the center were the same as the likelihood for a person with $x_2\%$ burn for any $0 < x_1, x_2 < 100$. In particular, the model implies that the likelihood of a person with 0.1% burn to appear at the center is the same as the likelihood for a person with 20% burn.

This model is not reasonable in general: people with tiny amounts of burn never go to the hospital, and people with massive burns never even make it to the hospital. The distribution of X probably should have higher likelihood in the middle values, smaller at the extremes.

12.

- A. $F_{4,20} = 2.74$, 0.05 critical value is 2.87, $pv = 0.058$.
- B. In this example H_0 is false, and we failed to reject H_0 ($pv > 0.05$) so this is a Type II error.
- C. 34 out of 100 samples yielded rejections¹; power is estimated to be 0.34.
- D. 68 out of 100 samples yielded rejections; power is estimated to be 0.68.
- E. The F -statistic is given in Equation (16.14); replacing the estimates with the true values gives the ncp δ . In the case where the n_i are equal to n_w , you get $\delta = n_w \Sigma(\mu_i - \bar{\mu})^2 / \sigma^2$. In this example, $\bar{\mu} = (50 + 50 + 50 + 60 + 60)/5 = 54$. When $n_w = 5$, $\delta = 5 \{(50 - 54)^2 + (50 - 54)^2 + (50 - 54)^2 + (60 - 54)^2 + (60 - 54)^2\} / 10^2 = 6.0$. The power is then $\Pr(V > 2.87)$, where $V \sim F_{4,20,6}$, or Power = 0.3785. When $n_w = 10$, $\delta = 10 \{(50 - 54)^2 + (50 - 54)^2 + (50 - 54)^2 + (60 - 54)^2 + (60 - 54)^2\} / 10^2 = 12.0$. The power is then $\Pr(V > 2.58)$, where $V \sim F_{4,45,12}$, or Power = 0.7535.

19. Robustness and Nonparametric Methods

Solutions to Exercises

1.

A.

Return	Time	Rank	Return	Time	Rank
-0.00114	BEFORE	12	0.002651	AFTER	18
-0.01571	BEFORE	2	0.012301	AFTER	26
0.006127	BEFORE	24	-0.00332	AFTER	10
0.0055	BEFORE	23	-0.00542	AFTER	6
0.001091	BEFORE	15	0.001494	AFTER	17
0.002917	BEFORE	19	0.005037	AFTER	22
0.016819	BEFORE	28	0.00397	AFTER	20
0.001246	BEFORE	16	-0.0049	AFTER	8
0.004446	BEFORE	21	-0.00946	AFTER	5
-0.00152	BEFORE	11	-0.00112	AFTER	13
0.000492	BEFORE	14	-0.01	AFTER	4
-0.00521	BEFORE	7	-0.01653	AFTER	1
-0.00346	BEFORE	9	0.007035	AFTER	25
			0.01415	AFTER	27
			-0.01055	AFTER	3

B. $T_{26} = -0.60$ (after minus before), $pv = 0.55$. The difference between averages is easily explainable by chance alone.

C. $T_{r,26} = -0.57$, $pv = 0.57$. The t -statistic and pv for the rank-transformed two-sample t -test are nearly identical to those of the ordinary two-sample t -test.

D. Returns are not normally distributed. The distribution that produces returns produces more extreme outliers than the normal distribution would predict; i.e., the distribution is heavy-tailed.

E. Using 10,000,000 permuted data sets, the proportion of samples yielding an absolute two-sample rank statistic greater than 0.57 is 0.59. It appears the t -distribution, which gives an approximate p -value of 0.57, provides an adequate approximation.

2. The mean difference is 0.089 and the standard error is 0.3551. The 5th percentile of the distribution of the bootstrap t -statistic (based on 100,000 bootstrap with replacement samples) is -1.627 and the 95th percentile is 2.187. So the bootstrap percentile-t interval is $0.089 - 2.187(0.3551) \leq \mu_D \leq 0.089 + 2.187(0.3551)$, or $-0.687 \leq \mu_D \leq 0.667$.

3.

A. Raw data; $f = 0.82$. Ranked data; $f_r = 1.10$. Critical value ($\alpha = 0.05$) for both is $F_{4,45,0.95} = 2.579$. Both tests made the correct decision.

B. Based on 1000 simulated data sets, the estimated levels are $48/1000 = 0.048$ and $45/1000 = 0.045$. The tests appear to have approximately the correct levels (0.05).

- C. Based on 1000 simulated data sets, the estimated powers are $777/1000 = 0.777$ and $750/1000 = 0.750$. The F test based on the raw data appears to be slightly more powerful.
- D. Based on 1000 simulated data sets, the estimated levels are $14/1000 = 0.014$ and $37/1000 = 0.037$. The rank test appears to have closer to the correct level (0.05).
- E. Based on 1000 simulated data sets, the estimated powers are^{li} $686/1000 = 0.686$ and $1000/1000 = 1.000$. The F test based on the ranked data appears to be far more powerful, hence is more robust for power.

4.

- A. $N(\mu, \sigma^2)$. Clearly the normal model is wrong because the data are discrete. Identical distributions is questionable because different times of the year (e.g., holidays) have higher sales. Independence is questionable because of small term cycles in the economy might cause correlations in adjacent sales days.
- B. $\bar{y} = 2.434$. Plug-in (n formula) $\hat{\sigma}^2 = 3.9963$, Unbiased ($n - 1$ formula) $\hat{\sigma}^2 = (365/364)3.9963 = 4.0073$, giving $\hat{\sigma} = 2.002$. The interval is $2.434 \pm 1.967(2.002/365^{1/2})$, or $2.23 \leq \mu \leq 2.64$.
- C. Based on 100,000 iid samples^{lii} of $n = 365$ each from $\hat{p}(y)$, the 2.5th and 97.5th percentiles of the bootstrap t -statistic are -1.976 and 1.940 , respectively. So the bootstrap percentile- t interval is $2.434 - 1.940(2.002/365^{1/2}) \leq \mu \leq 2.434 - (-1.976)(2.002/365^{1/2})$, or $2.23 \leq \mu \leq 2.64$. The bootstrap and ordinary confidence intervals provide the same results, to two decimals, in this example, so normality is not a concern.
- D. $Y_1, Y_2, \dots, Y_{365} \sim_{\text{iid}} p(y)$. Identical distributions is questionable because different times of the year (e.g., holidays) have higher sales. Independence is questionable because of small term cycles in the economy might cause correlations in adjacent sales days.

5.

A.

	Red	Gray	Green
Younger	5	1	1
Older	1	2	4

Pearson $\chi^2 = 4.8$, the approximate p -value calculated using the chi-square distribution with $df = 2$ is $p_v = 0.091$.

- B. The sample sizes are too small according to Ugly Rule Thumb 17.2, 80% of the expected values should be 5.0 or more. In the given table, all of the six expected values are less than 5.0, so the condition is not satisfied.
- C. 17% of the randomized tables have a Pearson chi-square statistic ≥ 4.8 , so the exact p -value is 0.17.

Appendix. SAS Code for Selected Problems

```

i
data pup;
call streaminit(12345);
do sample = 1 to 10;
do sc = 1 to 100;
fish = rand('Poisson', 1.0);
output;
end; end;
run;

ods output crosstabfreqs = tab;
proc freq;
tables sample*fish/ norow nocol nopct;
run;

ii
data q;
do x = 0 to 12 by .1;
f = (2-x)**2 + (10 - x)**2;
output;
end;
run;

ods rtf style = journal;
proc sgplot;
series y = f x = x;
yaxis label = "f(x)" values = (0 to 100 by 20);
run;
ods rtf close;

iii
data q;
do x = 0 to 30 by .1;
f = 6 + x -0.03*x**2;
output;
end;
run;

ods rtf style = journal;
proc sgplot;
series y = f x = x;
yaxis label = "Preference" ;
xaxis label = "Complexity" ;
run;
ods rtf close;

iv
data q;
do x = 0 to 100 by .1;
f = .01;
output;
end;
run;

ods rtf style = journal;
proc sgplot;
series y = f x = x;
yaxis label = "p(y)" values = (0 to .02 by .01) ;
xaxis label = "y" offsetmin =.1 offsetmax =.1 ;
run;
ods rtf close;

v
data q;
do x = 0 to 2 by .01;
f = (exp(-2)/2 + exp(-x));
output;
end;
run;

ods rtf style = journal;
proc sgplot;
series y = f x = x;
yaxis label = "p(y)" values = (0 to 1.2 by .2);
xaxis label = "y" offsetmin =.1 offsetmax =.1 ;

```

```

run;
ods rtf close;

vi
ods rtf style = journal;
Data d;
input y p;
cards;
0.00    0.50
0.50    0.00
1.20    0.10
1.70    0.20
1.90    0.15
1.95    0.05
;
proc sgplot noautolegend;
scatter y = p x = y;
needle y = p x = y;
xaxis offsetmin = .05 offsetmax = 0.05;
yaxis values = (0 to 1 by .2) label = "p(y)";
run;
ods rtf close;

vii
data g;
input y p distribution;
cards;
1 .1 1
2 .1 1
3 .2 1
4 .3 1
5 .3 1
1 .05 2
2 .15 2
3 .4 2
4 .2 2
5 .2 2
1 .1 3
2 .15 3
3 .35 3
4 .25 3
5 .15 3
;
ods rtf style = journal;
proc sgpanel;
panelby distribution / rows = 3 columns=1 noborder spacing=20;
needle y = p x = y;
* colaxis values = (-.1 to .1 by .05) label = "Today's Return";
run;
ods rtf close;

viii
ods rtf style = journal;
data qq;
do i = 1 to 6;
y = 1.4 + 1.4478*rannor(12345);
output;
end;
run;

proc univariate;
qqplot y;
run;
ods rtf close;

ix
data r;
set isqs5347.djial;
rn = 0.00020019 + 0.01118705*rannor(12345);
run;

ods rtf style = journal;

proc univariate data=r;
var rn;
histogram rn /normal(mu=est sigma=est);
run;
ods rtf close;

```

```

x
data time;
  input t @@;
  cards;
0.48    1.15    0.26    0.05    0.06    0.02    2.12    0.45    0.07    0.99    0.70    1.55    1.72
0.90    0.76    2.03    0.63    0.53    0.30    0.51    0.49    0.52    0.05    0.38    0.43    0.60
0.01    0.11    0.00    0.68    0.02    1.46    0.17    0.10    0.01    0.38    0.60    0.14    0.52
0.13    1.30    0.81    1.37    0.51    0.36    0.34    0.49    0.01    1.60    0.73    2.65    0.04
1.15    0.68    0.13    0.19    0.11    0.16    1.23    1.01
;

proc means;
run;

proc sort data = time;
  by t;
run;

data qq;
  set time;
  q = 0.6158333*quantile('exponential', (_n_-.5)/60);
run;

ods rtf style = Journal;
proc sgplot data = qq noautolegend;
  scatter y = t x = q;
  series y = q x = q;
  xaxis label= "Theoretical Quantile";
  yaxis label = "Ordered Data";
run;
ods rtf close;

xi
data angle;
  input a @@;
  cards;
149    174    309    1    82    9    218    231    49    76    0    219    215
119    119    148    187    231    7    2    3    10    25    86
;

proc sort data = angle;
  by a;
run;

data qq;
  set angle;
  q = 360*(_n_-.5)/23;
run;

ods rtf style = Journal;
proc sgplot data = qq noautolegend;
  scatter y = a x = q;
  series y = q x = q;
  xaxis label= "Theoretical Quantile";
  yaxis label = "Ordered Data";
run;
ods rtf close;

proc corr data = qq;
  var a q;
run;

xii
data u;
  do i = 1 to 1000;
    u1 = ranuni(12345); u2 = ranuni(0);
    y1 = -log(u1); y2 = -log(u1)-log(u2);
    if y1<=1 then subset = "A1"; else subset = "A2";
    output;
  end;
run;

ods rtf style = journal;
proc sgpanel;
  panelby subset / rows = 2 columns=1 noborder spacing=20;
  histogram y2;
run;

```

```
ods rtf close;
```

```
xiii
ods rtf style =journal;
data h;
do x = 40 to 120 by 1;
  do y = 0 to 80 by .5;
    p_xy = pdf('normal', y, .3*x, .04*x)
      * 1/80;
    if y <61 then p_xy=0;
    output; end; end;
run;
```

```
data h;
set h;
label p_xy = 'p(x,y)';
label y = 'House Expense';
label x = 'Income';
run;
```

```
proc g3d data=h;
plot y*x=p_xy ;
run;
quit;
ods rtf close;
```

```
xiv
ods rtf style =journal;
data h;
do x = 40 to 220 by 1;
  do y = 60 to 60;
    p_xy = pdf('normal', y, .3*x, .04*x)
      * pdf('normal', x, 80, 15);
*   if y <61 then p_xy=0;
    output; end; end;
run;
```

```
data h;
set h;
label p_xy = 'p(x,y)';
label y = 'House Expense';
label x = 'Income';
run;
```

```
proc sgplot;
series y = p_xy x = x;
run;
ods rtf close;
```

```
xv
ods rtf style = journal;
data sim;
call streaminit(12345);
do i = 1 to 1000;
  x = rand("normal", 80, 15);
  y = rand("normal", 0.3*x, 0.04*x);
  output;
end;
run;
```

```
proc sgplot;
scatter y = y x = x;
run;
ods rtf close;
```

```
xvi
data m;
do i = 1 to 200 by 1;
  if i < 101 then do; y = i; music = "Off"; py = pdf("normal", y, 50, 15); output; end;
  else do; music = "On "; y = i-100; py = pdf("normal", y, 55, 15); output; end;
end;
run;

ods rtf style = journal;
proc sgpanel;
panelby music / rows = 2 columns=1 spacing=20;
series y = py x = y;
run;
```

```
ods rtf close;
```

```
xvii
data d;
call streaminit(12345);
do i = 1 to 10000;
  d1 = rand('table', 1/6, 1/6, 1/6, 1/6, 1/6, 1/6);
  d2 = rand('table', 1/6, 1/6, 1/6, 1/6, 1/6, 1/6);
  d3 = rand('table', 1/6, 1/6, 1/6, 1/6, 1/6, 1/6);
  d4 = rand('table', 1/6, 1/6, 1/6, 1/6, 1/6, 1/6);
  d5 = rand('table', 1/6, 1/6, 1/6, 1/6, 1/6, 1/6);
  do; Ave = d1; n = 1; output; end;
  do; Ave = mean(of d1-d2); n=2; output; end;
  do; Ave = mean(of d1-d3); n=3; output; end;
  do; Ave = mean(of d1-d4); n=4; output; end;
  do; Ave = mean(of d1-d5); n=5; output; end;
end;
```

```
run;

ods rtf style=journal;
proc sgpanel;
  panelby n/ rows = 5 columns = 1;
  vbar Ave / stat = freq;
run;
ods rtf close;
```

```
xviii
%let nsimu = 200000;
data simu;
call streaminit(929045);
ybar_n = 0;
do n = 1 to &nsimu;
  x1 = rand('table', 1/6,1/6,1/6,1/6,1/6,1/6);
  x2 = rand('table', 1/6,1/6,1/6,1/6,1/6,1/6); y = x1/x2;
  true_mean = 1.429;
  Ybar_n = ((n-1)/n)*ybar_n + (1/n)*y; /* running average of the observations */
  output; end;
run;
```

```
ods rtf style = journal;
proc sgplot data=simu noautolegend;
  series x = n y = ybar_n;
  series x = n y = true_mean / lineattrs= (pattern=dot) ;
  xaxis type=log logbase=10 label = "n";
  yaxis values = (0 to 6 by .5) label = "Average of First n Die1/Die2";
run;
ods rtf close;
```

```
xix
data l;
call streaminit(12345);
do rep = 1 to 10000;
  sum = 0;
  do i = 1 to 1000;
    y = rand('table', .05, .15, .15, .15, .50);
    sum = sum + y;
  end;
  ybar = sum/1000; output;
end;
run;
```

```
data l; set l;
chk1 =(ybar >3.85889039)*(ybar <3.94110961);
chk2 =(ybar >3.817780781)*(ybar <3.982219219);
chk3 =(ybar >3.776671171)*(ybar <4.023328829);
run;
```

```
proc means; var chk1 chk2 chk3; run;
```

```
ods rtf style=journal;
proc sgplot data=l;
  histogram ybar;
run;
ods rtf close;
```

```
xx
data q;
do i = 1 to 10000;
```

```

    Z1 = rannor(12345); Z2 = rannor(0);
output; end;
run;

ods rtf style = journal;
proc sgplot;
    scatter y = z2 x = z1/markerattrs = (size = 2);
run;

proc corr;
    var z1 z2;
run;
ods rtf close;

xxi
data q;
    do i = 1 to 10000;
        Z1 = rannor(12345); Z2 = rannor(0); Z3 = rannor(0);
        SumZsq = z1**2 + z2**2 + z3**2;
    output; end;
run;

ods rtf style = journal;
proc sgplot;
    histogram SumZsq;
run;

ods rtf close;

xxii
ods rtf style = journal;
data clt;
    do ybar = 0 to 1 by .001;
        p_ybar = pdf('normal', ybar, .45, sqrt(.45*(1-.45)/1000));
    output;
    end;
run;

proc sgplot;
    series y = p_ybar x = ybar;
    xaxis label = "Estimated Proportion" values = (0 .2 .45 .60 .8 1);
    yaxis label = "Approximate pdf";
run;
ods rtf close;

xxiii
data rat;
call streaminit(12545);
sx = 0 ; sy = 0;
do n = 1 to 200000;
    x = rand('table', 1/6,1/6,1/6,1/6,1/6,1/6);
    y = rand('table', 1/6,1/6,1/6,1/6,1/6,1/6);
    sx = sx+x; xbar = sx/n;
    sy = sy+y; ybar = sy/n;
    est = 100*(ybar/xbar - 1);
    target = 0;
    output;
end;
run;

ods rtf style = journal;
proc sgplot data=rat noautolegend;
    series x = n y = est;
    series x = n y = target / lineattrs= (pattern=dot) ;
    xaxis type=log logbase=10 label = "n";
    yaxis values = (-100 to 100 by 50) label = "Estimated Percent Difference";
run;
ods rtf close;

xxiv
data exp;
call streaminit(12545);
do rep = 1 to 10000;
    y1 = 4*rand('exponential');
    y2 = 4*rand('exponential');
    y3 = 4*rand('exponential');
    y4 = 4*rand('exponential');
    y5 = 4*rand('exponential');

```

```

        y6 = 4*rand('exponential');
        y7 = 4*rand('exponential');
        y8 = 4*rand('exponential');
        y9 = 4*rand('exponential');
        y10 = 4*rand('exponential');
        y11 = 4*rand('exponential');
        y12 = 4*rand('exponential');
        y13 = 4*rand('exponential');
        y14 = 4*rand('exponential');
        y15 = 4*rand('exponential');
        y16 = 4*rand('exponential');
        y17 = 4*rand('exponential');
        y18 = 4*rand('exponential');
        y19 = 4*rand('exponential');
        y20 = 4*rand('exponential');
        ave = mean(of y1-y20); std = std(of y1-y20);
        output;
end;
run;

proc means n mean std var;
    var ave std;
run;

xxv
data one;
sum = 0;
do n = 1 to 100000;
    y1 = rannor(12345);
    sum = sum + y1;
    est = sum/n + 1/n;
    true_mean = 0;
    output; end;

ods rtf style = journal;
proc sgplot data =one noautolegend;
series x = n y = est;
series x = n y = true_mean / lineattrs= (pattern=dot) ;
xaxis type=log logbase=10 label = "n";
yaxis label = "Biased Estimate";
run;
ods rtf close;

xxvi
data l;
do pi = 0 to 1 by .001;
    L = pi**4 * (1 - pi)**5;
    output;
end;
run;

ods rtf style = journal;
proc sgplot;
series y = L x = pi;
run;
ods rtf close;

xxvii
ods rtf style = journal;
data f;
do pi2 = 0 to 1 by .0001;
    pi1 = 1 - pi2;
    output;
end;
proc sgplot;
band upper = pi2 lower = 0 x = pi1;
run;
ods rtf close;

xxviii
ods rtf style = journal;
data f;
do lambda = 0 to 5 by .001;
    L = lambda**7 * exp(-11*lambda)/48;
    output;
end;
proc sgplot;
series y = L x = lambda;

```

```

run;
ods rtf close;

xxix
ods rtf style = journal;
data f;
do lambda = .05 to 10 by .001;
mean = 1/lambda;
L = lambda* exp(-lambda*2);
output;
end;
proc sgplot;
series y = L x = mean;
run;
ods rtf close;

xxx
ods rtf style = journal;
data l;
b01 = -1; b11 = 0.5;
b02 = -1; b12 = 0;

do x = 0 to 10 by .01;
p1 = exp(b01 + b11*x)/(1+exp(b01 + b11*x));
p2 = exp(b02 + b12*x)/(1+exp(b02 + b12*x));
output;
end;
proc sgplot;
series y = p1 x = x;
series y = p2 x = x;
yaxis values = (0 to 1 by .2) label = "Probability";
run;
ods rtf close;

xxxi
data h;
input x y @@;
yhat1 = 2.683 + .799*x;
yhat2 = -.224 + 2.815*x;
cards;
6.11 8.99 1.8 5.5 2.32 7.14 1.17 5.59 5.28 1.58 0.62 2.93 0.68
      0.81 0.43 0.47 1.18 3.73 2.2 9.64 1.24 0.62 1.92 6.03 0.63
      0.93 1.18 4.97
;

ods rtf style = journal;
proc sgplot noautolegend;
series y = yhat1 x = x;
series y = yhat2 x = x;
scatter y = y x = x;
run;
ods rtf close;

xxxii
ods rtf style = journal;
data l;
do theta = 0 to 4 by .001;
if theta < 2.39 then Likelihood = 0; else Likelihood = 1/theta**30;
output;
end;
proc sgplot;
series y = Likelihood x = theta;
run;
ods rtf close;

xxxiii
ods rtf style = journal;
data l;
do y = 0 to 1 by .001;
py = 39.72 * y**(39.72 - 1);
output;
end;
proc sgplot;
series y = py x = y;
run;
ods rtf close;

```

```

xxxiv data h;
do x = 1 to 5 by 1;
m1 = exp(0 + 0*x);
m2 = exp(0 - .5*x);
m3 = exp(1 + 1*x);
m4 = exp(2 - .5*x);
mmle = exp(2.259136484 - 0.71626629*x);
output;
end;
run;

ods rtf style = journal;
proc sgplot;
series y = m1 x = x;
series y = m2 x = x;
series y = m3 x = x;
series y = m4 x = x;
series y = mmle x = x;
yaxis values = (0 to 10 by 1) label = "Mean Number in Line";
xaxis label = "lines";
run;
ods rtf close;

xxxv data r;
input x y;
yt = y/x;
xt = 1/x;
cards;
6.11 8.99
1.8 5.5
2.32 7.14
1.17 5.59
5.28 1.58
0.62 2.93
0.68 0.81
0.43 0.47
1.18 3.73
2.2 9.64
1.24 0.62
1.92 6.03
0.63 0.93
1.18 4.97
;

proc mcmc data=r outpost=classout nmc=50000 thin=5 seed=246810;
parms beta0 0 betal 0;
parms sigma2 1;
prior beta0 betal ~ normal(mean = 0, var = 1e6);
prior sigma2 ~ general(-log(sigma2));
mu = beta0 + betal*x;
var1 = x**2 * sigma2;
model y ~ n(mu, var = var1);
run;

data c;
set classout;
cb = betal > 0;
m10 = beta0 + betal*10;
CH10 = (BETA0+BETA1*10 > 10);
run;

proc univariate;
var cb m10 ch10;
run;

xxxvi data sep;
input x y;
cards;
1 4
1 0
4 0
2 1
2 4
1 8
1 4
3 1

```



```

4      1
1      6
2      4
4      0
;
proc genmod;
  model y = x / dist = poisson;
  bayes coeffprior=jeffreys /* A vague, or non-informative prior */
  seed = 12342321 outpost=posterior(rename=(intercept=beta0 x=beta1));
run;

data p;
  set posterior;
  chk = beta1 > 0;
  mean2 = exp(beta0 + beta1*2);
  Ystar = ranpoi(12345, mean2);
run;

proc univariate;
  var beta1 chk mean2;
  qqplot beta1;
run;

proc freq;
  tables ystar;
run;

xxxvii
  data n;
  do z = -3 to 3 by .001;
    p = pdf('normal', z, 0 ,1);
    output;
  end;
run;

ods rtf style = journal;
proc sgplot;
  series y = p x = z;
  yaxis label = "N(0,1) density";
  refline 1.6445 1.86 2.578 / axis = x;
run;
ods rtf close;

xxxviii
  data n;
  do z = -3 to 3 by .001;
    p = pdf('normal', z, 0 ,1);
    output;
  end;
run;

ods rtf style = journal;
proc sgplot;
  series y = p x = z;
  yaxis label = "N(0,1) density";
  refline 1.6445 1.86 2.578 / axis = x;
run;
ods rtf close;

xxxix
  data ind;
  call streaminit(12345);
  do i = 1 to 1000;
    do j = 1 to 12;
      x = rand('table', 4/12, 0/12, 2/12, 2/12, 4/12);
      y = rand('table', 4/12, 4/12, 3/12, 0/12, 1/12);
      output;
    end; end;
run;

ods listing close;
ods output pearsoncorr = pc;
proc corr data = ind ;
  by i;
  var x y;
run;
ods listing;

ods rtf style = journal;

```

```

proc sgplot data = pc(where = (Variable = 'y'));
    histogram x;
    xaxis label = "Correlation";
run;
ods rtf close;

data chk;
    set pc;
    if variable = 'y';
    more_extreme = (x >= 0.145) + (x <= -0.145);
run;

proc means data = chk;
    var more_extreme;
run;

xl

data ind;
    call streaminit(12345);
    do i = 1 to 1000;
        do j = 1 to 33;
            x = rand('table', 6/33, 3/33, 7/33, 12/33, 5/33);
            y = rand('table', 6/33, 5/33, 7/33, 10/33, 5/33);
            output;
        end; end;
run;

ods listing close;
ods output pearsoncorr = pc;
proc corr data = ind ;
    by i;
    var x y;
run;
ods listing;

ods rtf style = journal;
proc sgplot data = pc(where = (Variable = 'y'));
    histogram x;
    xaxis label = "Correlation";
run;
ods rtf close;

data chk;
    set pc;
    if variable = 'y';
    more_extreme = (x >= 0.74613) + (x <= -0.74613);
run;

proc means data = chk;
    var more_extreme;
run;

xli
ods rtf style= journal;
proc iml;
    call randseed(12345);
    a = ranperm( { 36      23      22      27      26      24      28      23      30      25      22
                  26      22      35      24      36 }, 100000);
    create p from a;
    append from a;
    * print a;
    quit;

data p;
    set p;
    v1 = var(of col1 - col8);
    v2 = var(of col9 - col16);
    r = v1/v2;
    sig = (r >= 1/.6615) + (r<=.6615);
run;

proc means;
    var sig;
run;

proc sgplot data=p;
    histogram r;
    xaxis label = "Estimated Variance Ratio";

```

```

run;
ods rtf close;

xlii
data sim;
  call streaminit(12345);
  do sim = 1 to 1000;
    do i = 1 to 60;
      age = rand('exponential');
      output;
    end;
  end;
run;

proc sort data = sim out=simout;
  by sim age;
run;

data simout;
  set simout;
  n = mod(_n_,60);
  if n = 0 then n=60;
  q = quantile('exponential', (n-.5)/60);
run;

ods listing close;
ods output pearsoncorr=corr;
proc corr data=simout;
  by sim;
  var q age;
run;

data corr; set corr;
  if variable = "q";
  corr = age;
  chk = corr <= 0.99373;
  keep corr chk;
run;

ods listing;

proc means data = corr;
  var chk;
run;

ods rtf style=journal;
proc sgplot data=corr;
  histogram corr;
  xaxis label = "Correlation Coefficient" offsetmax=0;
  refline .99373 / axis = x;
run;
ods rtf close;

xliii
data t;
  input y;
  time = _n_;
  cards;
12.3
16.5
56.7
45.3
29.5
23.8
67.9
100.1
140.2
86.6
76.6
188.7
146.0
;
ods rtf style = journal;
proc sgplot data = t noautolegend;
  scatter y = y x = time;
  series y = y x = time;
  xaxis values = (0 to 14 by 2);
run;
ods rtf close;

```

```

proc corr data = t;
  var y time;
run;

xliv
data;
  do i = 1 to 1000;
    y1 = rannor(0)<0; y2 = rannor(0)<0; t = y1 + y2; output;
  end;
run;

proc sgplot;
  vbar t / stat = freq;
run;

xlv
ods rtf style=journal;
data p;
  input p pr;
  cards;
    .5 .1089
    .6 .1812
    .7 .3815
    .8 .6877
    .9 .9305
    1.0 1
  ;
proc sgplot noautolegend;
  scatter y = pr x =p;
  series y = pr x =p;
  xaxis label = "Probability of Heads";
  yaxis label = "Power";
run;
ods rtf close;

xlvi
ods rtf style=journal;
data p;
  input p pr;
  cards;
    .5 .1089
    .6 .1812
    .7 .3815
    .8 .6877
    .9 .9305
    1.0 1
  ;
proc sgplot noautolegend;
  scatter y = pr x =p;
  series y = pr x =p;
  xaxis label = "Probability of Heads";
  yaxis label = "Power";
run;
ods rtf close;

xlvii
ods rtf style = journal;
data p;
  m0 = 310;
  do m1 = 305 to 315 by .01;
    p2 = cdf('normal',10**(.5*(m0 -m1)/2 -1.96 , 0, 1) +
      (1 - cdf('normal',10**(.5*(m0 -m1)/2 +1.96 , 0, 1)));
    p4 = cdf('normal',10**(.5*(m0 -m1)/4 -1.96 , 0, 1) +
      (1 - cdf('normal',10**(.5*(m0 -m1)/4 +1.96 , 0, 1)));
    p6 = cdf('normal',10**(.5*(m0 -m1)/6 -1.96 , 0, 1) +
      (1 - cdf('normal',10**(.5*(m0 -m1)/6 +1.96 , 0, 1)));
    p8 = cdf('normal',10**(.5*(m0 -m1)/8 -1.96 , 0, 1) +
      (1 - cdf('normal',10**(.5*(m0 -m1)/8 +1.96 , 0, 1)));
    output;
  end;
run;

proc sgplot data=p noautolegend;
  series y = p2 x = m1 / lineattrs = (color = black pattern = 1);
  series y = p4 x = m1 / lineattrs = (color = black pattern = 1);
  series y = p6 x = m1 / lineattrs = (color = black pattern = 1);
  series y = p8 x = m1 / lineattrs = (color = black pattern = 1);

```

```

    yaxis label = "Power" offsetmin = 0 offsetmax=0 values = (0 .05 .25 .5 .8 1);
    xaxis label = "Process mean";
    reflate .05 / axis = y lineattrs = (color = black pattern = 2);
run;
ods rtf close;

xlviii
    ods rtf style = journal;
data p;
    m0 = 310;
    do m1 = 305 to 315 by .01;
        q20 = probit(.90);
        p20 = cdf('normal',10*.5*(m0-m1)/4.5-q20, 0, 1) +
            (1 - cdf('normal',10*.5*(m0-m1)/4.5+q20, 0, 1));
        q10 = probit(.95);
        p10 = cdf('normal',10*.5*(m0-m1)/4.5-q10, 0, 1) +
            (1 - cdf('normal',10*.5*(m0-m1)/4.5+q10, 0, 1));
        q05 = probit(.975);
        p05 = cdf('normal',10*.5*(m0-m1)/4.5-q05, 0, 1) +
            (1 - cdf('normal',10*.5*(m0-m1)/4.5+q05, 0, 1));
        q01 = probit(.995);
        p01 = cdf('normal',10*.5*(m0-m1)/4.5-q01, 0, 1) +
            (1 - cdf('normal',10*.5*(m0-m1)/4.5+q01, 0, 1));
        output;
    end;
run;

proc sgplot data=p noautolegend;
    series y = p20 x = m1 / lineattrs = (color = black pattern = 1);
    series y = p10 x = m1 / lineattrs = (color = black pattern = 1);
    series y = p05 x = m1 / lineattrs = (color = black pattern = 1);
    series y = p01 x = m1 / lineattrs = (color = black pattern = 1);
    yaxis label = "Power" offsetmin = 0 offsetmax=0 values = (0 .05 .25 .5 .8 1);
    xaxis label = "Process mean";
    reflate .05 / axis = y lineattrs = (color = black pattern = 2);
run;
ods rtf close;

xlix
data l;
call streaminit(12345);
do rep = 1 to 1000;
do i = 1 to 100;
    x = 100*RAND('uniform');
    pi = exp(-4.3 + .08*x)/(1+ exp(-4.3 + .08*x));
    Y = rand('uniform') < pi;
    output;
end;end;
run;

ods listing close;
ods output parameterestimates = b;
proc logistic;
    by rep;
    model y(event='1') = x;
run;

proc transpose data =b out=b1;
    var estimate;
    by rep;
run;

data b1;
    set b1;
    pi50 = exp(col1 + col2*50)/(1+ exp(col1 + col2*50));
run;

ods listing;
ods rtf style = journal;
proc sgplot;
    histogram pi50;
    xaxis values = (0 to 1 by .2);
run;
ods rtf close;

|
data f;
call streaminit(12345);
do rep =1 to 2000;

```

```

do g = 1 to 5;
  do i = 1 to 5;
    if g < 4 then y = rand('normal', 50, 10);
    else y = rand('normal', 60, 10);
    output;
  end;
end; end;

run;
ods listing close;
ods output modelanova=ft;
proc glm data=f;
  by rep;
  class g;
  model y = g;
run; quit;

daTA ft;
set ft;
sig = probf <.05;
run;
ods listing;
proc means;
  var sig; run;

li
  data f;
do rep = 1 to 1000;
  do g = 1 to 5;
    do i = 1 to 10;
      if g < 4 then y = 50 + randn(12345); else y = 60 + randn(12345);
      output;
    end;
  end;
end;
run;

proc rank data =f out=fr;
  by rep;
  var y;
  ranks yr;
run ;
ods listing close;
ods output modelanova =ma;
proc glm data = fr;
  by rep;
  class g;
  model y yr = g;
run;
quit;

data resy;
  set ma;
  if HypothesisType = "1";
  if Dependent = "y";
  sigy = probf < .05;
run;
ods listing;
proc means; var sigy; run;
data resyr;
  set ma;
  if HypothesisType = "1";
  if Dependent = "yr";
  sigyr = probf < .05;
run;
proc means; var sigyr; run;

lii
  data b;
call streaminit(12345);
do rep = 1 to 100000;
  sum = 0;
  sum2 = 0;
  do i = 1 to 365;
    s = rand('table', .082, .219, .219, .247, .129, .104);
    s = s-1;
    sum = sum + s;
  sum2 = sum2 + s**2;
  end;
  means = sum/365;

```

```
means2 = sum2/365;
varp = means2 - means**2;
var = (365/364)*varp;
t = (means - 2.434)/sqrt(var/365);
output;

end;
run;

proc univariate; var t;
  output pctlpts = 2.5, 97.5 pctlpre=boot_t_crit_ pctlname=l u out=p;
run;
```




CRC Press

Taylor & Francis Group
an informa business

www.taylorandfrancisgroup.com

6000 Broken Sound Parkway, NW
Suite 300, Boca Raton, FL 33487

270 Madison Avenue
New York, NY 10016

2 Park Square, Milton Park
Abingdon, Oxon OX14 4RN, UK

K22460

